MATH 142: Elementary Algebraic Topology

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Contents

8/26/26 Lecture 1: Metric Spaces . . . 1
Definition 1.1: Metrics . . . 1
Definition 1.2: Balls . . . 1
Definition 1.3: Open Sets . . . 1
8/28/26 Lecture 2: More Topology . . . 2
Definition 2.3: Closed Sets . . . 2
Definition 2.5: Metric Subspaces . . . 3
Definition 2.7: Product Spaces . . . 4
8/31/26 Lecture 3: Topology Continued . . . 5
Definition 3.1: Topologies . . . 5
Definition 3.2: Subspace Topology . . . 5
Definition 3.3: Neighborhoods . . . 5
Definition 3.4: Closed Sets . . . 5
Definition 3.5: Limit Points . . . 6
Definition 3.7: Closure of a Set . . . 6
Definition 3.10: Dense Sets . . . 7
Definition 3.11: Interior . . . 7
Definition 3.12: Frontier . . . 7
9/2/31 Lecture 4: Topological Spaces and Bases . . . 8
Definition 4.1: Basis . . . 8
Definition 4.2: Continuous Functions . . . 8
9/4/26 Lecture 5: Homeomorphisms and Space Filling Curves . . . 11
Definition 5.1: Homeomorphisms . . . 11
9/9/26 Lecture 6: More Topological Properties . . . 13
Definition 6.1: Hausdorff Spaces . . . 13
Definition 6.2: Distance to a Set . . . 13
Theorem 6.6: Tietze Extension Theorem . . . 14
Definition 6.7: Covers . . . 14
Definition 6.8: Compactness . . . 14
Theorem 6.9: Heine-Borel Theorem . . . 14
9/11/26 Lecture 7: Compact and Hausdorff Spaces . . . 16
Definition 7.2: Compact Subspace . . . 16
Theorem 7.8: Mini Heine-Borel Theorem . . . 18
Theorem 7.9: Extreme Value Theorem . . . 18
9/14/26 Lecture 8: Product Spaces . . . 19
Definition 8.2: Product Spaces . . . 19
Definition 8.3: Projections . . . 19
9/16/26 Lecture 9: Compactness and Connectedness . . . 22
Theorem 9.1: Tychonoff's Theorem . . . 22
Definition 9.3: Connectedness . . . 23
9/19/26 Lecture 10: Connectedness . . . 25
Definition 10.3: Separated Sets . . . 25

Definition 10.6: Connected Components . . . 26
9/21/26 Lecture 11: Path Connectedness . . . 28
Definition 11.1: Paths . . . 28
Definition 11.2: Path-Connectedness . . . 28
Definition 11.6: Path Components . . . 30
Definition 11.8: $\pi_0(X)$ . . . 30
9/23/26 Lecture 12: Identification Spaces . . . 32
Definition 12.1: Identification Space . . . 32
Definition 12.2: Identification Topology . . . 33
Definition 12.3: Identification Maps . . . 33
Definition 12.8: Cone over a Topological Space . . . 35
9/25/26 Lecture 13: Identification Spaces (cont.) and Topological Groups . . . 36
Definition 13.1: Gluing Maps Together . . . 37
Lemma 13.2: The Gluing Lemma . . . 38
Definition 13.3: Topological Groups . . . 38
9/28/26 Lecture 14: More on Topological Groups . . . 40
Definition 14.1: Left and Right Multiplication Maps . . . 40
Definition 14.4: Homogeneous Space . . . 40
9/30/26 Lecture 15: Group Actions . . . 43
Definition 15.2: Orbits of Group Actions . . . 44
Definition 15.3: Orbit Spaces . . . 44
Definition 15.4: Topological Groups acting on Topological Spaces . . . 44
10/2/26 Lecture 16: Topological Groups cont. and Intro to Algebraic Topology . . . 47
Definition 16.2: Loops . . . 49
Definition 16.3: Homotopies . . . 50
10/5/26 Lecture 17: The Fundamental Group . . . 51
Definition 17.3: Homotopy Classes . . . 52
Definition 17.4: Multiplication of Homotopy Classes . . . 53
Definition 17.7: The Fundamental Group . . . 53
10/7/26 Lecture 18: The Fundamental Group (cont.) . . . 54
Theorem 18.1: Homotopy Classes form a Group . . . 54
Definition 18.6: Diameter . . . 56
Lemma 18.7: Lebesgue Lemma . . . 56
10/9/26 Lecture 19: The Fundamental Group of the Circle . . . 58
Theorem 19.3: Fundamental Group of the Circle . . . 59
10/12/26 Lecture 20: Brouwer Fixed Point Theorem and Simply Connected Spaces . . . 61
Definition 20.1: Fixed Points . . . 61
Theorem 20.3: 2D Brouwer Fixed Point Theorem . . . 61
Definition 20.5: Simply Connected Spaces . . . 62
Definition 20.8: Free Actions . . . 64
10/14/26 Lecture 21: Fundamental Groups of Quotients and Products . . . 65
Theorem 21.2: Fundamental Group of a Product Space . . . 66
Definition 21.3: Homotopy Equivalence . . . 67
Definition 21.5: Algebraic Topology . . . 68
10/16/26 Lecture 22: Homotopy Equivalence and Fundamental Groups . . . 69
Definition 22.1: Manifolds . . . 69
Definition 22.2: Aspherical Manifolds . . . 69
Definition 22.3: Deformation Retractions . . . 70
Theorem 22.4: Homotopy Theorem . . . 70
Theorem 22.5: Generalized Homotopy Theorem . . . 71
Theorem 22.6: Fundamental Groups of Homotopy Equivalent Spaces . . . 71
10/19/26 Lecture 23: Contractible Spaces, Retractions, and Simplicial Complexes . . . 72
Definition 23.2: Contractible Spaces . . . 72
Definition 23.4: Retractions . . . 73

Definition 23.6: Hyperplanes . . . 74
Definition 23.7: General Position . . . 74
Definition 23.8: Simplices . . . 74
Definition 23.9: Faces . . . 74
Definition 23.10: Simplicial Complexes . . . 75
10/21/26 Lecture 24: Triangulations, Cones, and Barycentric Subdivision . . . 76
Definition 24.1: Triangulations . . . 76
Definition 24.2: Isomorphisms of Simplicial Complexes . . . 77
Definition 24.3: Interiors of Simplices . . . 77
Definition 24.5: Barycenters . . . 78
Definition 24.6: Barycentric Subdivision . . . 78
10/26/26 Lecture 25: Simplicial Maps and Simplicial Approximation . . . 80
Definition 25.1: Iterated Barycentric Subdivision . . . 80
Definition 25.2: Dimension . . . 80
Definition 25.3: Mesh . . . 80
Definition 25.5: Simplicial Vertex Maps . . . 80
Definition 25.6: Carriers . . . 81
Definition 25.7: Simplicial Approximations . . . 81
Definition 25.9: Stars . . . 82
Theorem 25.10: Simplicial Approximation Theorem (Alexander) . . . 82
10/28/26 Lecture 26: Free Groups, Presentations, and Maximal Trees . . . 83
Definition 26.1: Generators and Free Groups . . . 83
Definition 26.2: Words . . . 83
Definition 26.3: Finitely Presented Groups . . . 83
Definition 26.4: Free Products . . . 83
Definition 26.5: Trees . . . 84
10/30/26 Lecture 27: Edge Paths and the Edge Group . . . 86
Definition 27.1: Edge Paths . . . 86
Definition 27.2: Equivalent Edge Paths . . . 86
Theorem 27.3: Equivalence of $E(K, v)$ and $\pi_1(|K|, v)$ . . . 86
Definition 27.4: The Group $G(K, L)$ . . . 87
11/2/26 Lecture 28: Presentations of the Edge Group . . . 88
Theorem 28.1: Equivalence of $G(K, l)$ and $E(K, v)$ . . . 88
11/4/26 Lecture 29: Van Kampen's Theorem and Abstract Simplicial Complexes . . . 92
Theorem 29.4: Van Kampen's Theorem . . . 92
Definition 29.5: Finite Abstract Simplicial Complexes . . . 94
11/6/26 Lecture 30: Orientations, Chains, and Boundaries . . . 96
Definition 30.1: Orientation . . . 96
Definition 30.2: Induced Orientation . . . 96
Definition 30.3: Generating Sets . . . 97
Definition 30.4: Free Abelian Groups . . . 97
Definition 30.5: Abelian Words . . . 97
Definition 30.6: Chain Groups . . . 98
Definition 30.7: Chains . . . 98
Definition 30.8: Boundary . . . 98
Definition 30.9: Cycles and Boundaries . . . 99

8/26/26 Lecture 1: Metric Spaces

Course Overview: We will be "assigning numbers to shapes", that is; algebraic invariants to topological spaces. We start by introducing metric spaces.

Definition 1.1: Metrics

Let X be a set. A *metric* $d : X \times X \to \mathbb{R}$ is a map such that

1. $d(x, y) \geq 0$
2. $d(x, y) = 0$ iff $x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$

We then call (X, d) a *metric space*.

The example that most people are used to is $\mathbb{R}$ with the metric $d(x, y) = |x - y|$. Another common metric space is $\mathbb{R}^n$ with the metric

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Here are some other metrics:

- Let $X = \mathbb{R}^n$ and $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} |x_i - y_i|$. This is often called the taxicab metric, and proving that it is a metric is easy since each term in the sum is $|x - y|$, the usual Euclidean metric on $\mathbb{R}$.
- Let $X = \mathbb{R}^n$ and $d(\mathbf{x}, \mathbf{y}) = \max_i |x_i - y_i|$. This is a metric, and is called the ℓ^∞ metric.
- Let $X = C[a, b] = \{\text{continuous functions on } [a, b]\}$, and $d(f, g) = \max_{x \in [a,b]} |f(x) - g(x)|$. This is a metric.

Definition 1.2: Balls

Let (X, d) be a metric space. The *ball* with center $a \in X$ and radius $\delta > 0$ is

$$B(a, \delta) = \{x \in X \, : \, d(a, x) < \delta\}$$

The ball in $\mathbb{R}$ is an open interval, the ball in $\mathbb{R}^n$ with the ℓ^2 metric is a ball, the ball in $\mathbb{R}^n$ with the taxicab metric is a diamond, and the ball in $\mathbb{R}^n$ with the ℓ^∞ metric is a square.

Definition 1.3: Open Sets

Let (X, d) be a metric space. A subset $G \subseteq X$ is *open* if $\forall x \in G$, $\exists \delta > 0$ such that $B(x, \delta) \subseteq G$.

8/28/26 Lecture 2: More Topology

In this lecture, we prove some more results about the metric topology.

Proposition 2.1: Properties of Open Sets

Let (X, d) be a metric space. Then, the following hold.

1. X and $\emptyset$ are open.
2. $B(a, r)$ is open for $a \in X$ and $r \in \mathbb{R}$.
3. $\{x \in X : d(x, a) > r\}$ is open for any $a \in X$ and $r \in \mathbb{R}$.

Proof:
1) Obvious.

2) Let $x \in B(a, r)$. Since $d(a, x) < r$, we can consider the ball $B(x, r - d(a, x))$. For $y \in B(x, r - d(a, x))$

$$\begin{aligned} d(a, y) &\leq d(a, x) + d(x, y) \\ &< d(a, x) + r - d(a, x) \\ &= r \end{aligned}$$

Thus, we see that $y \in B(a, r)$. Thus, $B(a, r)$ is open.

3) Let $S(a, r) = \{x \in X : d(x, a) > r\}$, and let $x \in S$. Now, $d(x, a) > r$, so consider $B(x, d(x, a) - r)$. Then, for $y \in B(x, d(x, a) - r)$, we see that

$$\begin{aligned} d(y, a) &\geq d(s, a) - d(s, y) \\ &> d(s, a) - (d(s, a) - r) \\ &= r \end{aligned}$$

Thus, we see that $y \in S(a, r)$, so $S(a, r)$ is open.

■

Corollary 2.2

A subset $S \subseteq X$ is open if, and only if, it is the union of open balls.

Proof: The $(\Rightarrow)$ direction is obvious. The $(\Leftarrow)$ direction follows by noticing that for each point in S, we can find an open ball containing that point contained in S, so just take the union of all such balls.

■

Definition 2.3: Closed Sets

A subset $F \subseteq X$ is *closed* if the complement, $X \backslash F$, is open.

Proposition 2.4

The following are true about open and closed sets.

1. A union of open sets is open.
2. A finite intersection of open sets is open.
3. An intersection of closed sets is closed.
4. A finite union if closed sets is closed.

Proof:

1) Obvious.

2) For each point x, take the minimum over the δ_i for which $B(x, \delta_i) \subseteq A_i$.

3, 4) This follows from De Morgan's Laws.

■

Definition 2.5: Metric Subspaces

Let (X, d) be a metric space, and suppose $X_1 \subseteq X$. The metric d induces a metric d_1 on X_1 by

$$d_1(a, b) = d(a, b)$$

for $a, b \in X_1$. We see that (X_1, d_1) is indeed a metric space, and we call (X_1, d_1) a *subspace* of (X, d).

Now, notice that if $(X_1, d_1) \subseteq (X, d)$; that is, (X_1, d_1) is a subspace of (X, d), then $B_1(a, \delta) = B(a, \delta) \cap X_1$. Thus, the metric balls in X_1 are the intersections of metric balls in X with X_1. This fact will be useful in the following proposition.

Proposition 2.6

Let $(X_1, d_1) \subseteq (X, d)$. Then, the following are true.

1. If $G_1 \subseteq X_1$ is open, then $G_1 = G \cap X_1$ for some open $G \subseteq X$.
2. If $F_1 \subseteq X_1$ is closed, then $F_1 = F \cap X_1$ for some closed $F \subseteq X$.

Proof:

1) Suppose $G_1 \subseteq X_1$ is open. Write $G_1 = \cup_{x \in G_1} B_1(x, \delta_x)$ so that G_1 is the union of open balls. Then, we see that

$$\begin{aligned} G_1 &= \bigcup_{x \in G_1} B_1(x, \delta_x) \\ &= \bigcup_{x \in G_1} B(x, \delta_x) \cap X_1 \\ &= \left(\bigcup_{x \in G_1} B(x, \delta_x) \right) \cap X_1 \end{aligned}$$

Letting $G = \cup_{x \in G_1} B(x, \delta_x)$, we see that G is open in X and $G_1 = G \cap X$.

2) Suppose $F_1 \subseteq X_1$ is closed. Then, $X_1 \backslash F_1$ is open, so write $X_1 \backslash F_1 = G_1$. Then, writing $G_1 = G \cap X_1$ for a open G, let $F = X \backslash G$. We see that F is closed, and that

$$\begin{aligned} F \cap X_1 &= (X \backslash G) \cap X_1 \\ &= X_1 \cap G_1^c \\ &= X_1 \backslash G_1 \\ &= F_1 \end{aligned}$$

which was to be shown.

■

Definition 2.7: Product Spaces

Let (X_1, d_1) and (X_2, d_2) be metric spaces. If we consider the Cartesian product $X_1 \times X_2$, d_1 and d_2 induce a metric d on $X_1 \times X_2$ (there are many such induced metrics), and we call the metric space $(X_1 \times X_2, d)$ the *product space*.

Some examples of induced metrics on $X_1 \times X_2$ are:

1. $d((x_1, x_2), (x_1', x_2')) = \sqrt{d(x_1, x_1')^2 + d(x_2, x_2')^2}$
2. $d((x_1, x_2), (x_1', x_2')) = d(x_1, x_1') + d(x_2, x_2')$
3. $d((x_1, x_2), (x_1', x_2')) = \max\{d(x_1, x_1'), d(x_2, x_2')\}$

Interestingly, all of these metrics give the same open sets! (a fun way to think about this is that you can always fit a ball/square/diamond in a ball/square/diamond).

8/31/26 Lecture 3: Topology Continued

First, as a quick note, recall that on $\mathbb{R}^n$ we have 3 different metrics,

$$d(x,y) = \begin{cases} \sqrt{\sum (x_i - y_i)^2} \\ \sum |x_i - y_i| \\ \max_i |x_i - y_i| \end{cases}$$

It turns out all of these give the same open sets, so we say they have the same topology. This motivates the following definition

Definition 3.1: Topologies

Let X be any set. A *topology* τ on X is a collection of sets, which we call open, such that

1. $\emptyset, X \in \tau$
2. $\cup_{\alpha \in I} U_\alpha \in \tau$ if $U_\alpha \in \tau$
3. $\cap_{i=1}^n U_i \in \tau$ if $U_i \in \tau$.

A topological space is a pair (X, τ) where τ is a topology on the set X.

Note that every metric space is a topological space with it's induced metric topology.

Definition 3.2: Subspace Topology

Let X be a topological space and $Y \subseteq X$. The *subspace topology* on Y is formed by taking open sets of X and intersecting them with Y. That is, $U \subseteq Y$ is open if there exists an open $O \subseteq X$ such that

$$U = O \cap Y$$

Note that open sets of Y need not be open in X. For example, if $Y = [0,1]$, Y is an open set in Y but is not one in $\mathbb{R}$. Also, thankfully, the metric on a metric subspace induces the subspace topology, so these definitions "agree". Some examples of topologies on any set X are $\mathcal{P}(X)$, called the discrete topology, and $\{\emptyset, X\}$ called the indiscrete topology.

Question: Does every topology come from a metric?
Answer: No, and here's why. Suppose $X = \{x_1, \dots, x_n\}$ has a metric. What is the corresponding topology? The answer is the discrete topology, as it's possible to contain each x_i by itself in an open ball by taking a sufficiently small radius. Thus, all metrics on finite spaces induce the discrete topologies, but there are many non-discrete topologies on finite spaces (which must then not come from a metric!).

Now, a bunch of definitions that will be soon useful.

Definition 3.3: Neighborhoods

Let X be a topological space. Given $x \in X$, a subset $N \subseteq X$ is a *neighborhood* of x if there exists an open $\mathcal{U}$ containing x such that $\mathcal{U} \subseteq N$.

Definition 3.4: Closed Sets

If X is a topological space, then $F \subseteq X$ is *closed* if F^c is open.

Definition 3.5: Limit Points

Let $A \subseteq X$. Then p is a *limit point* of A if every neighborhood of p intersects $A \backslash \{p\}$.

Note that the limit points of a set do not necessarily need to be containing in that set.

Proposition 3.6

A set $A \subseteq X$ is closed iff A contains all of it's limit points.

Proof: ($\Rightarrow$) Suppose A is closed. Then $X \backslash A$ is open. Thus, if $x \in X \backslash A$, then there exists an open U containing x such that $U \subseteq X \backslash A$. Then, as $U \cap A = \emptyset$, we see that x is not a limit point of A. Thus, A contains all of it's limit points.

($\Leftarrow$) Suppose A contains all of it's limit points. Let $x \in X \backslash A$. Since x is not a limit point of A, there exists an open U_x containing x such that $U_x \cap A = \emptyset$, so $U_x \subseteq X \backslash A$. Then, as we can write

$$X \backslash A = \cup_{x \in X \backslash A} U_x$$

we see that $X \backslash A$ is open, so A is closed.

■

Definition 3.7: Closure of a Set

Let $A \subseteq X$. Then, $\overline{A} = A \cup \{\text{limit points of } A\}$ is called the *closure* of A

Proposition 3.8

If $A \subseteq X$, then $\overline{A}$ is the smallest closed set containing A. That is $\overline{A}$ is the intersection of all closed sets containing A.

Proof: First, we will show that $\overline{A}$ is closed. If $p \in X \backslash \overline{A}$, then $p \notin A$ and p is not a limit point of A, so there exists an open U_p such that $p \in U_p \subseteq X \backslash A$. If a limit point of A is containing in U_p, then a point of A must also be in U_p (because U_p would be a neighborhood of that limit point!), which cannot occur. Thus, we have that $p \in U_p \subseteq X \backslash \overline{A}$. Thus, as

$$X \backslash \overline{A} = \cup_{p \in X \backslash \overline{A}} U_p$$

we see that $X \backslash \overline{A}$ is open, so $\overline{A}$ is closed.

Now, if $F \supseteq A$ is a closed set, then it must contain A and limit points of A (as it contains limit points of itself), which implies that $\overline{A} \subseteq F$. Thus, $\overline{A}$ must be the smallest closed set containing A.

■

Corollary 3.9

A set A is closed iff $A = \overline{A}$

Now, some final definitions.

Definition 3.10: Dense Sets

Let X be a topological space. A subset $A \subseteq X$ is *dense* in X if $\overline{A} = X$.

Definition 3.11: Interior

If X is a topological space, the *interior* of an $A \subseteq X$ is the set

$$\text{int}(A) = \bigcup_{\mathcal{U} \subseteq A, \mathcal{U} \text{ is open}} \mathcal{U}$$

Definition 3.12: Frontier

The *frontier* of a set $A \subseteq X$ is $\overline{A} \cap \overline{X \backslash A}$.

9/2/31 Lecture 4: Topological Spaces and Bases

Definition 4.1: Basis

Let X be a topological space. A collection β of open sets is a *basis* for X if every open set can be written as a union of elements in β.

If $X = \mathbb{R}$ it can be shown that $\beta = \{(a, b) : a, b \in \mathbb{Q}\}$ is a basis. Similarly, $\beta = \{B(x, r) : x \in X, r \in \mathbb{Q}^+\}$ is a basis for a typical metric space.

Proposition 4.2

A collection β is a basis if, and only if, given $p \in X$ and a neighborhood N of p, there exists a $B \in \beta$ such that $p \in B \subseteq N$.

Proof: ($\Rightarrow$) Suppose that β is a basis. Given a point $p \in X$ and a neighborhood N of p, there exists an open $\mathcal{O}$ such that $p \in \mathcal{O} \subseteq N$. Since β is a basis, $\mathcal{O}$ is a union of elements in β, and one of them, namely B, contains p, so $p \in B \subseteq N$.

($\Leftarrow$) Let $\mathcal{O}$ be an open set. For all $p \in \mathcal{O}$, $\mathcal{O}$ is a neighborhood of p, so there exists a $B_p \in \beta$ such that $p \in B_p \subseteq \mathcal{O}$. Thus, we can write

$$\mathcal{O} = \bigcup_{p \in \mathcal{O}} B_p$$

and as $B_p \in \beta$ for all p, $\mathcal{O}$ can be written as a union of elements in β.

■

Definition 4.2: Continuous Functions

Let X, Y be topological spaces. A function $f : X \to Y$ is *continuous* if, and only if, for any open $\mathcal{U} \subseteq Y$, $f^{-1}(\mathcal{U})$ is open in X. We will use the word *map* for continuous functions.

Now, for the rest of this section we will give properties about maps.

Proposition 4.3

The composition of maps is a map.

Proof: Let $f : X \to Y$ and $g : Y \to Z$ be maps. If $\mathcal{U} \subseteq Z$ is open,

$$(g \circ f)^{-1}(\mathcal{U}) = f^{-1} \circ g^{-1}((U)) = f^{-1}(g^{-1}(\mathcal{U}))$$

Then, as $g^{-1}(\mathcal{U})$ is open, we see that $f^{-1}(g^{-1}(\mathcal{U}))$ is open in X.

■

Proposition 4.4

If $f : X \to Y$ is a map and $A \subseteq X$, then $f|_A : A \to Y$ is a map (in the subspace topology).

Proof: We see that if $\mathcal{U} \subseteq Y$ is open, then as

$$\begin{aligned} f|_A{}^{-1}(\mathcal{U}) &= \{x \in A : f(x) \in \mathcal{U}\} \\ &= f^{-1}(\mathcal{U}) \cap A \end{aligned}$$

Since $f^{-1}(\mathcal{U})$ is open in X, then $f|_A{}^{-1}(\mathcal{U})$ is open in A.

■

Now, since the identity function on X, id_X, is continuous, we see that for a subset $A \subseteq X$, the inclusion $i : A \to X$, which is equal to $\mathrm{id}_X|_A$ is continuous as well.

Proposition 4.5

The following are equivalent.

1. $f : X \to Y$ is a map.
2. If β is a basis for Y, then for all $B \in \beta$, $f^{-1}(B)$ is open in X.
3. For any $A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$
4. For any $B \subseteq Y$, $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$
5. The preimage of a closed set is closed.

Proof:
(1) $\implies$ (2) Obvious.

(2) $\implies$ (3) Suppose $A \subseteq X$. We have $f(A) \subseteq \overline{f(A)}$. For $x \in \overline{A}$, we want to show that $f(x) \in \overline{f(A)}$. If $f(x) \in f(A)$ we are done, so suppose that $f(x) \notin f(A)$. If $f(x) \notin \overline{f(A)}$, then $f(x) \in X \backslash \overline{f(A)}$, which is an open set. Thus, there exists some $B \in \beta$ such that $f(x) \in B \subseteq X \backslash \overline{f(A)}$. Now, $f^{-1}(B)$ is an open set in X containing x, and as for all $b \in B$, $f(b) \notin f(A)$, we see that $f^{-1}(B) \cap A = \emptyset$. This contradicts the fact that $x \in \overline{A}$.

(3) $\implies$ (4) Since $f(\overline{f^{-1}(B)}) \subseteq \overline{f(f^{-1}(B))} = \overline{B}$, we see that

$$\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$$

(4) $\implies$ (5) Suppose that B is closed. Then $f^{-1}(B) \supseteq \overline{f^{-1}(B)}$. However, $f^{-1}(B) \subseteq \overline{f^{-1}(B)}$, so $f^{-1}(B) = \overline{f^{-1}(B)}$. Thus, the preimage of B is closed.

(5) $\implies$ (1) Suppose that $\mathcal{U}$ is open. Then $Y \backslash \mathcal{U}$ is closed, so $f^{-1}(Y \backslash \mathcal{U})$ is closed. Now,

$$\begin{aligned} X \backslash f^{-1}(Y \backslash \mathcal{U}) &= \{x \in X : x \notin f^{-1}(Y \backslash \mathcal{U})\} \\ &= \{x \in X : f(x) \notin Y \backslash \mathcal{U}\} \\ &= \{x \in X : f(x) \in \mathcal{U}\} \\ &= f^{-1}(\mathcal{U}) \end{aligned}$$

Thus, $f^{-1}(\mathcal{U})$ is open, so f is a map.

■

9/4/26 Lecture 5: Homeomorphisms and Space Filling Curves

First, fun fact: $x \in \overline{A}$ if, and only if, for a neighborhood N of x, we have that $N \cap A \neq \emptyset$.

Now, recall from last time that continuous functions are those such that the preimage of an open set under that function is open as well. This seems different than the definition in real analysis, but it turns out they are equivalent! Let X, Y be metric spaces and let $f : X \to Y$ be a continuous map in the topological sense. Now, given $\epsilon > 0$, we see that $B(f(x), \epsilon)$ is open in Y, so as f is continuous,

$$f^{-1}(B(f(x), \epsilon)) = \mathcal{U}$$

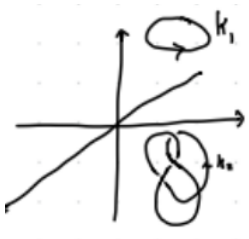
is open in X. Then, as $x \in \mathcal{U}$, there exists a $\delta > 0$ such that $B(x, \delta) \subseteq \mathcal{U}$. In summary, for all y with $d(y, x) < \delta$, we see that $d(f(y), f(x)) < \epsilon$. The opposite direction (metric continuity to topological continuity) is quite simple as well.

Now, a definition:

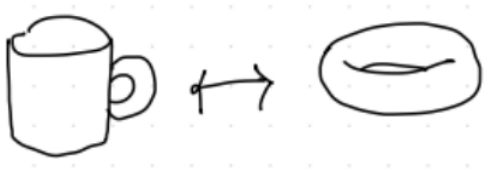
Definition 5.1: Homeomorphisms

A *homeomorphism* $f : X \to Y$ is a continuous bijection whose inverse is also continuous.

Here are some examples:



These knots in $\mathbb{R}^3$ are homeomorphic under the subspace topology



A coffee cup and a donut are actually homeomorphic!

Here's another important one: Let $f : X \to Y$ be a bijection with Y being a topological space. Then, we can induce a topology on X by saying that $\mathcal{U} \subseteq X$ is open iff $f(\mathcal{U})$ is open in Y. This makes f a homemorphism as well.

Now, let $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$, and let $p \in S^n$. Then, we claim that $S^n \backslash \{p\}$ is homeomorphic to $\mathbb{R}^n$. To prove this, first assume that p is the north pole (we can always reorient), and consider the map $\pi : S^n \backslash \{p\} \to \mathbb{R}^n$, called stereographic projection, as depicted below. It turns out this is a homeomorphism.

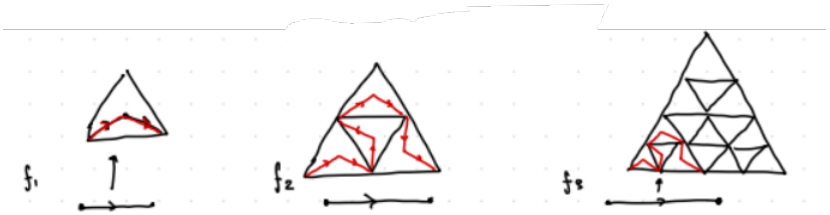


Projecting S^n onto R^n by "casting a light" at the north pole.

Space Filling Curves

The question we will answer with this section is: How weird can continuous functions be?

Let $\triangle$ be the solid equilateral triangle with side length $1/2$. We will construct a continuous function $f : [0,1] \to \triangle$ that is onto! To do this, define $f_n : [0,1] \to \triangle$ for $n = 1, 2, 3$ as follows:



The f_n zig-zag through smaller and smaller triangles breaking up $\triangle$

First, say $n \geq m$. From our construction, we have that for $t \in [0,1]$, $f_n(t)$ and $f_m(t)$ are both in a triangle of length 2^{-m}, so $\|f_n(t) - f_m(t)\|_{\mathbb{R}^2} \leq 2^{-m}$. Thus, we see that $f_n(t)$ is Cauchy, so it has a limit which we call $f(t)$. Now, as

$$\|f_n(t) - f_m(t)\|_{C^0} = \max_{t \in [0,1]} \|f_n(t) - f_m(t)\|_{\mathbb{R}^2} \leq 2^{-m}$$

we see that f_n is uniformly continuous. Thus, we see that $\lim_{n\to\infty} f_n(t) = f(t)$ is continuous. Now, we need to show that $f(t)$ is space filling. First, we show that $f([0,1])$ is dense in $\triangle$.

Let $x \in \triangle$. Suppose $x \notin f[0,1]$ and $\mathcal{U}$ is a neighborhood of x. Taking N sufficiently large, we can have

$$B(x, 2^{-(N-2)}) \subseteq \mathcal{U}$$

Now, choose $t_0 \in [0,1]$ such that $\|x - f_N(t_0)\|_{\mathbb{R}^2} \leq 2^{-N}$. Passing to an N large enough so that $\|f_N - f\|_{C^0} \leq 2^{-N}$, we see that, by the triangle equality

$$\|x - f(t_0)\| \leq 2^{-(N-1)} \implies f(t_0) \in B(x, 2^{-(N-2)})$$

thus, we see that $\mathcal{U} \cap f([0,1]) \neq \emptyset$, so $\overline{f[0,1]} = \triangle$. Now, we just need to prove that f is onto.

9/9/26 Lecture 6: More Topological Properties

We will come back to our function $f : [0, 1] \to \triangle$ and show that it is really space filling, but before that we will look at some other topological properties.

Definition 6.1: Hausdorff Spaces

A topological space X is *Hausdorff* if, for all $x \neq y \in X$, there exist disjoint open set U, V such that $x \in U$ and $y \in V$.

It is clear that all metric spaces are Hausdorff, and that if a space has more than one point, then with the indiscrete topology, the space is not Hausdorff.

Definition 6.2: Distance to a Set

Let X be a metric space and $A \subseteq X$, Then, we define

$$d(x, A) := \inf\{d(x, a) \, : \, a \in A\}$$

Proposition 6.3

The function $d(x, A)$ is continuous in x. In other words, the function $d_A : X \to \mathbb{R}$ by $d_A(x) = d(x, A)$ is continuous.

Proof: We will use the $\epsilon - \delta$ definition of continuity. Pick $x \in X$ and $\epsilon > 0$. Let $\delta = \epsilon/2$. Then, by the definition of $d(x, A)$, pick an a such that $d(x, a) < d(x, A) + \epsilon/2$. Then, for z such that $d(x, z) < \delta = \epsilon/2$, we have that

$$d(z, a) \leq d(z, x) + d(x, a) < d(x, A) + \epsilon$$

Thus, since $d(z, A) < d(z, a)$, we see that $d(z, A) < d(x, A) + \epsilon$. By a similar argument, we can get the other direction, which shows that $|d(x, A) - d(z, A)| < \epsilon$.

■

Lemma 6.4

$d(x, A) = 0$ if, and only if, $x \in \overline{A}$.

Proof: Suppose that $d(x, A) = 0$. Then, for all $\epsilon > 0$, there exists an $a \in A$ such that $d(x, a) < \epsilon$. Thus, either $x \in A$, or $x \notin A$ and every neighborhood of x intersects $A \backslash \{x\}$, which implies that $x \in \overline{A}$.

Similarly, if $x \in \overline{A}$, then either $x \in A$ for which $d(x, A)$ holds trivially, or that for all $\epsilon > 0$, there exists an $a \in A$ such that $a \in B(x, \epsilon)$, which implies that $d(x, a) < \epsilon$. Thus, $d(x, A) = 0$.

■

Proposition 6.5

If X is a metric space and $A, B \subseteq X$ are disjoint closed sets, then there exists a continuous function $f : X \to [-1, 1]$ such that $f|_A = 1$ and $|_B = -1$.

Proof: Let

$$f(x) = \frac{d(x, B) - d(x, A)}{d(x, B) + d(x, A)}$$

Now, if $x \in B$, then $x \notin A = \overline{A}$, so

$$f(x) = \frac{-d(x, A)}{d(x, A)} = -1$$

and if $x \in A$, then $x \notin B = \overline{B}$, so

$$f(x) = \frac{d(x, B)}{d(x, B)} = 1$$

Now, we need to make sure this function is continuous. Since it is just the combination of continuous functions, we just need to show that $d(x, A) + d(x, B) = 0$. However, this only occurs if $d(x, A) = 0$ and $d(x, B) = 0$, which only occurs when $x \in A \cap B$, which doesn't happen.

■

Theorem 6.6: Tietze Extension Theorem

If X is a metric space, $A \subseteq X$ is closed, and $f : A \to \mathbb{R}$ is continuous, then there exists a continuous $F : X \to \mathbb{R}$ such that $F|_A = f$.

Definition 6.7: Covers

Let X be a topological space. An *open cover* of X is a family $\mathcal{F}$ of open subsets so that

$$X = \bigcup_{\mathcal{O} \in \mathcal{F}} \mathcal{O}$$

A *subcover* of $\mathcal{F}$ is a subfamily $\mathcal{F}' \subseteq \mathcal{F}$ such that we still have $X = \bigcup_{\mathcal{O} \in \mathcal{F}'} \mathcal{O}$.

Definition 6.8: Compactness

A set X is *compact* if every open cover has a finite subcover.

An example of a non-compact set is $\mathbb{R}$, as the cover $\mathcal{F} = \{(i, i+2)\}_{i \in \mathbb{Z}}$. This is an open cover of X with no finite subcover. Thus, $\mathbb{R}$ is not compact.

Theorem 6.9: Heine-Borel Theorem

The set $[a, b]$ is compact.

Warning: For a general metric space, if a subset is compact in the subspace topology, then it is closed and

bounded. However, not all closed and bounded sets are compact (usually).

Here is such an example. Let $X = \mathbb{Z}$ and $d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$. We see this induces the discrete topology on $\mathbb{Z}$. Then, we see that $\mathbb{Z}$ is closed and bounded under this metric, but $\mathbb{Z} = \bigcup_{z \in \mathbb{Z}} \{z\}$ is an open cover with no finite subcover.

Lemma 6.10

Let $f : X \to Y$ be a function. Then,

1. $A \subseteq X \implies f^{-1}(f(A)) \supset A$ with equality when f is injective.
2. $B \subseteq Y \implies f(f^{-1}(B)) \subseteq B$ with equality when f is surjective.

9/11/26 Lecture 7: Compact and Hausdorff Spaces

Propositon 7.1

If $f : X \to Y$ is continuous and X is compact, then $f(X) \subseteq Y$ is compact.

Proof: We see that $f : X \to f(X)$ is continuous. Let $\{\mathcal{O}_\lambda\}$ be an open cover of $f(X)$. Then,

$$f(X) = \bigcup_\lambda \mathcal{O}_\lambda$$

Then, we see that $f^{-1}(f(X)) = X$ as X is onto, and

$$f^{-1}\left(\bigcup_\lambda \mathcal{O}_\lambda\right) = \bigcup_\lambda f^{-1}(\mathcal{O}_\lambda)$$

is an open cover as $f^{-1}(\mathcal{O}_\lambda)$ is open. Then, as X is compact, we can write

$$X = \bigcup_{i=1}^n f^{-1}(\mathcal{O}_{\lambda_i})$$

then, as $f(X) = f\left(\bigcup_{i=1}^n f^{-1}(\mathcal{O}_{\lambda_i})\right) = \bigcup_{i=1}^n \mathcal{O}_{\lambda_i}$, so $f(X)$ has a finite subcover. ■

Definition 7.2: Compact Subspace

Let X be a topological space. Then $Y \subseteq X$ is compact if it is compact in the subspace topology.

This means that if $Y \subseteq \bigcup_\lambda \mathcal{O}_\lambda$ for $\mathcal{O}_\lambda$ open in X, then we can write $Y = \bigcup_{i=1}^n \mathcal{O}_{\lambda_i}$.

Proposition 7.3

A closed subset of a compact space is compact.

Proof: Let $Y \subseteq X$ be closed and X be compact. Now, take an open cover $Y \subseteq \cup_\alpha \mathcal{O}_\alpha$. Then, notice that $X \backslash Y$ is open, so letting $X \backslash Y = \mathcal{O}_{\alpha'}$, we have

$$X = \bigcup_\alpha \mathcal{O}_\alpha$$

which allows us to pass to an open cover

$$X = \bigcup_{i=1}^n \mathcal{O}_{\alpha_i}$$

Now, this may or may not contain $\mathcal{O}_{\alpha_i}$, but removing that set, it must still cover Y, so we have that $Y \subseteq \bigcup_{i=1}^n \mathcal{O}_{\alpha_i}$, we see that Y has a finite subcover. ■

Proposition 7.4

If X is Hausdorff, $A \subseteq X$ is compact and $x \in X \backslash A$, then there exists disjoint neighborhoods of x and A. Thus, this implies that $X \backslash A$ is open, so A is closed.

Proof: Take $z \in A$. Since X is Hausdorff, we can find disjoint open sets U_z, V_z so that $x \in U_z$ and $z \in V_z$.

Then, we see that $A \subseteq \bigcup_{z \in A} V_z$, and as A is compact, pass to a finite subcover $A \subseteq \bigcup_{i=1}^{n} V_{z_i}$. Now, since $x \in \bigcap_{i=1}^{n} U_{z_i}$ is open, and $U_{z_i} \cap V_{z_i}$ are disjoint, we see that $\bigcup_{i=1}^{n} V_{z_i} \cap (\bigcap_{i=1}^{n} U_{z_i}) = \emptyset$, which gives us our two disjoint open neighborhoods of x and A.

■

Corollary 7.5

The curve $f : [0,1] \to \triangle$ from above is space filling.

Proof: Since $[0,1]$ is compact, $f[0,1]$ is compact, and as $\triangle$ is Hausdorff, we see that $f[0,1]$ is closed by Proposition 7.4. Thus, we see that $f[0,1] = \overline{f[0,1]} = \triangle$.

Proposition 7.6

If $f : X \to Y$ is a continuous bijection, X is compact, and Y is Hausdorff, then f is a homeomorphism.

Proof: We just need to show that $f^{-1} = g : Y \to X$ is continuous. Let C be closed in X. Then all there is to show is that $g^{-1}(C) = f(C)$ is closed.

However, this is easy because $C \subseteq X$ is closed, so it is compact by Proposition 7.3, and as Y is Hausdorff, we see that as $f(C)$ is compact, it is also closed.

■

Now, an interesting question: Can the space filling curve from before, $f : [0,1] \to \triangle$, be injective? Well, as $[0,1]$ is compact, and $\triangle$ is Hausdorff, we see that if f is injective, then f is a homeomorphism by Proposition 7.6. However, we will later show that this is a contradiction, as $[0,1]$ and $\triangle$ are not homeomorphic.

Proposition 7.7

If X is compact, every infinite subset has a limit point.

Proof: Let $Y \subseteq X$ be infinite and suppose that Y has no limit points. Then, for each $x \in X$, x is not a limit point of Y, so there exists an open U_x containing x such that the intersection of U_x and Y contains at most one point. Then, as $\cup_{x \in X} U_x$ is an open cover for X, and X is compact, consider the open cover

$$X \subseteq \bigcup_{i=1}^{n} U_{x_i}$$

Now, notice that Y must be contained in this union, but each element of this union contains at most one

point of Y, which implies that Y must be finite.

Theorem 7.8: Mini Heine-Borel Theorem

A compact subset of $\mathbb{R}^n$ is closed and bounded.

Proof: Since $\mathbb{R}^n$ is Hausdorff, we see that a compact set C is also closed by Proposition 7.4. Thus, we just need to show that this is bounded. However, notice that we can write $\mathbb{R}^n = \bigcup_{k=1}^{\infty} B(0,k)$, so as $C \subseteq \bigcup_{i=1}^{\infty} B(0,k)$ is comapact, we can write $C \subseteq \bigcup_{i=1} B(0,k_i)$, we can take the maximal k_i and see that $C \subseteq B(0,k_i)$, which implies that it is bounded.

■

Theorem 7.9: Extreme Value Theorem

If $f : X \to \mathbb{R}$ is a continuous function and X is compact, then f is bounded above and below, and achieves its supremum and infimum.

Proof: Since $f(X)$ is compact, we see that $f(X)$ is closed and bounded. Then, as $f(X)$ is closed, we see that $\sup f, \inf f \in f(X)$, so there exists x_1, x_2 such that

$$f(x_1) = \sup f \quad \text{and} \quad f(x_2) = \inf f$$

9/14/26 Lecture 8: Product Spaces

Question: Given a collection of subsets of X, β, when is it the basis for some topology?

Lemma 8.1

A collection β of subsets is the basis for some topology on X if, and only if $X = \bigcup_{B\in\beta} B$, and finite intersections of elements in β is a union of elements in β.

Proof: ($\Rightarrow$) This follows from the definition of a basis.

($\Leftarrow$) Suppose that β is a collection of subsets that have the above property, and say that U is open iff it is the union of elements in β. We will show that this is a topology.

First, we see that $\emptyset$ and X are open. Now, if $\{A_\lambda\}$ are open, then we see that $\bigcup_\lambda A_\lambda$ is open, because each of the A_λ are unions of elements in β, so $\bigcup_\lambda A_\lambda$ is too.

Finally, if A_1, A_2 are open, with $A_1 = \bigcup_{B_{\lambda,1}\in\beta} B_{\lambda,1}$ and $A_2 = \bigcup_{B_{\lambda,2}\in\beta} B_{\lambda,2}$, then we see that

$$A_1 \cap A_2 = \bigcup_{\lambda_1,\lambda_2} (B_{\lambda,1} \cap B_{\lambda,2})$$

and as each of the $B_{\lambda,1}, B_{\lambda,2}$ are unionts of elements in β, we see that so is $A_1 \cap A_2$. Thus, by induction, all finite intersections are as well.

■

Product Spaces

Let X, Y be topological spaces. We will consider $X \times Y = \{(x, y) : x \in X, y \in Y\}$ and see what natural topologies we can put on it.

First Attempt: What if we just let the open sets be of the form $\mathcal{U} \times \mathcal{V}$ for open sets $\mathcal{U}, \mathcal{V}$ of X and Y respectively?
Problem: Quite easy to show that this is not closed under unions.

Second Attempt: How about, in the fashion of Lemma 8.1, we define the basis of our topology to be $\beta = \{\mathcal{U} \times \mathcal{V} : \mathcal{U} \text{ open in } X, \mathcal{V} \text{ open in } Y\}$. It turns out this works, because

1. $X \times Y$ is in β
2. $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$, so intersections of basis elements are basis elements!

Thus, we now have our naturally arising topology.

Definition 8.2: Product Spaces

Let X, Y be topological spaces. The *product space* is the set $X \times Y$ equipped with the topology generated by the basis $\beta = \{\mathcal{U} \times \mathcal{V} : \mathcal{U} \text{ open in } X, \mathcal{V} \text{ open in } Y\}$, called the *product topology*.

Definition 8.3: Projections

The maps $p_1 : X \times Y \to X$ and $p_2 : X \times Y \to Y$, called *projections* are given by

$$p_1(x, y) = x \quad \text{and} \quad p_2(x, y) = y$$

These are continuous and are also open maps (meaning the image of an open set is open).

The proof of the claims made above are quite simple. First, just consider p_1. Then, $p_1^{-1}(U) = U \times Y$, so if U is open, the preimage is open as well. Similarly, if $\bigcup_\alpha U_\alpha \times V_\alpha$ is open in $X \times Y$, then we see that

$$p_1\left(\bigcup_\alpha U_\alpha \times V_\alpha\right) = \bigcup_\alpha U_\alpha$$

which is open.

Proposition 8.4

A function $f : Z \to X \times Y$ if, and only if $p_1 \circ f : Z \to X$ and $p_2 \circ f : Z \to Y$ are continuous.

Proof: First, as continuity is preserved under composition, we see that if f is continuous, so are $p_1 \circ f$ and $p_2 \circ f$.

Now, suppose that $p_1 \circ f$ and $p_2 \circ f$ are continuous. Then, as

$$f^{-1}\left(\bigcup U_\alpha \times V_\alpha\right) = \bigcup f^{-1}(U_\alpha \times V_\alpha)$$

it suffices to show that the preimage of basis elements are open. Well, we see that

$$p_1^{-1}(U_\alpha) = U_\alpha \times Y \quad \text{and} \quad p_2^{-1}(V_\alpha) = X \times V_\alpha$$

so

$$U_\alpha \times V_\alpha = p_1^{-1}(U_\alpha) \cap p_2^{-1}(V_\alpha)$$

Thus,

$$\begin{aligned} f^{-1}(U_\alpha \times V_\alpha) &= f^{-1}(p_1^{-1}(U_\alpha) \cap p_2^{-1}(V_\alpha)) \\ &= (p_1 \circ f)^{-1}(U_\alpha) \cap (p_2 \circ f)^{-1}(V_\alpha) \end{aligned}$$

which is open.

■

Proposition 8.5

$X \times Y$ is Hausdorff if, and only if, X and Y are Hausdorff.

Proof: ($\Leftarrow$) Suppose that X and Y are Hausdorff, and let $(x_1, y_1) \neq (x_2, y_2)$. Then, WLOG, assume that $x_1 \neq x_2$. Then, let U_1 and U_2 be disjoint open neighborhoods of X containing x_1 and x_2 respectively. Then, we see that $U_1 \times Y$ and $U_2 \times Y$ are disjoint open neighborhoods containing (x_1, y_1) and (x_2, y_2) respectively. Thus, $X \times Y$ is Hausdorff.

($\Rightarrow$) Suppose that $X \times Y$ is Hausdorff. We will show that X is Hausdorff, and the proof for Y follows similarly. Let $x_1 \neq x_2$. If Y is empty, then $X \times Y = X$, so it follows. Otherwise, pick a $y \in Y$, and consider (x_1, y) and (x_2, y). These are nonequal, so let $(U_1 \times V_1)$ and $(U_2 \times V_2)$ be disjoint open basis elements containing (x_1, y) and (x_2, y) respectively (recall that we can always pass open neighborhoods to basis elements). Then, notice that

$$x_1 \in U_1, x_2 \in U_2$$

and U_1 and U_2 are open. Also, if some p lies in both, then (p, y) lies in both $U_1 \times V_1$ and $U_2 \times V_2$, which is a contradiction. Thus, they are disjoint open neighborhoods, so X is Hausdorff.

■

Lemma 8.6

If X is a topological space and β is a basis, then X is compact if, and only if, every covering of X by basis elements has a finite subcover.

Proof: ($\Rightarrow$) Obvious.

($\Leftarrow$) Suppose that every covering of X by basis elements has a finite subcover. Now, let $\{U_\lambda\}$ be an open cover of X. Then, we can write $U_\lambda = \bigcup_\alpha B_{\alpha,\lambda}$ for some $B_{\alpha,\lambda} \in \beta$. Thus, as

$$X = \bigcup_{\lambda,\alpha} B_{\alpha,\lambda}$$

we can pass to a finite subcover

$$X = \bigcup_{i=1}^{n} B_{\alpha_i,\lambda_i}$$

Then, as $B_{\alpha_i,\lambda_i} \subseteq U_{\lambda_i}$, we see that

$$X = \bigcup_{i=1}^{n} U_{\lambda_i}$$

which is a finite subcover.

■

9/16/26 Lecture 9: Compactness and Connectedness

We start by proving a weaker form of Tychonoff's theorem.

Theorem 9.1: Tychonoff's Theorem

X and Y are compact topological spaces if, and only if, $X \times Y$ is compact.

Proof:
($\Rightarrow$) If $X \times Y$ is compact, then as the projections p_1 and p_2 are continuous and onto, X and Y are continuous.

($\Leftarrow$) Let $\mathcal{F}$ be an open cover of $X \times Y$ by basis elements $\mathcal{U}_\alpha \times \mathcal{V}_\alpha$.

Now, fix $x \in X$. Then, we see that $\{x\} \times Y$ is compact as it is the image of the map $y \mapsto (x, y)$. Thus, as $\{U_\alpha \times V_\alpha\}$ covers it, pass to a subcover $\{U_i^x \times V_i^x\}_{i=1}^{n_x}$.

Now, consider $U^x = \bigcap_{i=1}^{n_x} U_i^x$. This is an open set containing x and the union of $\{U_i^x \times V_i^x\}_{i=1}^{n_x}$ covers $U^x \times Y$.

Consider the collection $\{U^x : x \in X\}$. This is a cover of X, so pass to a subcover $\{U^{x_i}\}_{i=1}^{n}$. Then, we see that

$$\begin{aligned} X \times Y &= \bigcup_{i=1}^{n} (U^{x_i} \times Y) \\ &= \bigcup_{i=1}^{N} \left(\bigcup_{j=1}^{n_{x_i}} (U_j^{x_i} \times V_j^{x_i}) \right) \end{aligned}$$

Thus, we see that $\mathcal{F}$ has a finite subcover.

■

Fortunately, the product topology on $\mathbb{R}^n$ is equivalent to the topology induced by the Euclidean metric. This can be seen because $\{(a_1, b_1) \times \cdots \times (a_n, b_n)\}$ forms a basis for the product topology, and we can write these as a union of metric balls.

Theorem 9.2: Heine-Borel Theorem

A subset of $\mathbb{R}^n$ is compact if, and only if, it is closed and bounded.

Proof:
($\Rightarrow$) We prove this last time!

($\Leftarrow$) Suppose $X \in \mathbb{R}^n$ is closed and bounded. Then, we can find an $s > 0$ so that $X \subseteq [-s, s] \times \cdots \times [-s, s]$. Then, as $[-s, s]$ is compact in $\mathbb{R}$, we see that $[-s, s]^n$ is compact in $\mathbb{R}^n$. Thus, as X is a closed subset of a compact set, it is compact as well.

■

As emphasized previously, we will see that in the general metric space, compact implies closed and boundedness, but the converse does not typically hold.

Connectedness

Definition 9.3: Connectedness

A topological space X is *connected* if, whenever $X = A \sqcup B$ with A and B nonempty, then $\overline{A} \cap B = \emptyset$ or $A \cap \overline{B} = \emptyset$.

An example of a disconnected space is $X = [0,1] \cup [2,3]$ with the subspace topology from $\mathbb{R}$.

Fact

$\mathbb{R}$ is connected and a nonempty subset of $\mathbb{R}$ is connected if, and only if, it is an interval.

Proposition 9.4

The following are equivalent about a topological space X:

1. X is connected.
2. The only clopen subset of X are $\emptyset$ and X.
3. X is not the union of two nonempty disjoint open sets.
4. There is no surjective map to a discrete space with more than one point.

Proof:

(1) $\implies$ (2): If X is connected and A is clopen, then A^c is clopen. Namely, they are both closed, so as $X = A \cup A^c$, we must have that either A or A^c are nonempty (their closures, which are themselves, don't intersect). Thus, either $A = \emptyset$ or $A = X$.

(2) $\implies$ (3): Suppose that $X = A \sqcup B$ with A and B being open. Then, notice that $A^c = B$. Thus as A is open and A^c is open, A is clopen, so it is either $\emptyset$ or X, which means either A is empty or B is empty.

(3) $\implies$ (4): Suppose that $f : X \to Y$ is a surjective map onto a discrete space with more than one point. Let $y_1 \in Y$. Then, we see that $f^{-1}(\{y_1\})$ is open in X and $f^{-1}(\{y_1\}^c) = f^{-1}(\{y_1\})^c$ is also open in X. Thus, one of them must be empty, but this either contradicts the fact that f is onto, or that Y has more than one point. Thus, no such f exists.

(4) $\implies$ (1): Suppose that X is disconnected. Then, write $X = A \cup B$ where A and B are nonempty and $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$. Now, if $b \in B$, then $b \notin \overline{A}$ so $b \in X - \overline{A}$, and vice versa, so $B = X - \overline{A}$. Similarly $A = X - \overline{B}$. This implies they are both open.

Now, define $f : X \to \{0,1\}$ with $f|_A = 0$ and $f|_B = 1$. Then, the preimage of any subset of $\{0,1\}$ is either $\emptyset, B, A$ or X, which is continuous. Thus, a surjective map clearly does exist.

■

The upshot of this theorem is that X is connected if, and only if, writing X as a disjoint union of open set implies that one of the open sets must be empty.

Proposition 9.5

If X is connected and $f : X \to Y$ is a map, then $f(X)$ is connected.

Proof: Suppose that $f(X) = U \cup V$ for disjoint open (in $f(X)$!) sets U and V. Then, as

$$X = f^{-1}(U) \cup f^{-1}(V)$$

and we have that $f^{-1}(U)$ and $f^{-1}(V)$ are disjoint open sets, this implies that one of them is empty. This then implies that either U or V is empty as f is onto $f(X)$.

■

Corollary 9.6

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) \leq r \leq f(b)$, then $\exists c \in [a, b]$ for which $f(c) = r$

Proof: We know that $f[a, b]$ is connected, which implies that it is an interval. Then, as it contains $f(a)$ and $f(b)$, it must also contain any r between them.

■

Now, we wil prove this next theorem next time, but I will state it here so I can give an example.

Proposition 9.7

If X is a topological space and Z is a dense subset that is connected, then X is connected.

Example

An interesting example of this is

$$X = \{(x, \sin(1/x)) \,:\, 0 < x < \infty\} \cup \{(0, y) \,:\, -1 \leq y \leq 1\}$$

the set on the left is dense and connected, which implies that X is connected!

9/19/26 Lecture 10: Connectedness

Proposition 10.1

If X is a topological space and Z is a dense, connected subset, then X is connected.

Proof: Suppose that $A \subseteq X$ is a nonempty clopen subset. We want to show that $A = X$. We want to see how $A \cap Z$ sits in Z.

First, as Z is dense, it intersects every open set in X, so $A \cap Z$ is nonempty. Then, we see that $A \cap Z$ is open and closed in Z as A is open and closed. Thus, A is clopen in Z, and as it is nonempty and Z is connected, we must have that $A \cap Z = Z$. Then, as A is a closed set containing Z, A must equal X as Z is dense.

Thus, as every nonempty clopen set in X must equal X, we see that X is connected.

■

Example

Let $X = \{x \in \mathbb{R}^n : |x| \leq 1\}$ and $Z = \{x \in \mathbb{R}^n : |x| < 1\}$. We see that Z is a dense connected subset, so X is connected.

Corollary 10.2

If Z is a connected subset of a topological space X, then $\overline{Z}$ is connected.

Proof: We see that Z is dense in $\overline{Z}$.

■

Definition 10.3: Separated Sets

We say $A, B \subseteq X$ are *separated* if $\overline{A} \cap \overline{B} = \emptyset$.

Proposition 10.4

Suppose that X is a topological space and let $\mathcal{F}$ be a family of subsets such that X is the union of elements in $\mathcal{F}$. Then, if each $Z \in \mathcal{F}$ is connected and no $Z, Z' \in \mathcal{F}$ are separated, then X is connected.

Proof: Suppose $A \subseteq X$ is a clopen set. Now, pick a $Z \in \mathcal{F}$. Then, Z is connected, so as $A \cap Z$ is clopen in Z, then we must have that $A \cap Z = Z$ or $A \cap Z = \emptyset$.

Now, if $A \cap Z = \emptyset$ for all $Z \in \mathcal{F}$, then $A = \emptyset$.

Now suppose that for some Z, we have that $A \cap Z = Z$. Then $Z \subseteq A$. Let $W \in \mathcal{F}$ be another set (if no

other set exists, then $Z = X$ and $A = X$). Now, if $W \cap A = \emptyset$, then we see that $W \subseteq A^c$, which is a closed set and $Z \subseteq A$, which is a closed set, so $\overline{W} \subseteq A^c$ and $\overline{Z} \subseteq A$. Thus, $\overline{W} \cap \overline{Z} = \emptyset$, so W and Z are separated, which is a contradiction.

Thus, for all $W \in \mathcal{F}$, we see that $W \cap A = W$, so $X = \bigcup_{W \in \mathcal{F}} W \subseteq A$, which implies that $A = F$.

■

The following is a useful application of this above proposition.

Proposition 10.5

If X and Y are connected, then so is $X \times Y$.

Proof: If $x \in X$, then $\{x\} \times Y$ is connected as there is a map from Y onto $\{x\} \times Y$. Similarly, $X \times \{y\}$ is connected by the same fact. Then,

$$Z(x, y) = (\{x\} \times Y) \cup (X \times \{y\})$$

is connected by Proposition 10.4

Now, as we can write

$$X \times Y = \bigcup_{(x,y) \in X \times Y} Z(x, y)$$

and the $Z(x_1, y_1)$ intersects $Z(x_2, y_2)$ (namely at (x_1, y_2) and (x_2, y_1)), then we see that this is a family of connected sets which are pairwise not separate and whose union is $X \times Y$, so we see that $X \times Y$ is connected.

■

Definition 10.6: Connected Components

A *component* or *connected component* of X is a maximal connected subset C. That is, if $C \subseteq Y \subseteq X$ and Y is connected, then $Y = C$.

Example

For example, $\mathbb{R}$ has one component as $\mathbb{R}$ is connected.

Interesting, $\mathbb{Q}$'s connected components are points. Suppose $A \subseteq \mathbb{Q}$ with A a component, and suppose that $x, x' \in A$ with $x \neq x'$. Say $x < x'$. Choose an irrational for which $x < \alpha < x'$. Then, we can write

$$A = (A \cap (-\infty, \alpha)) \cup (A \cap (\alpha, \infty))$$

These are disjoint open subsets of A which are nonempty, which contradicts the assumption that A is connected.

Proposition 10.7

Each component of X is closed, and disjoint components are separated.

Proof: Let C be a component. Then, as C is connected, $\overline{C}$ is connected. Now, if $C \neq \overline{C}$, then as $C \subseteq \overline{C}$, this contradicts the maximality of C. Thus, we must have that $C = \overline{C}$, so C is closed.

Now, let D be another component. If $D \cap C \neq \emptyset$, then we see that $D \cup C$ is connected, contradicting that D is a component. If $D \cap C = \emptyset$, then, as $D = \overline{D}$ and $C = \overline{C}$, so $\overline{D} \cap \overline{C} = \emptyset$, so they are separated. ■

Proposition 10.8

Every connected subset of X is in some component.

Proof: Let $A \subseteq X$ with $A \neq \emptyset$, and say A is connected.

Now, let C be the union of all nonempty connected sets containing A. Then, C is connected by Proposition 10.4. Then, we see that C is a component, so that C is a component containing A.

Corollary 10.9

X is the disjoint union of it's connected components.

Proof: Every point is connected, and components are disjoint. ■

9/21/26 Lecture 11: Path Connectedness

Definition 11.1: Paths

A *path* in a topological space X is a map $\gamma : [0,1] \to X$. We call $\gamma(0)$ the *beginning* of γ and $\gamma(1)$ the *end* of γ. We say γ *joins* $\gamma(0)$ and $\gamma(1)$.

Here is an example of a path



Definition 11.2: Path-Connectedness

A topological space X ia *path-connected* if two paths in X can be joined by a path.

Proposition 11.3

A path-connected space is connected.

Proof: Let A be path-connected. By contradiction, suppose that $A \subseteq X$ is clopen that is nonempty and not equal to X. Because of this, we can pick $x \in A$ and $y \in X \backslash A$.

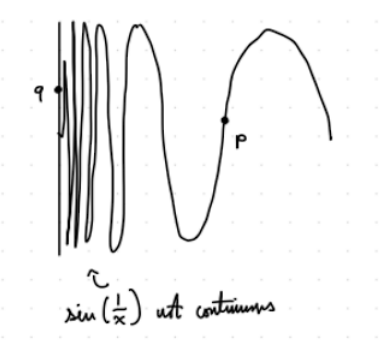
Now, by assumption, let $\gamma : [0,1] \to X$ be a path joining x and y. Then, $\gamma^{-1}(A)$ is clopen in $[0,1]$, but as $[0,1]$ is connected, we see that either $\gamma^{-1}(A) = \emptyset$ or $\gamma^{-1}(A) = [0,1]$. It cannot be empty because $0 \in \gamma^{-1}(A)$. It also cannot be $[0,1]$ as $y \notin A$, but $y = \gamma(1)$.

■

Example

Let $\{(t, \sin(t)) : 0 < t < 1\} \cup \{(0,s) : -1 \leq s \leq 1\}$. We see it is connected as it is the union of two connected sets that intersect on their limit points, but it is not path connected. As any point on the "sin" part of the curve has no continuous path to the "line" part of the curve.

Here is a picture of the above example.



Given $x, y, z \in X$, and paths α joining x to y and β joining y to z. Then, we can join these paths to get a path joining x to z by

$$\gamma(t) = \begin{cases} \alpha(2t) & 0 \leq t \leq 1/2 \\ \beta(2t-1) & 1/2 \leq t \leq 1 \end{cases}$$

We do need to check γ is a map from x to z.

Proposition 11.4

A connected open subset of $\mathbb{R}^n$ is path-connected.

Proof: Assume that $X \neq \emptyset$. Pick $x \in X$. Now, let

$$U(x) = \{y \in X \, : \, y \text{ can be joined to } x\}$$

We want to show that $U(x) = X$, which is to show that $U(x)$ is open and closed in X, since $U(x)$ is nonempty.

To show that $U(x)$ is open, let $y \in U(x)$. Now, as X is open, let B be a metric ball such that $y \in B \subseteq X$. Now, for all $z \in B$, then we can join y to z with a straight line, which implies we can join $x \to z$. Thus, we see that $z \in U(x)$. Thus, $U(x)$ is open in X.

Now, to show that $U(x)$ is closed, we want to show that

$$X - U(x) = \bigcup_{y \in X - U(x)} U(y)$$

First, we see immediately that $X - U(x) \subseteq \bigcup_{y \in X-U(x)} U(y)$. Now, we want to show the other direction, which is to show that for all $y \in X - U(x)$, $U(y) \subseteq X - U(x)$. Suppose not. Then, there is some $z \in U(y) \cap U(x)$. However, this implies there is a path between x and z, and a path between z and y, which implies that $y \in U(x)$, which is a contradiction. Thus, we have that

$$X - U(x) = \bigcup_{y \in X - U(x)} U(y)$$

This implies that $X - U(x)$ is closed, which means it is clopen and nonempty. Thus, as X is connected, we see that $U(x) = X$, which was to be proved.

■

Proposition 11.5

If X is path connected and $f : X \to Y$ is a map, then $f(X)$ is path connectedness.

Proof: Let $y_1, y_2 \in f(X)$. Then, pick $x_1, x_2 \in X$ for which $f(x_1) = y_1$ and $f(x_2) = y_2$. Then, as X is path connected, let $\gamma : [0,1] \to X$ be a path connecting x_1 and x_2. Then, it is clear that $f \circ \gamma : [0,1] \to Y$ is a path connecting y_1 and y_2.

Definition 11.6: Path Components

A *path component* of X is a maximal path connected set.

Proposition 11.7:

X is a disjoint union of it's path components.

Proof: Obvious, any intersection contradicts maximality.

Definition 11.8: $\pi_0(X)$

Define

$$\pi_0(X) = \{\text{path components of } X\}$$

Example

For the topologists sine curve, we have

$$\pi_0(S) = \{\{(t, \sin(t)) \,:\, 0 < t < 1\}, \{(0, s) \,:\, -1 \leq s \leq 1\}\}$$

Now, given a map $f : X \to Y$, this induces a map $f_* : \pi_0(X) \to \pi_0(Y)$ given as follows: Let C be a path component of X. Then $f(X)$ is path connected in Y, so it lies in some unique path component D of Y. Then, define $f_*(C) = D$.

Proposition 11.9

If we have $X \xrightarrow{f} Y \xrightarrow{g} Z$, then

$$(g \circ f)_* = g_* \circ f_*$$

sending $\pi_0(X)$ to $\pi_0(Z)$.

Proof: Given $C \in \pi_0(X)$, choose a point $x \in C$. Now, notice that $f_*(C)$ is a path component of Y containing $f(x)$, and so $g_*(f_*(C))$ is a path component of Z containing $g(f(x))$.

Similarly, we also see that $(g \circ f)_*(C)$ is also a path component of Z containing $g(f(x))$. Thus, these must

be the same, so

$$(g \circ f)_* = g_*(f_*(C))$$

■

Proposition 11.10

If $f : X \to Y$ is a homeomorphism, then $\pi_0(X)$ and $\pi_0(Y)$ are isomorphic as sets.

Proof: Consider $g = f^{-1} : Y \to X$. Then, we see that $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. Then, this implies that

$$g_* \circ f_* = (\mathrm{id}_X)_* = \mathrm{id}_{\pi_0(X)} \quad \text{and} \quad f_* \circ g_* = (\mathrm{id}_Y)_* = \mathrm{id}_{\pi_0(Y)}$$

so g_* is the inverse of f_*.

■

Corollary 11.11

If $n > 1$, then $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^n$.

Proof: Suppose $f : \mathbb{R} \to \mathbb{R}^n$ is a homeomorphism. Then, we see that

$$f\big|_{\mathbb{R}-\{0\}} : \mathbb{R} - \{0\} \to \mathbb{R}^n - \{f(0)\}$$

is also a homeomorphism. However, notice that $\pi_0(\mathbb{R} - \{0\})$ has two elements, but $\pi_0(\mathbb{R}^n - \{f(0)\})$ has only one element. This contradicts the fact that $f\big|_{\mathbb{R}-\{0\}}$ is a homeomorphism, so no such f can exist.

■

Corollary 11.12

The interval $[0, 1]$ is not homeomorphic to $\triangle$.

Proof: If $f : [0, 1] \to \triangle$ is such a homeomorphism, then

$$f\big|_{[0,1]-\{0.5\}} : [0, 1] - \{0.5\} \to \triangle - \{f(0.5)\}$$

is also a homeomorphism, but notice that $[0, 1] - \{0.5\}$ has two path components whereas $\triangle - \{f(0.5)\}$ has only two.

■

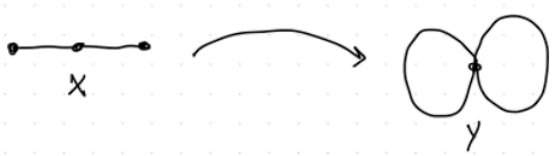
Thus, as we saw above, no space filling curve $f : [0, 1] \to \triangle$ can be injective, as any such function would be a homeomorphism.

9/23/26 Lecture 12: Identification Spaces

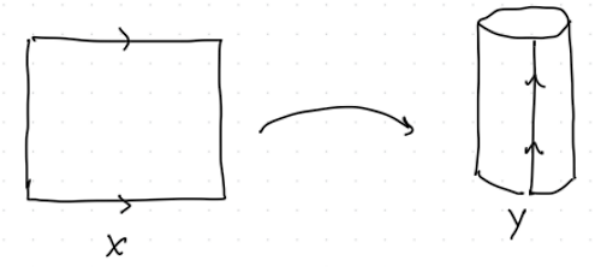
Consider $X = [0, 1]$ where we want to identify $\{0\}$ and $\{1\}$. What happens is that it becomes a circle!



Now, consider $X = [0, 1]$ where we want to identify $\{0\}, \{1/2\}$ and $\{1\}$. This gives us a figure 8!



Finally, consider $X = [0, 1] \times [0, 1]$. We want to identify the top and bottom edges. That is, for each $x \in [0, 1]$, we want to identify $(x, 0)$ and $(x, 1)$.



Question: If X is a topological space, how to put a topology on Y?

Definition 12.1: Identification Space

Let X be a set and P is a family of disjoint subsets whose union is X (that is, P is a partition). Then, the *identification space* Y is the set of elements in P.

This definition induces an onto function $f : X \to Y$ by $f(x) = [x]$ where $[x]$ is the element of P containing x.

Conversely, given an onto function $\pi : X \to Y$, then we can let $P = \{\pi^{-1}(y)\}_{y \in Y}$, and let that be an identification space.

Equivalently, we can also let X be a set and $\sim$ be an equivalence relation on X. Then, we can let P be the set of equivalence classes of $\sim$, and this gives us another identification space. In this case, we call Y the *quotient space* of $\sim$, denoted $X/\sim$.

Identification Topology (a.k.a Quotient Topology)

Suppose that X has a topology $\mathcal{T}$ and we want to put a topology on Y, an identification space. To do this, all we need is to force the quotient map $\pi : X \to Y$ to be continuous.

Definition 12.2: Identification Topology

Given an identification space Y of X, we say $\mathcal{O} \subseteq Y$ is open if $\pi^{-1}(\mathcal{O})$ is open in X.

Definition 12.3: Identification Maps

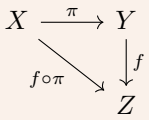
Suppose that X and Y are topological spaces. We say that an onto map $f : X \to Y$ is an *identification map* if $\mathcal{O} \subseteq Y$ is open if, and only if $f^{-1}(\mathcal{O})$ is open in X. Then, Y is homeomorphic to the identification space induced by $P = \{f^{-1}(y)\}_{y \in Y}$.

Warning: If $f : X \to Y$ is an identification map, it need not send open sets to open sets. Consider the following example: Let $X = \{a, b, c, d\}$ with the topology $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, b, c, d\}\}$, and let $Y = \{e, f\}$ consider the onto map $f : X \to Y$ by $\phi(a) = \phi(c) = e$ and $\phi(b) = \phi(d) = f$.

Then, the quotient topology induced by f is $\{\emptyset, Y\}$ as $\phi^{-1}(e)$ and $\phi^{-1}(d)$ are not open in X. We see immediately that ϕ is not an open map as $\phi(\{a\}) = \{e\}$ which is not open.

Proposition 12.4

Say $\pi : X \to Y$ is an ID map. Given a topological space Z and a map $f : Y \to Z$, then f is continuous iff $f \circ \pi : X \to Z$ is continuous. That is, π satisfies the universal property for a quotient map



Proof:

($\Rightarrow$) This is immediate as π is continuous.

($\Leftarrow$) Suppose that $f \circ \pi$ is continuous, and let $U \subseteq Z$ be open. Then, we see that

$$(f \circ \pi)^{-1}(U) = \pi^{-1}(f^{-1}(U)) \text{ is open}$$

But as π is an identification map, we see that $f^{-1}(U)$ is open.

■

Proposition 12.5

Suppose that $f : X \to Y$ is an onto map from $X \to Y$. Then, if f is an open map (resp. closed map); that is, it sends open sets to open sets (resp. closed sets to closed sets), then it is an ID map.

Proof: Suppose that f is an open map. We need to show that $O \subseteq Y$ is open if, and only if, $f^{-1}(O)$ is open.

Now, if $O \subseteq Y$ is open, then $f^{-1}(O)$ is open as f is a map.

Now, suppose that $f^{-1}(O)$ is open. Then, as $f(f^{-1}(O)) = O$ because f is surjective, and f is an open map, we see that O is open, which was to be shown.

■

Proposition 12.6:

Suppose that $f : X \to Y$ is an onto map. Then, if X is compact and Y is Hausdorff, then f is an ID map. Hence, Y is homeomorphic to the ID space corresponding to $P = \{f^{-1}(Y)\}_{y \in Y}$.

Proof: It is enough to show that f sends closed sets to closed sets by Proposition 12.5. Let C be closed. Then C is compact, so $f(C)$ is compact in Y. Then, $f(C)$ is closed as Y is Hausdorff.

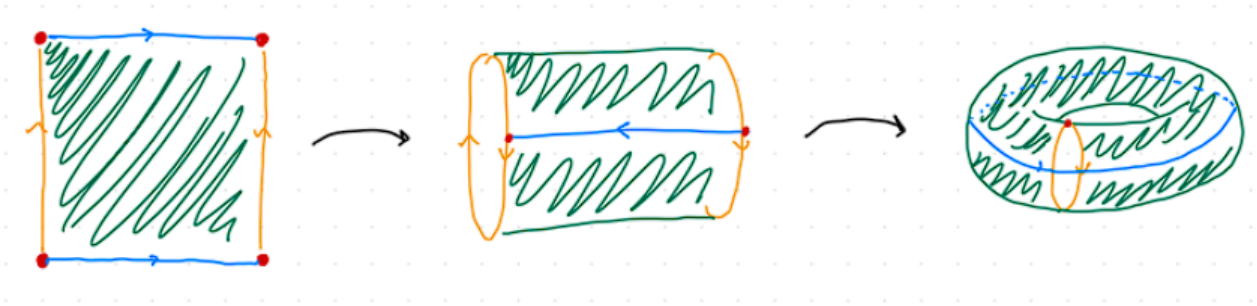
■

Now with the formal rigamarole, here are some fun examples of identification spaces:

Let $X = [0, 1] \times [0, 1]$. Say P consists of:

1. all of the 4 corner points of the square.
2. the top and bottom edges of the square.
3. the left and right edges of the square.
4. all the interior points.

This is what this looks like:



We call this surface T^2, or the two-torus.

Lemma 12.7

The 2-torus, T^2 is homeomorphic to $S^1 \times S^1$.

Proof: Consider the map $f : [0,1] \times [0,1] \to S^1 \times S^1$ by $f(r,s) = (e^{2\pi ir}, e^{2\pi is})$. We see that f is surjective, $[0,1] \times [0,1]$ is compact, and $S^1 \times S^1$ is Hausdorff.

The preimages of f will consist of either all the corners of the square, the top and bottom edges of the square, the left and right edges of the square, or some interior open set. Thus, by above, there exists a homeomorphism h so that this diagram commutes.

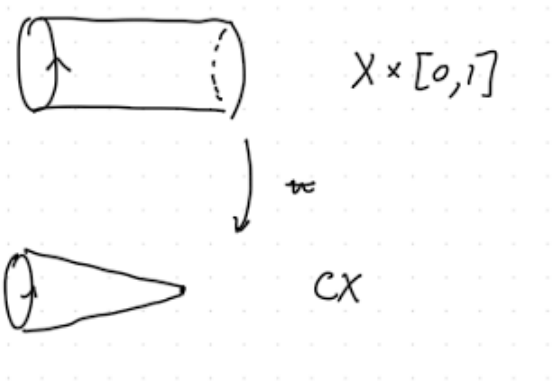
$$\begin{array}{ccc} [0,1] \times [0,1] & & \\ \downarrow{\scriptstyle \pi} & \searrow^{f} & \\ T^2 & \xrightarrow[h]{} & S^1 \times S^1 \end{array}$$

■

Definition 12.8: Cone over a Topological Space

Suppose that X is a topological space. Start with $X \times [0,1]$, and let $P = \{X \times \{1\}\} \cup \{(x,t)\}_{x \in X, t \in [0,1)}$. Then, the ID space is CX called the *cone* over X.

Here's an example to show why it is called a cone: Suppose $X = S^1$. Then, the cone over X is



9/25/26 Lecture 13: Identification Spaces (cont.) and Topological Groups

Here are some more examples of identification spaces:

Example: n-Spheres and n-Balls

Consider B^n/S^{n-1} which consists of $X = B^n = \{x \in \mathbb{R}^n : |x| \leq 1\}$ identified by the partition be

$$\mathcal{P} = \{\{x \in \mathbb{R}^n : |x| = 1\}\} \cup \{\{x\} : |X| < 1\}$$

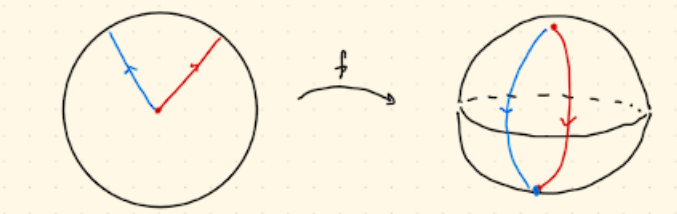
We want to show the identifcation space is homeomorphic to S^n. To prove this, we want to produce a map $f : B^n \to S^n$ which is onto and the preimage of points are our partition. From our theorem, that would imply that S^n is homeomorphic to our ID space since B^n is compact and S^{n-1} is Hausdorff.

The idea is to send the interior of the n-ball onto the sphere minus a point, and then send the boundary of the ball all to a single point.

Let

$$f(x_1, \ldots, x_n) = \left(\pi \frac{\sin r}{r} x_1, \ldots, \pi \frac{\sin r}{r} x_n, \cos r\right)$$

where $r = \pi\sqrt{x_1^2 + \cdots + x_n^2}$. Adding up the squares, this is a map $f : B^n \to S^n$. This visual depicts what it does (don't mind the white background, i forgot my example boxes are yellow).



Indeed preimage of the south pole is the boundary of B^n, and any other point is just a point in the ball.

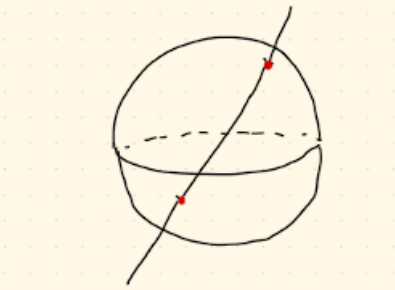
Example: Projective Space

Now, here is another example. There are three ways to construct it:

1. Consider $X = S^n$ and let $P = \{\{x, -x\} : x \in S^n\}$. We are identifying antipodal points on the sphere, and the corresponding identification space is called n-dimensional *projective space*, denoted P^n.
2. Another way to construct this is starting with $\mathbb{R}^{n+1} - \{0\}$, and let $P = \{\text{lines through the origin}\}$.
3. One more way to construct it is letting $X = B^n$ and letting

$$P = \{\{x, -x\} : |x| = 1\} \cup \{\{x\} : |x| < 1\}$$

We want to show that these are equivalent first. To show that 2 is equivalent to 1, consider this picture. Notice that our lines intersect antipodal points on the sphere. Thus, there is an obvious homeomorphism between these two identification spaces.



Now, we want to show that 1 is equivalent to 3. Notice that we can think of 3 as the upper half of S^n where we also identify antipodal points of the equator (consider the idea of the first example). Thus, we can construct a homeomorphism as follows: For antipodal points on S^n that don't lie on the equator, pick the one that lies above the equator



Then, for antipodal points on S^n that lie on the equator, just pick both of those points (recall that we need to identify antipodal points on the equator)



Now, we move onto gluing.

Definition 13.1: Gluing Maps Together

Say T is a topological space and suppose that X and Y are subsets with $T = X \cup Y$. Now, given another topological space Z, with maps $f : X \to Z$ and $g : Y \to Z$ such that $f|_{X\cap Y} = g|_{X\cap Y}$. Then, we can glue these maps together to get a function $F : T \to Z$ with

$$F(p) = \begin{cases} f(p) & p \in X \\ g(p) & p \in Y \end{cases}$$

Lemma 13.2: The Gluing Lemma

If X and Y are closed in Y, then the map F above is continuous.

Proof: Suppose that $C \subseteq Z$ is closed. Then, we can write

$$F^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$$

Since f is continuous, we have that $f^{-1}(C)$ is closed in X, so we can write

$$f^{-1}(C) = X \cap D$$

for some closed D in T. Thus, $f^{-1}(C)$ is closed in T as it is the intersection of two closed sets. Similarly, $g^{-1}(C)$ is closed in T, so $F^{-1}(C)$ is closed in T. Thus, F is continuous.

■

Example: Gluing Paths Together

Let $T = [0, 1]$, $X = [0, 1/2]$ and $Y = [1/2, 1]$. Then, if $f : [0, 1/2] \to Z$ and $g : [1/2, 1]$ are continuous and $f(1/2) = g(1/2)$, then we have a continuous map $F : [0, 1] \to Z$.

Example: Not an Example!

Let $T = \mathbb{R}$, $X = \mathbb{Q}$, and $Y = \mathbb{R}\backslash\mathbb{Q}$, and let $f : X \to \mathbb{R}$ with $f = 1$ and $g : Y \to \mathbb{R}$ by $g = -1$. They agree on their empty intersection, but there is no gluing map. The problem here is that X and Y are not closed.

Topological Groups

Recall that a group is a set with an associative operation $m : G \times G \to G$, an inverse operation $i : G \to G$, and an identity for m.

Definition 13.3: Topological Groups

A *topological group* is a Hausdorff space G with a group structure; that is, functions m, i that are continuous.

Example: Examples of Topological Groups

Here are some examples of topological groups.

1. One basic example is $G = (\mathbb{R}, +)$. Since addition is continuous and taking the negative is continuous, G is a topological group. This also means that $(\mathbb{R}, +)$ is a topological group with the induced group structure.
2. Another example is $G = S^1$ with the operation $m(e^{i\theta}, e^{i\phi}) = e^{i(\theta+\phi)}$ and $i(e^{i\theta}) = e^{-i\theta}$. We see these operations are continuous (under subspace topology), so this makes G a topological group. This also makes $T^2 = S^1 \times S^1$ a topological group by using the induced group structure on $S^1 \times S^1$.
3. A trivial example is any group G with the discrete topology (obviously).
4. One interesting example is

$$S^3 \simeq \{a + bi + cj + dk \in \mathbb{H} : a^2 + b^2 + c^2 + d^2 = 1\}$$

where $\mathbb{H}$ are the quaternions. Recall that the quaternions is the set $\{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ with the relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k,\ ji = -k,\ jk = i,\ kj = -i,\ ik = j,\ ki = -j$$

We see that $\mathbb{H}$ is a non-commutative groups. It turns out S^3 is a topological group under this operation with the inverse

$$(a + bi + cj + dk)^{-1} = a - bi - cj - dk$$

(for $a^2 + b^2 + c^2 + d^2 = 1$).

5. Finally, one useful example is $\mathrm{GL}(n, \mathbb{R}) = \{\text{invertible } n \times n \text{ real matrices}\}$, where the topology comes from the subspace topology of $\mathbb{R}^{n^2}$, and the group structure comes from the group structure under matrix multiplication.

9/28/26 Lecture 14: More on Topological Groups

Definition 14.1: Left and Right Multiplication Maps

For a topological group G and $x \in G$, define $L_x : G \to G$ by

$$L_x(g) = m(x, g) = x \cdot g$$

and similarly, define $R_x : G \to G$ by

$$R_x(g) = m(g, x) = g \cdot x$$

Lemma 14.2

The maps L_x and R_x defined above are homeomorphisms.

Proof: First, we will show that L_x is continuous.

Define $\phi : G \to G \times G$ by $\phi(g) = (x, g)$. If U and V are open in G, then $U \times V$ is a basis element for $G \times G$. Then, we see that

$$\phi^{-1}(U \times V) = \begin{cases} \emptyset & \text{if } x \notin U \\ V & \text{if } x \in U \end{cases}$$

which is open. Thus, ϕ is continuous. Now, notice that $L_x = m \circ \phi$, so thus, L_x is continuous as m is.

Then, as $(L_x)^{-1} = L_{x^{-1}}$, we see that L_x has a continuous inverse, so it is a homeomorphism. The same proof works for R_x.

Corollary 14.3

For a topological group G and any $g, g' \in G$, there exists a homeomorphism $f : G \to G$ for which $f(g) = g'$.

Proof: Let $f = L_{g'g^{-1}}$. This is a homeomorphism satisfying the above property.

Definition 14.4: Homogeneous Space

A topological space X is *homogeneous* if, for any pair of points $x, y \in X$, there is a homeomorphism $f : X \to X$ such that $f(x) = y$.

Proposition 14.5

If G is a topological group, and K is the connected component containing e, then K is a closed normal subgroup of G.

Proof: First, we see that K is closed as all components are closed. Now, we need to show that K is a normal subgroup of G.

Notice that for any $x \in K$, $Kx^{-1} = R_{x^{-1}}(K)$, so Kx^{-1} is connected. However, since $x \in K$, $e = xx^{-1} \in R_{x^{-1}}(K)$, so $R_{x^{-1}}(K)$ is the connected component containing e, which is contained in K. Then, as $e \in K$, we see that $x^{-1} \in R_{x^{-1}}(K)$, so $x^{-1} \in K$. Thus, K is closed under inverses.

Now, if $y \in K$, then $yx^{-1} \in K$, as $yx^{-1} \in R_{x^{-1}}(K)$, which implies that K is closed under products.

Finally, we need to show that K is a normal subgroup. Let $g \in G$. Then, we see that

$$gKg^{-1} = L_g(R_{g^{-1}}(K))$$

This is connected as L_g and $R_{g^{-1}}$ are continuous, but also $e \in gKg^{-1}$ as $e \in K$. Thus, as gKg^{-1} is a connected component containing e, it must be in K. Thus, $gKg^{-1} \subseteq K$, so K is normal.

■

Proposition 14.6

Suppose that G is a connected topological group, and V is an open neighborhood of e. Then, G is generated by V.

Proof: Let $H = \langle V \rangle$ be the subgroup generated by V. Then, for all $h \in H$, we can write

$$h = v_1^{n_1} v_2^{n_2} \cdots v_k^{n_k}$$

with $v_i \in V$ and $n_i \in \mathbb{Z}$. We want to show that H is both open and closed. To see that H is open, let $h \in H$ and consider hV. We have that $hV \subseteq H$ by definition of H. Well, $hV = L_h(V)$, so as L_h is a homeomorphism, $L_h(V)$ is an open neighborhood of h ($h \in hV$ as $e \in V$).

Now, to show that H is closed. We want to show that $G \setminus H$ is open. Let $g \in G \setminus H$. Now, $gV = L_g(V)$ is an open neighborhood of g by above, so we want to show that $gV \subseteq G \setminus H$. To do this, notice that if $x \in H$ and $x \in gV$, then we can write $x = gv$ for some $v \in V$, but when we see that $g = xv^{-1}$, and as $xv^{-1} \in H$, this implies that $g \in H$, which is a contradiction. Thus, we must have that $gV \subseteq G \setminus H$, so $G \setminus H$ is open.

Thus, as G is connected, either $H = G$ or $H = \emptyset$, but H contains e, so it must be the former.

■

Proposition 14.7

Consider the group $\mathrm{GL}(n) = \{\text{invertible } n \times n \text{ real matrices}\}$ with the topology coming from the homeomorphism with $\mathbb{R}^{n^2}$. Then, $\mathrm{GL}(n)$ is a topological group.

Proof: First, we want to show that $m : M_n \times M_n \to M_n$ is continuous. Well, we have that

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

is a polynomial in a_{ij} and b_{ij}, which are continuous on $\mathbb{R}^2$. We also have that

$$(A^{-1})_{ij} = \frac{1}{\det(A)}(ji^{th} \text{ cofactor of A})$$

Then, as this is a rational function in the entires of (a_{ij}) with a determinant that is never 0, we see that inversion is also continuous.

■

Proposition 14.8

Consider the following subgroups of $\mathrm{GL}(n)$:

$$O(n) = \{A \in \mathrm{GL}(n) \,:\, A^{-1} = A^T\}$$

and

$$SO(n) = \{A \in O(n) \,:\, \det(A) = 1\}$$

Then, $O(n)$ and $SO(n)$ are compact.

Proof: Start with $O(n)$. To show that it is closed and bounded in M_n, as this is isomorphic to $\mathbb{R}^{n^2}$. Define $\phi : M_n \to M_n$ by $\phi(M) = MM^T$. Then, we see that

$$O(n) = \phi^{-1}(I_n)$$

so as ϕ is continuous, and as $\{I_n\}$ is closed, we see that $O(n)$ is closed. Now, to show that it is bounded, notice that

$$A \in O(n) \implies \mathrm{tr}(AA^T) = \mathrm{tr}(I) \implies \sum_{i,j=1}^{n} a_{ij}^2 = n$$

thus, we see that $O(n)$ is bounded in M_n.

Now, we will do $SO(n)$. First, notice that $\det : O(n) \to \mathbb{R}$ is continuous, so as $SO(n) = \det^{-1}(1)$, and $\{1\}$ is closed, we see that $SO(n)$ is closed in $O(n)$, so it is compact.

■

Here are some examples of these groups and what spaces they are similar to.

- $n = 1$: We see that $SO(n) = \{1\}$ and $O(1) = \{1, -1\}$
- $n = 2$: We have
$$SO(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \right\} = S^1$$
and $O(2)$ is homeomorphic to a disjoint union of two circles; $SO(2)$ and $SO(2)$ times the matrix
$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
- $n = 3$: We have that $SO(3)$ is homeomorphic to P^3, three dimensional projective space.

9/30/26 Lecture 15: Group Actions

Recall the quaternions

$$\mathbb{H} = \{a + bi + cj + dk \,:\, a, b, cd \in \mathbb{R}\}$$

with multiplication such that

$$i^2 = j^2 = k^2 = -1$$

and

$$ij = k,\ ji = -k\ jk = i\ kj = -i\ ki = j\ ik = -j$$

Then, the unit quaternions

$$U_{\mathbb{H}} = \{q \in \mathbb{H} \,:\, a^2 + b^2 + c^2 + d^2 = 1\}$$

turn out to be a group. Because we can identify a point in $U_{\mathbb{H}}$ with three coordinates satisfying a condition, we see that

$$U_{\mathbb{H}} \simeq S^3$$

Now, the pure quaternions

$$\{bi + cj + dk \,:\, b, c, d \in \mathbb{R}\} \simeq \mathbb{R}^3$$

are not a group under multiplication. However, it can be checked that if q is a unit quaternion and v is a pure quaternion, then qvq^{-1} is a pure quaternion. We will use the above discussion to prove the following lemma.

Proposition 15.1

The group $SO(3)$ is homeomorphic to P^3.

Proof: Define a map $\phi : S^3 \to SO(3)$ as follows: given a unit quaternion in S^3, define $\phi(q) = \phi_q \in SO(3)$ by

$$\phi_q(v) = qvq^{-1}$$

for any $v \in \mathbb{R}^3$ for any pure quaternion. Now $\phi(q)$ sends $\mathbb{R}^3$ to $\mathbb{R}^3$ linearly, so we can check that it is indeed a matrix in $SO(3)$. This is a homomorphism $S^3 \to SO(3)$. Thus also turns out to be continuous and onto. Interestingly, we see that

$$\ker(\phi) = \{\pm 1\} \subseteq U_{\mathbb{H}}$$

then, by the first isomorphism theorem, we see that

$$S^3/\{\pm 1\} \to SO(3)$$

is an isomorphism of groups, and it turns out this is also a homeomorphism. Finally, notice that

$$S^3/\{\pm 1\} \simeq \mathcal{P}^3$$

as points in $S^3/\{\pm 1\}$ identify antipodal points in S^3.

■

Group Actions

Let G be a group and let X be a set. Then, we say that G acts on X to mean that we have a map $f : G \times X \to X$, which we denote $(g, x) \to gx$ so that $g_1(g_2x) = (g_1g_2)x$ and $ex = x$. Equivalently, a group action is a homeomorphism $\phi : G \to \mathrm{Perm}(X)$, where $\mathrm{Perm}(X)$ is the permutation group on X.

For example, we see that $O(n)$ acts on $\mathbb{R}^n$ by $(A, v) \mapsto Av$.

Definition 15.2: Orbits of Group Actions

We define the *orbit* of $x \in X$ under a group action to be

$$\mathcal{O}(x) = \{gx : g \in G\}$$

This gives us a partition of x by orbits.

For $O(n)$ acting on $\mathbb{R}^n$, it turns out that $\mathcal{O}(x)$ is a sphere around the origin containing x, and $\mathcal{O}(0) = \{0\}$.

Definition 15.3: Orbit Spaces

An *orbit space* is a set denoted X/G, is the identification space induced by the partition of X by G-orbits.

For $O(n)$ acting on $\mathbb{R}^n$, we see that the orbit space is $[0, \infty)$, as each sphere can be identified by its radius. Here are some another example:

Example

Let $G = \mathbb{Z}$ and $X = \mathbb{R}$, and consider $(n, x) \mapsto x + n$ for $n \in \mathbb{Z}$ and $x \in \mathbb{R}$. Then, the orbit space is $[0, 1)$, as each orbit must have one element in $[0, 1)$. Then, as $[0, 1) \simeq S^1$, we see that that

$$S^1 \simeq \mathbb{R}/\mathbb{Z}$$

Now, consider $O(n)$ acting on $S^{n-1} \subseteq \mathbb{R}^n$. This action is transitive meaning that there is only a single orbit. That is, between any two points x, y in S^{n-1}, there exists a $g \in O(n)$ such that $gx = y$. To see this, we see that if $(e_1, \dots, e_n)$ is the standard basis of $\mathbb{R}^n$, we can complete this to an orthonormal basis $(v, f_2, \dots, f_n)$, and as there is some orthonormal matrix that maps $(e_1, \dots, e_n)$ to $(v, f_2, \dots, f_n)$. Thus, v and e_1 lie in the same orbit, and as v was arbitrary, all of S^{n-1} lies in the orbit of e_1.

Definition 15.4: Topological Groups acting on Topological Spaces

A topological group G *acts o*n a topological space X if G acts on X and

$$G \times X \to X \text{ is continuous}$$

The above definition gives us a homeomorphism $\phi : G \to \text{Homeo}(X)$ where

$$\text{Homeo}(X) = \{\text{Homeomorphisms from } X \to X\}$$

and gives us the ID space X/G from the partition into G orbits. Here are some examples.

Example

For example, if we consider $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ with $(n, x) \to x + n$ with $n \in \mathbb{Z}^2$ and $x \in \mathbb{R}^2$. Then, $\mathbb{R}^2/\mathbb{Z}^2$ is homeomorphic to T^2 as we can pick representatives in $[0, 1) \times [0, 1) \simeq S^1 \times S^1 \simeq T^2$.

Consider $\mathbb{Z}_2$ acting on S^n by $\phi(1) = \text{id}_{S^n}$ and $\phi(-1) = -\text{id}_{S^n}$. Then, we see that

$$S^n/\mathbb{Z}_2 = P^n$$

Now, let G be any topological group, and let H be any subgroup of G. We know H acts on G by left-

multiplication $(h, g) \to hg$. Now, g, g' is in the same orbit if there is some h such that $g' = hg$. Thus, the orbit space is the set of right-cosets

$$G/H = Hg$$

Similarly, we also know that H acts on G by right-multiplication $(h, g) \to gh^{-1}$. From the same arguments above, the orbit space is the set of left-cosets

$$G/H = gH$$

Proposition 15.5

If G acts on X, then the ID map

$$\pi : X \to X/G$$

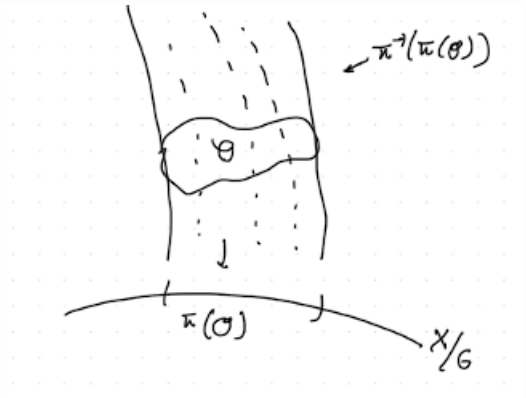
is an open map.

Proof: Say that $\mathcal{O} \subseteq X$ is open. We want to show that $\pi(\mathcal{O})$ is open in X/G. That is to show that $\pi^{-1}(\pi(\mathcal{O}))$ is open in X, by the definition of an identification map.

Let $x \in \pi^{-1}(\pi(\mathcal{O}))$. Then, as $\pi(x) \in \pi(\mathcal{O})$, this implies that there exists a $y \in \mathcal{O}$ so that $\pi(x) = \pi(y)$, so $x = gy$ for some $g \in G$. Thus, we have that

$$\pi^{-1}(\pi(\mathcal{O})) = \bigcup_{g \in G} g\mathcal{O}$$

Then, as this is a union of open sets in X, we see that $\pi^{-1}(\pi(\mathcal{O}))$ is open. Here is a visual proof:



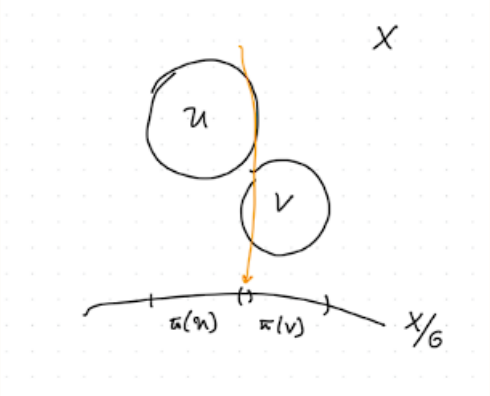
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Proposition 15.6

If G acts on X, and G and X/G are connected, then X is connected.

Proof: Suppose not. Then, we can write $X = U \cup V$ with U and V being disjoint nonempty open sets. Then, as the projection π is open, then $\pi(U)$ and $\pi(V)$ are nonempty open sets in X/G. However, as X/G is connected, then $\pi(U) \cap \pi(V)$ must not be empty.

Now, say $\pi(x) \in \pi(U) \cap \pi(V)$. Then, there exists an orbit Gx that intersects U and V. Since the map $G \to X$ by $g \to gx$ is continuous, which implies that the image of G is connected. Thus, Gx must be connected, but $Gx \cap U$ and $Gx \cap V$ are two nonempty open sets whose union is G. Thus, we have a contradiction. Here is a visual proof:



■

10/2/26 Lecture 16: Topological Groups cont. and Intro to Algebraic Topology

Let's consider some examples:

Consider $O(n)$ acting on S^{n-1}. If $A \in O(n)$ and v is a unit vector in $\mathbb{R}^n$, then $(A, v) \to Av$ is our group action. Now, consider $e_1 \in S^{n-1}$. Which $A \in O(n)$ send e_1 to e_1? Well, it must be a matrix of the form

$$\begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$$

where $B \in O(n-1)$. Thus, this gives us a map $\alpha = O(n)/O(n-1) \to S^{n-1}$ by $[g] \to ge_1$, which is well defined. The map α is onto since $O(n)$ acts transitively on S^{n-1}, injective since we quotiented by the isotropy group of e_1. Then, $O(n)$ is compact, so the quotient space is compact, and as S^{n-1} is Hausdorff, we see that α is a homeomorphism between $O(n)/O(n-1)$ and S^{n-1}.

By the same argument $SO(n)/SO(n-1)$ is homeomorphic to S^{n-1}.

Proposition 16.1

For any n, $SO(n)$ is connected.

Proof: We will prove this by induction. First $SO(1) = \{(1)\}$ is a point, so it is connected.

Now, suppose that for some $n > 1$, $SO(n-1)$ is connected. Then, taking $G = SO(n-1)$ and $X = SO(n)$, where G acts on X by "multiplication" (make an n by n block matrices with elements of G), so as G is connected, and $X/G = SO(n)/SO(n-1)$ is connected, we must have that $X = SO(n)$ is connected. ■

Example: Lens Spaces

Let

$$\begin{aligned} S^3 &= \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} \\ &= \{(a, b, c, d) \in \mathbb{R}^3 : a^2 + b^2 + c^2 + d^2 = 1\} \end{aligned}$$

Then, pick relatively prime integers $q < p$. Now, consider $G = \mathbb{Z}_p$ and $X = S^3$. We can define how G acts on X by saying that $1 \in \mathbb{Z}_p$ sends

$$(z_1, z_2) \mapsto (e^{\frac{2\pi i}{p}} z_1, e^{\frac{2\pi i q}{p}} z_2)$$

We call X/G the *lens space* denoted $L(p, q)$. The map $S^3 \to L(p, q)$ turns out to be p-to-one. Note that if $p = 1$ and $q = 1$, then $L(2, 1) = P^3$.

Example:

Consider $X = \mathbb{R}^2$. An *isometry* of $\mathbb{R}^2$ is a distance preserving map $\mathbb{R}^2 \to \mathbb{R}^2$. The group $\text{Isom}(\mathbb{R}^2)$ of isometries of $\mathbb{R}^2$ are a group under composition, and it turns out

$$\text{Isom}(\mathbb{R}^2) = \{x \mapsto Ax + b : A \in O(2), b \in \mathbb{R}^2\}$$

Note that

$$x \mapsto A_2x + b_2 \mapsto A_1(A_2x + b_2) + b_1 = A - 1A_2x + A_1b_2 + b_1$$

so if we represent $\mathrm{Isom}(\mathbb{R}^2)$ as pairs (A, b), then

$$(b_1, A_1) \cdot (b_2, A_2) = (b_1 + b_2A, A_1A_2)$$

This is called the *semidirect product* of $\mathbb{R}^2$ and $O(2)$. More generally, if we have groups G_1 and G_2 and a homomorphism $\alpha : G_2 \to \mathrm{Aut}(G_1)$, then the semidirect product $G_1 \tilde{\times} G_2$ is the group over $G_1 \times G_2$ with the following operation

$$(g_1, g_2)(g_1', g_2') = (g_1 \cdot \alpha(g_2)(g_1'), g_2g_2')$$

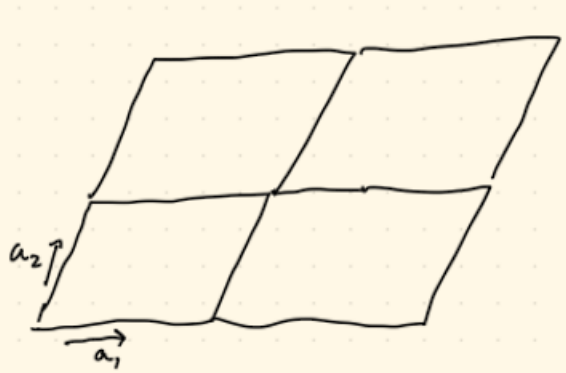
so in our case, we have $G_1 = (\mathbb{R}^2, +)$, $G_2 = O(2)$, and $\alpha : O(2) \to \mathrm{Aut}(G_1)$ is the inclusion $\alpha : O(2) \to \mathrm{GL}_2(\mathbb{R})$.

Going back, consider the topological space, $\mathrm{Isom}(\mathbb{R}^2) = \mathbb{R}^2 \times O(2)$. Then, if G is a discrete subgroup of $\mathrm{Isom}(\mathbb{R}^2)$ (discrete in the subspace top), then G acts on $\mathbb{R}^2$ by isometries, and we call G a *plane crystallographic group* or *wallpaper group* if $\mathbb{R}^2/G$ is compact.

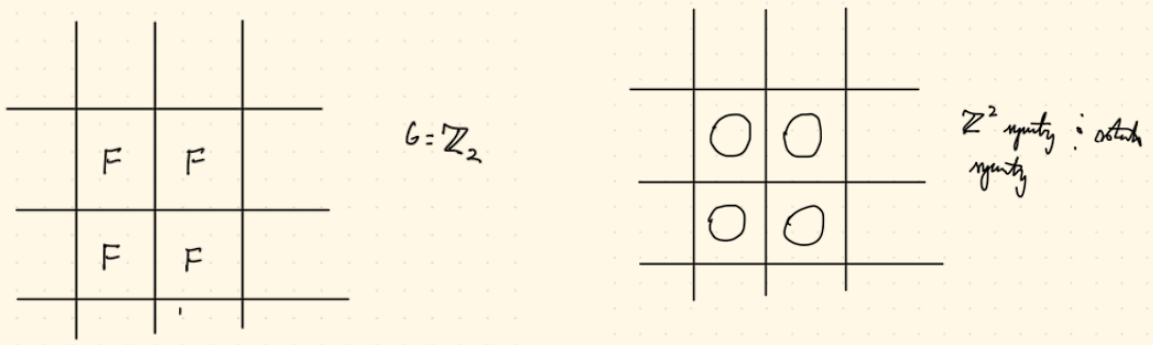
For instance, if $G = \mathbb{Z}^2 \subseteq \mathbb{R}^2 \subseteq \mathrm{Isom}(\mathbb{R}^2)$ by

$$G = \{x \mapsto x + n_1a_1 + n_2a_2 : n_1, n_2 \in \mathbb{Z}\}$$

where a_1, a_2 are fixed vectors in $\mathbb{R}^2$.



It turns out that there are only 17 distinct wallpaper groups (up to isomorphism!). Here are some examples and their corresponding symmetries:



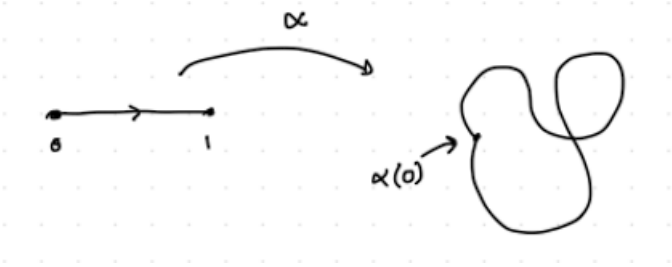
We can also do the same thing with $\mathbb{R}^2$ giving the spatial crystallographic groups, of which there are 230 unique.

Moving Towards Algebraic Topology

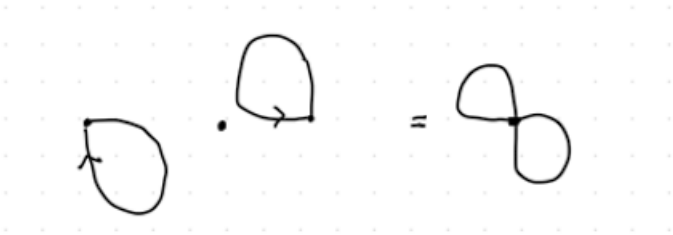
Let X be a topological space.

Definition 16.2: Loops

A loop in X is a map $\alpha : [0,1] \to X$ such that $\alpha(0) = \alpha(1)$. We say that the loop is *based* at $\alpha(0)$.



Idea: We fix a point $p \in X$, and we can form a group out of loops based at p where multiplication is the composition of loops, and where the inverse is traversing the loop backwards.



Here are two problems with this construction:

1. Too many loops!
2. Multiplication is not associative, traversal speeds change!
3. A loop times its inverse is not the identity!

Fix: Consider loops up to deformation. What is this? Given two maps, we want to define a continuous way of transforming f into g. That is, we want this collection $\{f_t\}_{t\in[0,1]}$ of continuous maps so that $f_0 = f$ and $f_1 = g$, and they transition smoothly.

We will define this more rigorously here.

Definition 16.3: Homotopies

Let $f, g : X \to Y$ be maps. Then, we say f is *homotopic* to g, denoted $f \overset{F}{\sim} g$, if there exists a map $F : X \times I \to Y$ with $I = [0, 1]$ so that $f(x) = F(x, 0)$ and $g(x) = F(x, 1)$ for all $x \in X$. We then call F a *homotopy* from f to g.

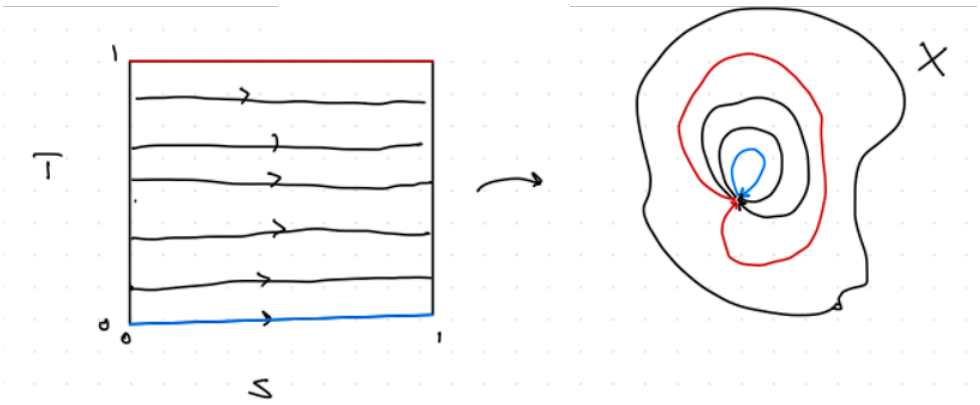
10/5/26 Lecture 17: The Fundamental Group

If $A \subseteq X$ and $f|_A = g|_A$, then we may want a homotopy that preserves the values on A. That is, for our homotopy F, we want $F(a,t) = f(a) = g(a)$ for all $t \in I$. In this case, we say that f, g are *homotopic relative to* A, and we write $f \overset{F}{\simeq} g$ (rel A).

For instance, if $\alpha, \beta : I \to X$ are two loops based at p, then we can ask if they are homotopic relative to $\{0,1\}$. If so, then we have a map $F : I \times I \to X$ with

$$F(s,0) = \alpha(s),\ F(s,1) = \beta(s),\ F(0,t) = p\ F(1,t) = p$$

for all $s \in [0,1]$ and $t \in [0,1]$.



Example: Convex Sets

Suppose that $C \subseteq \mathbb{R}^n$ is convex; that is, for any $p, q \in C$, the straight line segment joining them lies in C. Then, any two maps $f, g : X \to C$ are homotopic.

Proof: Take $F(x,t) = (1-t)f(x) + tg(x)$. Then, we see that $F(x,0) = f(x)$ and $F(x,1) = g(x)$ for all x, and as C is convex, F is indeed a map into C. Thus, f is homotopic to g.

Example: Maps into the Sphere

Let $f, g : X \to S^n$ and $\{f(x), g(x)\}$ are never antipodal points. Then, f is homotopic to g.

Proof: Consider $f, g : X \to S^n \subseteq \mathbb{R}^{n+1}$. By assumption, for all $x \neq 0$, we have that

$$f(x) + g(x) \neq 0$$

as $f(x)$ and $g(x)$ are never antipodal points. Now, in $\mathbb{R}^{n+1}$, we have the homotopy

$$(1-t)f(x) + tg(x)$$

which is never 0 for all x, as $f(x)$ and $g(x)$ are never antipodal points. Thus, we can consider the homotopy

$$F(x,t) = \frac{(1-t)f(x) + tg(x)}{|(1-t)f(x) + tg(x)|} \in S^n$$

which is well defined. Thus, f is homotopic to g.

Lemma 17.1

Homotopy is an equivalence relation on maps from X to Y.

Proof: For reflexivity, if taking $F(x,t) = f(x)$ for all t, x, then we see that $f \overset{F}{\simeq} f$.

Now, if $f \overset{F}{\simeq} g$, then if $F(x,t)$ is such a homotopy, letting

$$G(x,t) = F(x, 1-t)$$

we get the reverse homotopy $g \overset{G}{\simeq} f$.

Finally, suppose that $f \overset{F}{\simeq} g$ and $g \overset{G}{\simeq} h$. Then, letting F and G be the corresponding homotopies, we can let

$$H(x,t) = \begin{cases} F(x,2t) & 0 \le t \le 1/2 \\ G(x,2t-1) & 1/2 \le t \le 1 \end{cases}$$

we see that $f \overset{H}{\simeq} h$.

■

Lemma 17.2

If $A \subseteq X$, then homotopy relative to A is an equivalence relation on maps $f : X \to Y$ with a given value on A.

Proof: Same as before.

Lemma 17.3

If $f, g : X \to Y$ are homotopic, and $h : Y \to Z$, then $h \circ f$ is homotopic to $h \circ g$.

Proof: If $f \overset{F}{\simeq} g$, then $h \circ f \overset{h \circ F}{\simeq} h \circ g$.

Note that this extends to homotopies relative to A. Also, if $f : X \to Y$ and $g, h : Y \to Z$ are homotopic (rel B) for $B \subseteq Y$, then $g \circ f$ and $h \circ f$ are homotopic (rel $f^{-1}(B)$).

Going back to loops, let X be a topological space and $p \in X$.

Definition 17.3: Homotopy Classes

Consider all loops based at p and consider the equivalence relation given by homotopy (rel $\{0,1\}$). We call the equivalence classes of this relation the *homotopy classes* of loops based at p. If α is a loop based at p, then denote it's homotopy class $\langle a \rangle$.

Definition 17.4: Multiplication of Homotopy Classes

Define $\langle \alpha \rangle \cdot \langle \beta \rangle = \langle \alpha \circ \beta \rangle$ where

$$(\alpha \circ \beta)(s) = \begin{cases} \alpha(2s) & 0 \leq s \leq 1/2 \\ \beta(2s-1) & 1/2 \leq s \leq 1 \end{cases}$$

We need to show that the above definition is well defined.

Lemma 17.5

The above definition is well-defined.

Proof: Suppose that $\alpha' \overset{F}{\simeq} \alpha$ (rel $\{0,1\}$) and $\beta' \overset{G}{\simeq} \beta$ (rel $\{0,1\}$). Then, $\alpha' \circ \beta' \overset{H}{\simeq} \alpha \circ \beta$ where

$$H(s,t) = \begin{cases} F(2s,t) & 0 \leq s \leq 1/2 \\ G(2s-1,t) & 1/2 \leq s \leq \end{cases}$$

Theorem 17.6: Homotopy Classes form a Group

The homotopy classes of loops based at p form a group.

Proof: We need to show that this product is associative, and that we have inverses. We will prove this next lecture.

Definition 17.7: The Fundamental Group

The *fundamental group* of X at p is

$$\pi_1(X,p) = \{\text{homotopy classes of loops at } p\}$$

This is similar notation to $\pi_0(X) = \{\text{path components of } X\}$, but not that one is a set but the other is a group.

Example

If $X = S^1$ and $\alpha(s) = 1$ and $\beta(s) = e^{2\pi i s}$ we will show that these are not homotopic relative to $\{0,1\}$.

10/7/26 Lecture 18: The Fundamental Group (cont.)

Theorem 18.1: Homotopy Classes form a Group

The homotopy classes of loops based at p form a group.

Proof: Let $\langle\alpha\rangle, \langle\beta\rangle, \langle\gamma\rangle$. We first need to show that

$$(\langle\alpha\rangle \cdot \langle\beta\rangle) \cdot \langle\gamma\rangle = \langle\alpha\rangle \cdot (\langle\beta\rangle \cdot \langle\gamma\rangle)$$

which is to show that

$$\langle(\alpha \cdot \beta) \cdot \gamma\rangle = \langle\alpha \cdot (\beta \cdot \gamma)\rangle$$

To do this, notice that

$$(\alpha \cdot \beta) \cdot \gamma = (\alpha \cdot (\beta \cdot \gamma)) \circ f$$

where $f : I \to I$ by

$$f(s) = \begin{cases} 2s & 0 \le s \le 1/4 \\ s + 1/4 & 1/4 \le s \le 1/2 \\ \frac{s+1}{2} & 1/2 \le s \le 1 \end{cases}$$

Now, since $I \subseteq \mathbb{R}$ is convex, and $f(0) = 0$ with $f(1) = 1$, we see that $f \simeq 1_I$ (rel $\{0, 1\}$). Thus, as

$$(\alpha \cdot \beta) \cdot \gamma = (\alpha \cdot (\beta \cdot \gamma)) \circ f \simeq (\alpha \cdot (\beta \cdot \gamma)) \circ 1_I$$

so

$$(\alpha \cdot \beta) \cdot \gamma \simeq \alpha \cdot (\beta \cdot \gamma)$$

Now, we want to show that the identity of $\pi_1(X, p)$ is $\langle e\rangle$ where $e(s) = p$ for all $s \in I$. It is clear that by reparameterization, $\langle e\rangle \cdot \langle a\rangle = \langle a\rangle \cdot \langle e\rangle = \langle a\rangle$.

Finally, we need to show that $\langle\alpha^{-1}\rangle$ is the inverse of $\langle\alpha\rangle$. That is, we want to show

$$\langle\alpha^{-1}\rangle \cdot \langle\alpha\rangle = \langle\alpha\rangle \cdot \langle\alpha^{-1}\rangle = \langle e\rangle$$

which is equivalent to

$$\langle\alpha^{-1} \cdot \alpha\rangle = \langle\alpha \cdot \alpha^{-1}\rangle = \langle e\rangle$$

Now, we see that $\alpha \cdot \alpha^{-1} = \alpha \circ f$ where

$$f(s) = \begin{cases} 2s & 0 \le s \le 1/2 \\ 2 - 2s & 1/2 \le s \le 1 \end{cases}$$

Now, we see that $f(0) = f(1) = 0$, so $f \simeq 0_I$ (rel $\{0, 1\}$), so

$$\alpha \cdot \alpha^{-1} = \alpha \circ f \simeq \alpha \circ 0_I = e$$

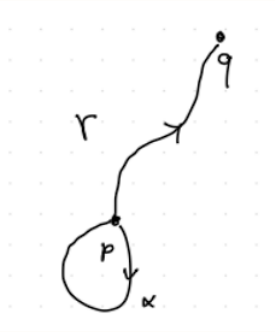
so $\langle\alpha \cdot \alpha^{-1}\rangle = \langle e\rangle$.

■

Proposition 18.2

If X is path connected, then for any $p, q \in X$, $\pi_1(X, p)$ is isomorphic to $\pi_1(X, q)$.

Proof: Let γ be a path between p and q. Then, if α is a loop at p, then $\gamma^{-1} \cdot \alpha \cdot \gamma$ is a loop at q. Thus, define $\pi_1(X, p) \to \pi_1(X, q)$ by $\langle \alpha \rangle \mapsto \langle \gamma^{-1} \alpha \gamma \rangle$. Now, this is well defined in the sense that if $\alpha \simeq \alpha'$ (rel $\{0, 1\}$) then $\gamma^{-1}\alpha\gamma \simeq \gamma^{-1}\alpha'\gamma$ (rel $\{0, 1\}$), which can be shown by just applying the homotopy after γ^{-1} and before γ.



Now, this is a homomorphism since

$$\langle \gamma^{-1}\alpha\alpha'\gamma \rangle = \langle \gamma^{-1}\alpha\gamma\gamma^{-1}\alpha'\gamma \rangle = \langle \gamma^{-1}\alpha\gamma \rangle \cdot \langle \gamma^{-1}\alpha'\gamma \rangle$$

Finally, since we have the inverse homomorphism $\pi_1(X, q) \to \pi_1(X, p)$ by $\langle \beta \rangle \mapsto \langle \gamma\beta\gamma^{-1} \rangle$.

■

Proposition 18.3

Suppose that $f : X \to Y$ is a map. Then, given $\alpha : I \to X$, then we see that $f \circ \alpha : I \to Y$. Thus, we get a homomorphism $f_* : \pi_1(X, p) \to \pi_1(Y, f(p))$ by $f_*(\langle \alpha \rangle) = \langle f \circ \alpha \rangle$.

Proof: We see that

$$\begin{aligned} f_*(\langle \alpha \rangle \cdot \langle \beta \rangle) &= f_*(\langle \alpha \cdot \beta \rangle) \\ &= \langle f \circ (\alpha \cdot \beta) \rangle \\ &= \langle (f \circ \alpha) \cdot (f \circ \beta) \rangle \\ &= \langle f \circ \alpha \rangle \cdot \langle f \circ \beta \rangle \\ &= f_*(\langle \alpha \rangle) \cdot f_*(\langle \beta \rangle) \end{aligned}$$

■

Proposition 18.4

Given maps $f : X \to Y$ and $g : Y \to Z$,

$$(g \circ f)_* = g_* \circ f_*$$

So the diagram below commutes

$$\pi_1(X,p) \xrightarrow{f_*} \pi_1(Y, f(p)) \xrightarrow{g_*} \pi_1(Z, g(f(p)))$$
$$\pi_1(X,p) \xrightarrow[(g\circ f)_*]{} \pi_1(Z, g(f(p)))$$

Corollary 18.5

If $h : X \to Y$ is a homeomorphism, then $h_* : \pi_1(X,p) \to \pi_1(Y, h(p))$ is a isomorphism.

Proof: Starting with the diagrams

$$X \xrightarrow{h} Y \xrightarrow{h^{-1}} X \qquad\qquad Y \xrightarrow{h^{-1}} X \xrightarrow{h} Y$$
$$X \xrightarrow[\mathrm{id}_X]{} X \qquad\qquad Y \xrightarrow[\mathrm{id}_Y]{} Y$$

and applying the chain rule as above, we get the diagrams

$$\pi_1(X,p) \xrightarrow{h_*} \pi_1(Y,h(p)) \xrightarrow{h_*^{-1}} \pi_1(X,p) \qquad\qquad \pi_1(Y,h(p)) \xrightarrow{h_*^{-1}} \pi_1(X,p) \xrightarrow{h_*} \pi_1(Y,h(p))$$
$$\pi_1(X,p) \xrightarrow[(\mathrm{id}_X)_*]{} \pi_1(X,p) \qquad\qquad \pi_1(Y,h(p)) \xrightarrow[(\mathrm{id}_Y)_*]{} \pi_1(Y,h(p))$$

and from the definition, since we have that $(\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X,p)}$ and $(\mathrm{id}_Y)_* = \mathrm{id}_{\pi_1(Y,h(p))}$, we see that h_* and h_*^{-1} are inverses, and thus, we have an isomorphism.

■

Now, we will quickly prove some useful facts about metric spaces.

Definition 18.6: Diameter

Let X be a metric space and $S \subseteq X$. Then, the diameter of S is

$$\mathrm{diam}(S) = \sup_{s_1, s_2 \in S} d(s_1, s_2)$$

Lemma 18.7: Lebesgue Lemma

If X is a compact metric space and $\mathcal{F}$ is an open cover, then there exists a $\delta > 0$ called the *Lebesgue number* of $\mathcal{F}$ so that any subset $A \subseteq X$ with $\mathrm{diam}(A) < \delta$ is contained in some member of $\mathcal{F}$.

Proof: Suppose not. Then, there exists $A_1, A_2, \cdots$ with diameter $\mathrm{diam}(A_n) < 1/n$ but A_n is not contained in any member of the cover.

For all n, choose an $x_n \in A_n$. Then, as X is a compact metric space, we can find a subsequence x_{n_k} converging to some $p \in X$. Now, p must be inside an element of $\mathcal{F}$, so let $p \in U \in \mathcal{F}$. Then, there exists an $\epsilon > 0$ for which $p \in B(p, \epsilon) \subseteq U$. Then, pick sufficiently large k so that $\operatorname{diam}(A_{n_k}) < 1/2\epsilon$, and $x_{n_k} \in B(p, 1/2\epsilon)$. Then, for all $y \in A_{n_k}$, we see that $d(y, x_{n_k}) < \epsilon/2$ and $d(x_{n_k}, p) < 1/2\epsilon$, so

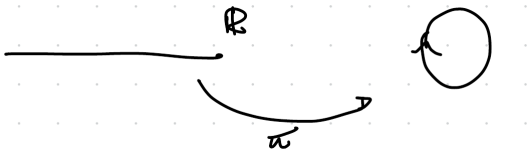
$$d(y, p) < \epsilon$$

by the triangle inequality. Thus, $A_{n_k} \subseteq B(p, \epsilon) \subseteq U$, which is a contradiction.

■

10/9/26 Lecture 19: The Fundamental Group of the Circle

We will show $\pi_1(S^1, p) \simeq \mathbb{Z}$. Heuristically, a loop is sent to its winding number around 0. Define $\pi : \mathbb{R} \to S^1$ by $\pi(x) = e^{2\pi i x}$.



Take $1 \in S^1$ as the basepoint. Given $n \in \mathbb{Z}$, say

$$\gamma_n(s) = ns, \quad 0 \le s \le 1$$

Now, $\pi \circ \gamma_n$ is a loop in S^1 that starts and ends at 1.

Note: If γ is any path in $\mathbb{R}$ from 0 to n, then $\gamma \simeq \gamma_n$ (rel $\{0,1\}$) since $\mathbb{R}$ is convex in itself. We will use this result a lot.

Lemma 19.1: Path Lifting Lemma

If σ is a path in S^1 starting at 1, then there exists a unique path $\tilde{\sigma}$ in $\mathbb{R}$ such that $\tilde{\sigma}(0) = 0$ and $\pi \circ \tilde{\sigma} = \sigma$.

Proof: Let $U = S^1 - \{-1\}$ and $V = S^1 - \{1\}$. Consider

$$\pi^{-1}(U) = \bigcup_{n \in \mathbb{Z}} \left(n - \frac{1}{2}, n + \frac{1}{2}\right) \quad \text{and} \quad \pi^{-1}(V) = \bigcup_{n \in \mathbb{Z}} (n, n+1) \subseteq \mathbb{R}$$

U and V are an open cover of S^1, so $\sigma^{-1}(U)$ and $\sigma^{-1}(V)$ are an open cover of $[0,1]$. By the Lebesgue lemma, find an $\ell > 0$ such that every interval A with $|A| < \ell$ lies in $\sigma^{-1}(U)$ or $\sigma^{-1}(V)$, so $\sigma(A) \subseteq U$ or $\sigma(A) \subseteq V$.

Let $0 = T_0 < T_1 < \cdots < T_m = 1$ so that $|T_{j+1} - T_j| < \ell$ for all j. Since $\sigma(0) = 1 \in U$ and $\sigma(0) \notin V$, we have $\sigma[T_0, T_1] \subseteq U$. Then, as $\pi\big|_{(-\frac{1}{2}, \frac{1}{2})} : (-\frac{1}{2}, \frac{1}{2}) \to U$ is a homeomorphism, let f be the inverse. Then, define

$$\tilde{\sigma}(s) = f(\sigma(s)) \text{ for } s \in [T_0, T_1]$$

Inductively, suppose $\tilde{\sigma}$ is defined for $s \in [0, T_k]$. We want to define $\tilde{\sigma}$ on $[T_k, T_{k+1}]$.

Suppose $\sigma([T_k, T_{k+1}]) \subseteq U$ and $\tilde{\sigma}(T_k) \in (n - \frac{1}{2}, n + \frac{1}{2})$. Now, $\pi\big|_{(n-\frac{1}{2}, n+\frac{1}{2})} : (n - \frac{1}{2}, n + \frac{1}{2}) \to U$ is a homeomorphism, so say g is its inverse. Then, define

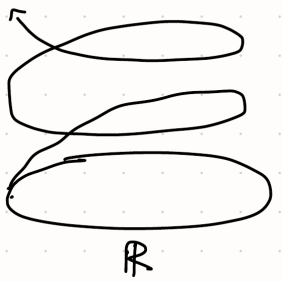
$$\tilde{\sigma}(s) = g(\sigma(s)) \text{ for } s \in [T_k, T_{k+1}]$$

so $\tilde{\sigma}$ is defined on $[0, T_{k+1}]$.

Now, suppose $\sigma([T_k, T_{k+1}]) \subseteq V$ and $\tilde{\sigma}(T_k) \in (n, n+1)$. Let h be the inverse of $\pi\big|_{(n,n+1)} : (n, n+1) \to V$ and define

$$\tilde{\sigma}(s) = h(\sigma(s)) \text{ for } s \in [T_k, T_{k+1}]$$

so $\tilde{\sigma}$ is defined on $[0, T_{k+1}]$. Repeat until we lift all of σ.

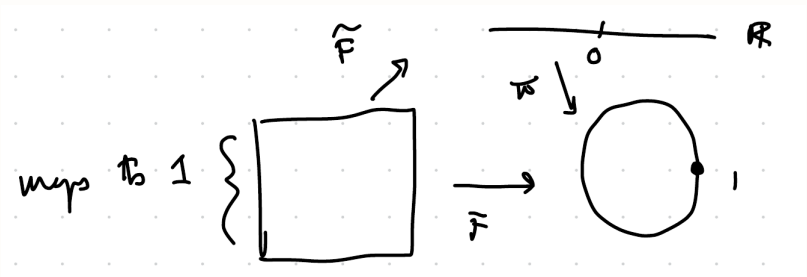


■

Lemma 19.2: Homotopy Lifting Lemma

If $F : I \times I \to S^1$ is such that $F(0,t) = 1 \in S^1$, then there exists a unique $\widetilde{F} : I \times I \to \mathbb{R}$ such that $\widetilde{F}(0,t) = 0$ for all $t \in [0,1]$ and $F = \pi \circ \widetilde{F}$.

Proof: Subdivide $[0,1] \times [0,1]$ into squares so F sends each square to U or V (Lebesgue lemma), and lift the squares starting with the lower left.



Theorem 19.3: Fundamental Group of the Circle

The function $\phi : \mathbb{Z} \to \pi_1(S^1, 1)$ given by $\phi(n) = \langle \pi \circ \gamma_n \rangle$ is an isomorphism.

Proof: First, we show ϕ is a homomorphism. Given $m, n \in \mathbb{Z}$, say $\sigma(s) = \gamma_n(s) + m$ for $0 \le s \le 1$. Then, $\pi \circ \sigma = \pi \circ \gamma_n$, and $\gamma_m \cdot \sigma$ joins 0 to $m + n$, so

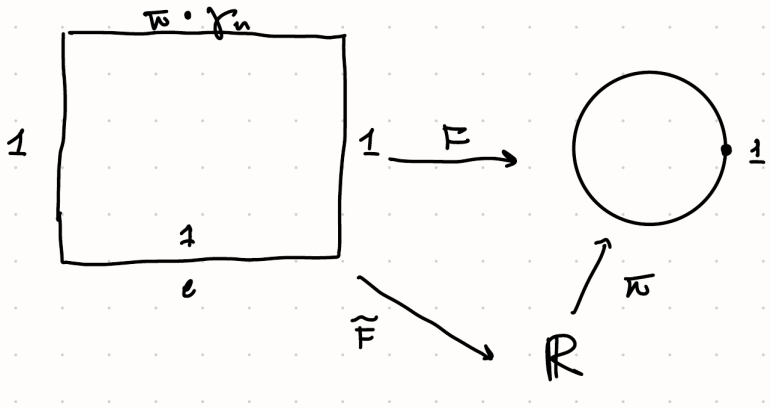
$$\begin{aligned} \phi(m+n) = \langle \pi \circ \gamma_{m+n} \rangle &= \langle \pi \circ (\gamma_m \cdot \sigma) \rangle = \langle \pi \circ \gamma_m \rangle \cdot \langle \pi \circ \sigma \rangle \\ &= \langle \pi \circ \gamma_m \rangle \cdot \langle \pi \circ \gamma_n \rangle \\ &= \phi(m) \cdot \phi(n) \end{aligned}$$

Now, we need to show ϕ is onto. Given $\langle \alpha \rangle \in \pi_1(S^1, 1)$, lift α to a path $\tilde{\alpha} : [0,1] \to \mathbb{R}$. Since $\alpha(1) = 1$,

$\tilde{\alpha}(1) \in \mathbb{Z}$. Say $\tilde{\alpha}(1) = n$. Then, by convexity, $\tilde{\alpha} \simeq \gamma_n$ (rel $\{0,1\}$). Thus, $\alpha = \pi \circ \tilde{\alpha} \simeq \pi \circ \gamma_n$, so

$$\langle \alpha \rangle = \langle \pi \circ \gamma_n \rangle = \phi(n)$$

Now, we show ϕ is one-to-one. Suppose $\phi(n) = \langle e \rangle$. Then $\pi \circ \gamma_n \simeq e$. Say $F : I \times I \to S^1$ is the homotopy from e to $\pi \circ \gamma_n$.



By the homotopy lifting lemma, lift F to $\widetilde{F} : [0,1] \times [0,1] \to \mathbb{R}$. Then, $\widetilde{F}(0,t) = 0$ for all $t \in [0,1]$.

Now, let $P = \{\text{left edge}\} \cup \{\text{right edge}\} \cup \{\text{bottom edge}\}$. Now, $\pi \circ \widetilde{F}(P) = F(P) = 1$ and P is connected, so $\widetilde{F}(P) = N$ for a single integer N (otherwise, it must go to another integer, contradicting connectedness).

However, $\widetilde{F}(0,0) = 0$, so $N = 0$, so $\widetilde{F}\,|_P = 0$. Now, on the top edge, $\{\widetilde{F}(s,1)\}_{s \in [0,1]}$ is a lift of $\pi \circ \gamma_n$ to $\mathbb{R}$, so $\widetilde{F}(s,1) = \gamma_n(s) + m$ for some $m \in \mathbb{Z}$. Plugging in $s = 0$, we get $m = 0$.

Thus, $\widetilde{F}(s,1) = \gamma_n(s)$ for all $s \in [0,1]$, so $\widetilde{F}(1,1) = \gamma_n(1)$. But $\gamma_n(1) = n$, and as $\widetilde{F}(1,1) = 0$, $n = 0$. Thus, ϕ is one-to-one.

■

10/12/26 Lecture 20: Brouwer Fixed Point Theorem and Simply Connected Spaces

Last time, we showed $\pi_1(S^1, 1) \simeq \mathbb{Z}$ by $\alpha \mapsto$ "winding number of α", by lifting α to $\tilde{\alpha}$ on $\mathbb{R}$ and defining its winding number to be $\tilde{\alpha}(1)$.

Definition 20.1: Fixed Points

If X is a set and $f : X \to X$ is a function, we say f has a *fixed point* if there exists $x \in X$ such that $f(x) = x$.

Lemma 20.2

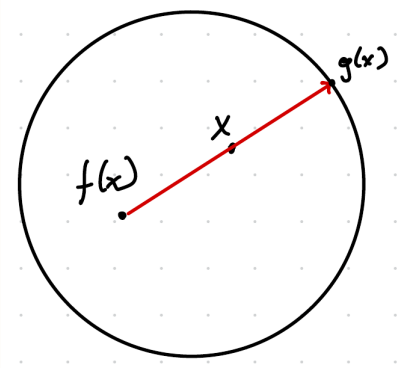
If $f : [0, 1] \to [0, 1]$, then f has a fixed point.

Proof: Let $g(x) = f(x) - x$. Now, $g(0) \geq 0$ and $g(1) \leq 0$, so there exists $c \in [0, 1]$ such that $g(c) = 0$, so $f(c) = c$.

Theorem 20.3: 2D Brouwer Fixed Point Theorem

Let D be the closed unit disk in $\mathbb{R}^2$, and $f : D \to D$ is a map. Then, f has a fixed point.

Proof: Suppose f has no fixed point. Then, $f(x) \neq x$ for all $x \in D$. Thus, consider the map g as follows:



Now, $g : D \to S^1$ is continuous and surjective, as $g|_{S^1} = \mathrm{id}_{S^1}$. Now, if $i : S^1 \hookrightarrow D$ is the inclusion, then $g \circ i = \mathrm{id}_{S^1}$. Now, $g \circ i$ is one-to-one and onto as it maps points on the boundary to themselves, so

$$\pi_1(S^1, 1) \xrightarrow{i_*} \pi_1(D, 1) \xrightarrow{g_*} \pi_1(S^1, 1), \qquad \mathrm{id}_{S^1}$$

so $g_* \circ i_* = \mathrm{id}_{\pi_1(S^1,1)}$. However, $\pi_1(S^1, 1) = \mathbb{Z}$ and $\pi_1(D, 1) = \{e\}$, so no such i_* and g_* can exist!

■

Corollary 20.4

If X is homeomorphic to D, then a map $\alpha : X \to X$ has a fixed point.

Proof: Let $h : D \to X$ be a homeomorphism. Then, $f = h^{-1} \circ \alpha \circ h : D \to D$, so there exists $x \in D$ such that $f(x) = x$. Then,

$$h^{-1} \circ \alpha \circ h(x) = x \implies \alpha(h(x)) = h(x)$$

so α has a fixed point.

■

Definition 20.5: Simply Connected Spaces

We say X is *simply connected* if it is path connected and $\pi_1(X, p) = \{e\}$.

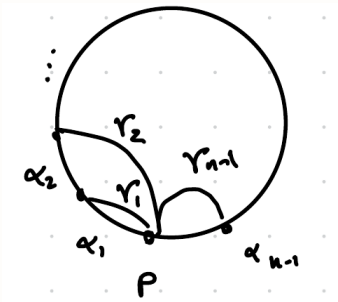
Proposition 20.6

Suppose $X = U \cup V$ with U and V open and simply connected, and $U \cap V$ is path connected. Then, X is simply connected.

Proof: It is clear that X is path connected. Choose $p \in U \cap V$ and let α be a loop based at p. We want to show α is homotopic to a product of loops based at p, each in U or V.

Now, $\{\alpha^{-1}(U), \alpha^{-1}(V)\}$ is an open cover of $[0, 1]$, so by the Lebesgue lemma, there exists a partition $0 = T_0 < T_1 < \cdots < T_n = 1$ so that $\alpha_k := \alpha\big|_{[T_{k-1}, T_k]}$ is contained in U or V.

Choose a path γ_k from p to $\alpha(T_k)$ that lies entirely in U if $\alpha(T_k) \in U$ and lies in V if $\alpha(T_k) \in V$. If $\alpha(T_k) \in U \cap V$, just pick γ_k lying in $U \cap V$ (we can always do this as U, V, and $U \cap V$ are path connected). Then,



So

$$\begin{aligned}\alpha = \alpha_1 \cdot \alpha_2 \cdots \alpha_n &\simeq \alpha_1 \cdot \gamma_1^{-1} \cdot \gamma_1 \cdot \alpha_2 \cdot \gamma_2^{-1} \cdot \gamma_2 \cdots \gamma_n \cdot \alpha_n \\ &= (\alpha_1 \cdot \gamma_1^{-1})(\gamma_1 \cdot \alpha_2 \cdot \gamma_2^{-1}) \cdots (\gamma_n \cdot \alpha_n) \\ &= \langle \alpha_1 \cdot \gamma_1^{-1} \rangle \cdot \langle \gamma_1 \cdot \alpha_2 \cdot \gamma_2^{-1} \rangle \cdots \langle \gamma_n \cdot \alpha_n \rangle \\ &\simeq \langle \mathrm{id} \rangle \cdot \langle \mathrm{id} \rangle \cdots \langle \mathrm{id} \rangle \\ &= \langle \mathrm{id} \rangle\end{aligned}$$

where the last homotopy comes from the fact that each term is a loop contained in U or V, so it follows from the simple connectedness of U and V.

Thus, $\pi_1(X, p)$ is trivial, which was to be shown.

■

Corollary 20.7

$\pi_1(S^n) = \{e\}$ if $n > 1$.

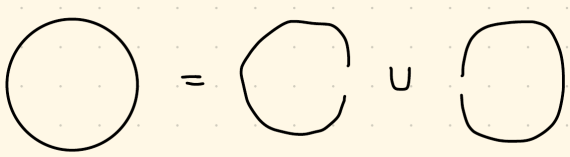
Proof: Let $U = S^n - \{\text{north pole}\}$ and $V = S^n - \{\text{south pole}\}$. U and V are homeomorphic to $\mathbb{R}^n$, so they are simply connected. Then, as $U \cap V = S^n - \{\text{north pole}, \text{south pole}\}$ is path connected, we see that S^n is simply connected.

■

We showed that $X = U \cup V$ with U and V open and simply connected, and $U \cap V$ path connected, implies X is simply connected. These hypotheses turn out to all be necessary. Here are example examples.

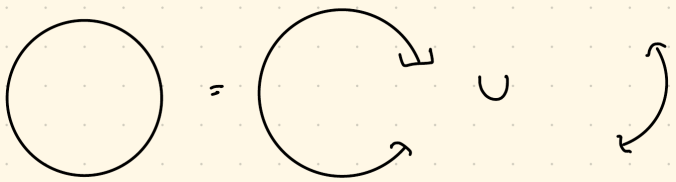
Example

If we forget that $U \cap V$ needs to be path connected, we have



Example

If we forget that U and V need to be open, we get



Definition 20.8: Free Actions

A group G *acts freely* on X if for all $g \neq e$ and $x \in X$, $gx \neq x$. That is, $\{gx\} \cap \{x\} = \emptyset$.

Example

$\mathbb{Z}$ acts freely on $\mathbb{R}$ by $(n, x) \mapsto x + n$.

Proposition 20.9

Suppose G is a discrete group that acts freely on a simply connected space X so that for all $x \in X$, there exists a neighborhood U of x so that $U \cap gU = \emptyset$ for all $g \neq e$. Then, $\pi_1(X/G) \simeq G$.

Proof: We will prove this next time.

Example

Let $G = \mathbb{Z}$ and $X = \mathbb{R}$. Then, for $x \in \mathbb{R}$, $U = (x - \frac{1}{6}, x + \frac{1}{6})$ satisfies the above property. Thus,

$$\pi_1(\mathbb{R}/\mathbb{Z}) \cong \mathbb{Z}$$

10/14/26 Lecture 21: Fundamental Groups of Quotients and Products

Proposition 21.1

Suppose G is a discrete group acting on a simply connected space X so that every $x \in X$ has a neighborhood $\mathcal{U}$ so that $g\mathcal{U} \cap \mathcal{U} = \emptyset$ whenever $g \neq e$. Then, $\pi_1(X/G) \simeq G$.

Proof Idea: Given $g \in G$, join x_0 to gx_0 by a path γ_g. If $\pi : X \to X/G$ is the quotient map, then $\pi \circ \gamma_g$ is a loop based at $\pi(x_0)$, since

$$(\pi \circ \gamma_g)(0) = \pi(x_0) \quad \text{and} \quad (\pi \circ \gamma_g)(1) = \pi(\gamma_g(1)) = \pi(gx_0) = \pi(x_0)$$

Then, define $\phi : G \to \pi_1(X/G, \pi(x_0))$ by sending $g \in G$ to $\langle \pi \circ \gamma_g \rangle$.

First, we should show this is independent of the choice of γ_g. If γ_g' is another path from x_0 to gx_0, then $\gamma_g' \cdot \gamma_g^{-1}$ is a loop at x_0, so as X is simply connected, $\langle \gamma_g' \cdot \gamma_g^{-1} \rangle = \langle e \rangle$, so

$$\gamma_g' \cdot \gamma_g^{-1} \simeq e_{x_0} \quad (\text{rel } \{0,1\})$$

Say $\{\eta_\tau\}_{\tau \in [0,1]}$ is a family of loops based at x_0 going from $\gamma_g' \cdot \gamma_g^{-1}$ to e_{x_0}. Then, $\{\eta_\tau \cdot \gamma_g\}_{\tau \in [0,1]}$ is a family of paths from γ_g' to γ_g (rel $\{0,1\}$), so $\gamma_g' \simeq \gamma_g$, so $\langle \pi \circ \gamma_g \rangle = \langle \pi \circ \gamma_g' \rangle$. Thus, the map is well defined.

Strategy: We want to show ϕ is a homomorphism, onto, and one-to-one.

We won't show this in detail, but the proof is similar to what we did with S^1. We would need to show the path lifting lemma, which is to show that if σ is a path in X/G that starts at $\pi(x_0)$, then there exists a unique $\tilde{\sigma}$ in X starting at x_0 so that $\pi \circ \tilde{\sigma} = \sigma$. This is to show ϕ is onto.

Similarly, to show ϕ is one-to-one, we would need a homotopy lifting lemma, which says that if $F : I \times I \to X/G$ is such that $F(0,t) = \pi(x_0)$ for all $t \in [0,1]$, then there exists a unique lift

$$\tilde{F} : I \times I \to X \text{ so that } \tilde{F}(0,t) = x_0 \ \forall t \in [0,1]$$

which shows ϕ is one-to-one.

■

Example: Torus

Consider $\mathbb{Z}^n$ acting on $\mathbb{R}^n$ by $(n, x) \mapsto (x + n)$. Balls of radius $1/2$ are moved off of themselves. Thus, as $\mathbb{R}^n$ is simply connected,

$$\pi_1(\mathbb{R}^n/\mathbb{Z}^n) \simeq \mathbb{Z}^n$$

where $\mathbb{R}^n/\mathbb{Z}^n = T^n$.

Example: Projective Space

Consider $\mathbb{Z}_2$ acting on S^n with $n > 1$. Since S^n is simply connected and -1 acts on S^n by the antipodal map. This clearly satisfies our requirements, so

$$\pi_1(S^n/\mathbb{Z}_2) \simeq \mathbb{Z}_2$$

where $S^n/\mathbb{Z}_2 = P^n$.

Example: Lens Space

$\mathbb{Z}_p$ acts on S^3 by choosing $q < p$ relatively prime positive integers, and letting $1 \in \mathbb{Z}_p$ send (z_1, z_2) to

$$\left(e^{\frac{2\pi i}{p}} z_1, e^{\frac{2\pi qi}{p}} z_2\right)$$

for $|z_1|^2 + |z_2|^2 = 1$. By our theorem,

$$\pi_1(S^3/\mathbb{Z}_p) \simeq \mathbb{Z}_p$$

where $S^3/\mathbb{Z}_p = L(p, q)$.

Note: In all these cases, $\pi_1(X)$ is abelian, but this isn't always true. We will see that

$$\pi_1(\infty) = F_2$$

, the free group on two elements, where ∞ is the figure 8.

Theorem 21.2: Fundamental Group of a Product Space

If X and Y are path connected, then

$$\pi_1(X \times Y) \simeq \pi_1(X) \times \pi_1(Y)$$

Proof: Choose $x_0 \in X$ and $y_0 \in Y$. Then, $(x_0, y_0) \in X \times Y$. Recall the projections p_1 and p_2. Now, ignoring base points for brevity,

$$(p_1)_* : \pi_1(X \times Y) \to \pi_1(X) \quad \text{and} \quad (p_2)_* : \pi_1(X \times Y) \to \pi_1(Y)$$

These are homomorphisms, so we get a homomorphism

$$\psi : \pi_1(X \times Y) \to \pi_1(X) \times \pi_1(Y)$$

by $\psi = ((p_1)_*, (p_2)_*)$, so $\psi(\langle\alpha\rangle) = (\langle p_1 \circ \alpha\rangle, \langle p_2 \circ \alpha\rangle)$.

First, we claim ψ is one-to-one. If α is a loop in $X \times Y$ such that $p_1 \circ \alpha \underset{F}{\simeq} e_1$ and $p_2 \circ \alpha \underset{G}{\simeq} e_2$, then $\alpha \underset{H}{\simeq} e_{(x_0,y_0)}$ by the homotopy

$$H(s,t) = (F(s,t), G(s,t))$$

so $\langle\alpha\rangle = \left\langle e_{(x_0,y_0)}\right\rangle$, so Ψ is one-to-one.

Now, we show ψ is onto. Given loops β in X and γ in Y, put $\alpha(s) = (\beta(s), \gamma(s))$, so

$$\psi(\langle\alpha\rangle) = (\langle\beta\rangle, \langle\gamma\rangle)$$

■

Example: Torus

We see that

$$\pi_1(T^2) = \pi_1(S^1 \times S^1) = \pi_1(S^1) \times \pi_1(S^1) = \mathbb{Z} \times \mathbb{Z} = \mathbb{Z}^2$$

Consider the trefoil knot:



This is a *torus knot*, as it can be drawn on the torus, and it represents the $(2, 3)$ class in $\pi_1(T^2)$.

Recall we've seen that if X and Y are homeomorphic, then $\pi_1(X) \simeq \pi_1(Y)$. In fact, we can weaken this and only require a "homotopy equivalence", which we define next.

Definition 21.3: Homotopy Equivalence

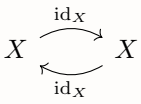
X and Y are *homotopy equivalent* if there exist maps $f : X \to Y$ and $g : Y \to X$ such that $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. Then, we say g is the *homotopy inverse* to f and f is a *homotopy equivalence*, and write $X \simeq Y$.

Lemma 21.4

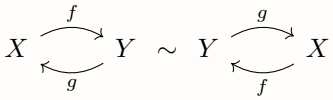
Homotopy equivalence is an equivalence relation on topological spaces.

Proof Idea:

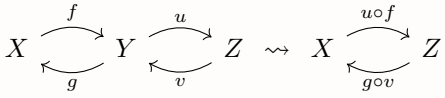
1. **Reflexive:**



2. **Symmetric:**



3. **Transitive:**



where transitivity holds because

$$\begin{aligned}(g \circ v) \circ (u \circ f) &= g \circ (v \circ u) \circ f \\ &\simeq g \circ \mathrm{id}_Y \circ f \\ &= g \circ f \\ &\simeq \mathrm{id}_X\end{aligned}$$

and

$$\begin{aligned}(u \circ f) \circ (g \circ v) &= u \circ (f \circ g) \circ v \\ &\simeq u \circ \mathrm{id}_Y \circ v \\ &= u \circ v \\ &\simeq \mathrm{id}_Z\end{aligned}$$

■

Definition 21.5: Algebraic Topology

Algebraic topology is the study of topological spaces up to homotopy equivalence.

10/16/26 Lecture 22: Homotopy Equivalence and Fundamental Groups

Example: Homeomorphic Spaces

If X and Y are homeomorphic, they are homotopy equivalent. Simply take f to be any homeomorphism and $g = f^{-1}$.

However, in general, homotopy equivalent $\not\Rightarrow$ homeomorphic.

Digression:

Definition 22.1: Manifolds

An *n-dimensional manifold* is a second-countable Hausdorff space locally homeomorphic to $\mathbb{R}^n$.

Q: If two compact manifolds are homotopy equivalent, are they homeomorphic?
A: No, $L(7,1)$ and $L(7,2)$ are homotopy equivalent but not homeomorphic.

Definition 22.2: Aspherical Manifolds

A manifold X is *aspherical* if every map $f : S^n \to X$, $n > 1$, is homotopic to a constant map.

Borel Conjecture: If X_1, X_2 are two homotopy equivalent compact aspherical manifolds, they are homeomorphic.

A: For dimension n of X_1, X_2:

n	1	2	3	4	5+
	Trivial, Yes	Classical, Yes	Perelman, Yes	(OpenAI) No	?

Example: Convex Sets

Any convex subset of $\mathbb{R}^n$ is homotopy equivalent to a point. This is why:

If C is convex in $\mathbb{R}^n$ and $p \in C$, take $X = \{p\}$, $Y = C$, and let $f : \{p\} \hookrightarrow C$ by inclusion and $g : C \to \{p\}$ by the only map.

Now, $g \circ f = \mathrm{id}_{\{p\}}$ and $f \circ g : C \to C$ by $C \to p$. Now let $F(x,t) = (1-t)x + tp$, which is a homotopy $\mathrm{id}_C \overset{F}{\simeq} p$.

Thus, $\mathbb{R}^n$ is homotopy equivalent to a point.

Example: Punctured $\mathbb{R}^n$

$\mathbb{R}^n - \{0\}$ is homotopy equivalent to S^{n-1}.

Take $f : S^{n-1} \hookrightarrow \mathbb{R}^n - \{0\}$ by the inclusion and $g : \mathbb{R}^n - \{0\} \to S^{n-1}$ by $g(x) = \frac{x}{|x|}$. Then, $g \circ f = \mathrm{id}_{S^{n-1}}$ and $f \circ g = g : \mathbb{R}^n - \{0\} \to \mathbb{R}^n - \{0\}$.

Let

$$F(x,t) = (1-t)x + t\left(\frac{x}{|x|}\right)$$

This is a homotopy $\mathrm{id}_{\mathbb{R}^n - \{0\}} \overset{F}{\simeq} g$.

Definition 22.3: Deformation Retractions

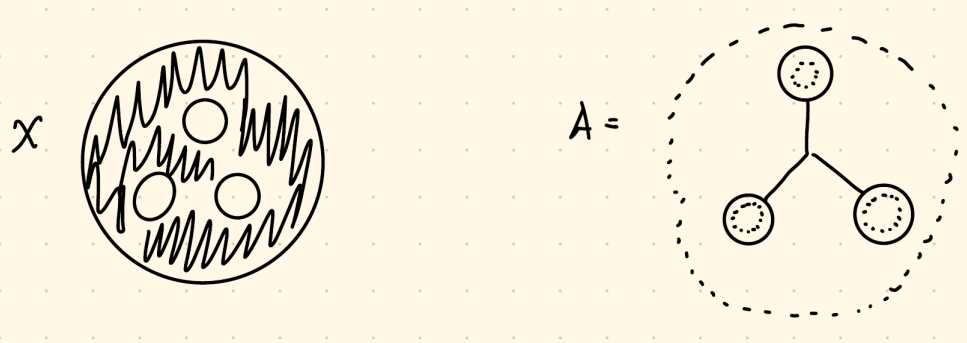
If $A \subseteq X$, a *deformation retract* of X to A is a homotopy $G : X \times I \to X$ (rel A) so that for all $x \in X$,

$$G(x,0) = x \quad \text{and} \quad G(x,1) \in A$$

If there exists a deformation retraction of X to A, then X and A are homotopy equivalent, and the two maps are $f : A \hookrightarrow X$ inclusion and $g : X \to A$ by $g(x) = G(x,1) \in A$.

Example: D^3 Minus Three Balls

Let $X = D^3 - \{3 \text{ open balls}\}$.



$X \to A$ is a deformation retraction.

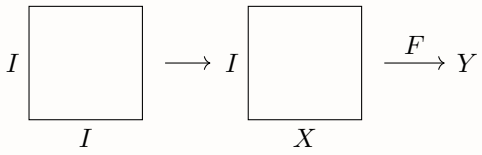
Theorem 22.4: Homotopy Theorem

If $f \overset{F}{\simeq} g : X \to Y$ (rel $\{p\}$), then $f_* = g_* : \pi_1(X,p) \to \pi_1(Y, f(p))$ ($f(p) = g(p)$, so it doesn't matter which base point).

Proof: Say α is a loop at p. Then, $f_* \langle \alpha \rangle = \langle f \circ \alpha \rangle$ and $g_* \langle \alpha \rangle = \langle g \circ \alpha \rangle$. We have a homotopy $F : X \times I \to Y$ so

$$F(x,0) = f(x), \quad F(x,1) = g(x), \quad F(p,t) = f(p) = g(p) \ \forall t \in [0,1]$$

We want



so define $G : I \times I \to Y$ by $G(s,t) = F(\alpha(s), t)$. Now $G(s,0) = F(\alpha(s),0) = (f \circ \alpha)(s)$ and

$$G(s,1) = F(\alpha(s),1) = g(\alpha(s)) = (g \circ \alpha)(s)$$

Finally, $G(0,t) = F(\alpha(0),t) = F(p,t) = f(p)$ and $G(1,t) = F(\alpha(1),t) = F(p,t) = f(p)$.

Thus, G is a homotopy between $f \circ \alpha$ and $g \circ \alpha$ relative to $f(p)$. Thus, $\langle f \circ \alpha \rangle = \langle g \circ \alpha \rangle \in \pi_1(Y, f(p))$. ■

Theorem 22.5: Generalized Homotopy Theorem

If $f \overset{F}{\simeq} g : X \to Y$, then $g_* : \pi_1(X,p) \to \pi_1(Y, g(p))$ is the same as the composition

$$\pi_1(X,p) \xrightarrow{f_*} \pi_1(Y, f(p)) \xrightarrow{\gamma_*} \pi_1(Y, g(p))$$

where γ is the path from $f(p)$ to $g(p)$ by $\gamma(s) = F(p,s)$, and we define

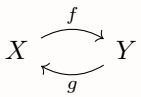
$$\gamma_* \langle \beta \rangle = \left\langle \gamma^{-1} \cdot \beta \cdot \gamma \right\rangle$$

Proof: See book.

Theorem 22.6: Fundamental Groups of Homotopy Equivalent Spaces

If two path-connected spaces X and Y are homotopy equivalent, then they have isomorphic fundamental groups.

Proof: We have



with $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. Choose $q \in Y$ and let $p = g(q) \in X$. Then, we will show $f_* : \pi_1(X,p) \to \pi_1(Y, f(p))$ is an isomorphism.

Like above, join p to $g(f(p))$ by $\gamma(s) = F(p,s)$, where F is the homotopy from id_X to $g \circ f$. Apply the previous theorem to $g \circ f$ and id_X, which gives

$$(g \circ f)_* = \gamma_* : \pi_1(X,p) \to \pi_1(X, (g \circ f)(p))$$

But γ_* is an isomorphism, so f_* is injective.

Similarly, we can show $(f \circ g)_*$ is an isomorphism, which implies f_* is onto. Thus, f_* is an isomorphism. ■

10/19/26 Lecture 23: Contractible Spaces, Retractions, and Simplicial Complexes

Recall that if two path connected spaces are homotopy equivalent, they have isomorphic fundamental groups.

Corollary 23.1

If $n > 2$, then $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$.

Proof: Say $f : \mathbb{R}^2 \to \mathbb{R}^n$ is a homeomorphism. Then, $f\big|_{\mathbb{R}^2-\{0\}} : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^n \setminus \{f(0)\}$ is a homeomorphism, and so it is a homotopy equivalence.

Hence $\pi_1(\mathbb{R}^2 - \{0\})$ is isomorphic to $\pi_1(\mathbb{R}^n \setminus \{f(0)\})$.

Now, $\mathbb{R}^2 - \{0\}$ deformation retracts to S^1 and $\mathbb{R}^n \setminus \{f(0)\}$ deformation retracts to a sphere S^{n-1} around $f(0)$.

Since fundamental groups are invariant under deformation retracts, we have

$$\pi_1(\mathbb{R}^2 \setminus \{0\}) = \pi_1(S^1) = \mathbb{Z}$$

and

$$\pi_1(\mathbb{R}^n \setminus \{f(0)\}) = \pi_1(S^{n-1}) = \{e\}$$

as $n - 1 > 1$. This is a contradiction. ■

Definition 23.2: Contractible Spaces

We say X is *contractible* if id_X is homotopic to a constant map $c_p : X \to \{p\}$.

Proposition 23.3

The following are true about contractible spaces:

a. X is contractible if, and only if, X is homotopy equivalent to a point.

b. X is contractible $\implies$ X is simply connected.

c. If X is contractible, then any two maps $f, g : Z \to X$ are homotopic.

d. If X is contractible, then $\mathrm{id}_X \simeq c_x$ for any $x \in X$.

Proof: **a)** ($\Rightarrow$) Say $\alpha_p : X \to \{p\}$ is the map and $i_p : \{p\} \hookrightarrow X$ is the inclusion.

$$X \underset{i_p}{\overset{\alpha_p}{\rightleftarrows}} \{p\}$$

Now, $\alpha_p \circ i_p = \mathrm{id}_p$ and $i_p \circ \alpha_p = c_p \simeq \mathrm{id}_X$. Thus, X is homotopy equivalent to $\{p\}$.

($\Leftarrow$) Suppose X is homotopy equivalent to a point $\{p\}$. Then, we have $f : X \to \{p\}$ and $g : \{p\} \to X$ such that

$$f \circ g \simeq \mathrm{id}_{\{p\}} \quad \text{and} \quad g \circ f \simeq \mathrm{id}_X$$

However, $g \circ f = c_{g(p)}$, so $\mathrm{id}_X \simeq c_{g(p)}$, so X is contractible.

b) Suppose X is contractible. Now, $\mathrm{id}_X \simeq c_p$, so there exists $F : X \times I \to X$ such that $F(x, 0) = x$ and $F(x, 1) = p$. Now, given $y \in X$, take $\gamma(t) = F(y, t)$. We see γ is a path from y to p. Thus, X is path connected (p connects to all points).

Now, since X is homotopy equivalent to $\{p\}$, and as $\pi_1(X) \simeq \pi_1(p) \simeq \{e\}$, we see X is simply connected.

c) Suppose X is contractible. Then, $\mathrm{id}_X \simeq c_p$. Thus,

$$f \simeq \mathrm{id}_X \circ f \simeq c_p \circ f$$
$$g \simeq \mathrm{id}_X \circ g \simeq c_p \circ g$$

but $c_p \circ f$ and $c_p \circ g$ are both the same map sending $Z \to \{p\}$. Thus, $f \simeq g$.

d) Follows from (c) immediately.

■

Definition 23.4: Retractions

If $A \subseteq X$, a *retraction* of X to A is a map $g : X \to A$ so $g|_A = \mathrm{id}_A$. We say A is a *retract* of X.

Example: Retraction to a Point

If $X = S^1$ and $A = \{1\}$, then the map $g : X \to A$ is a retraction.

Note: A deformation retraction gives a retraction $g(x) = G(x, 1)$.

Proposition 23.5

If $g : X \to A$ is a retraction, then $g_* : \pi_1(X, a) \to \pi_1(A, a)$ is onto for $a \in A$.

Proof: Say $i : A \to X$ is the inclusion. Then $g \circ i = \mathrm{id}_A$, so $g_* \circ i_* = \mathrm{id}_{\pi_1(A,a)}$. Thus, g_* must be onto. This can be seen below.

$$\pi_1(A, a) \xrightarrow{i_*} \pi_1(X, a) \xrightarrow{g_*} \pi_1(A, a), \qquad \mathrm{id}_{\pi_1(A,a)} : \pi_1(A, a) \to \pi_1(A, a)$$

■

Simplicial Complexes

These will be spaces built out of "generalized triangles".

Definition 23.6: Hyperplanes

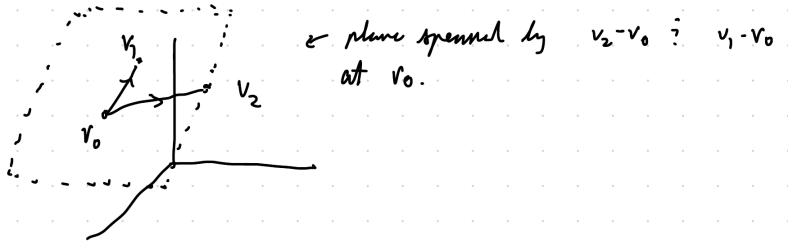
Say $v_0, \dots, v_k$ are points in $\mathbb{R}^n$. Then, the *hyperplane* they span is

$$\{\lambda_0 v_0 + \lambda_1 v_1 + \cdots + \lambda_k v_k \,:\, \lambda_0 + \cdots + \lambda_k = 1\}$$

This is the same as

$$\{v_0 + r_1(v_1 - v_0) + \cdots + r_k(v_k - v_0) \,:\, r_1, \dots, r_k \in \mathbb{R}\}$$

Example:



Definition 23.7: General Position

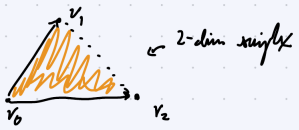
Let H be the hyperplane formed by $v_0, \dots, v_k$ in $\mathbb{R}^n$. We say $v_0, \dots, v_k$ are in *general position* if $\dim H = k$.

Definition 23.8: Simplices

If $v_0, \dots, v_k$ are in general position, then the smallest convex set containing them, called the *convex hull*, is the set

$$C = \{\lambda_0 v_0 + \cdots + \lambda_k v_k \,:\, \lambda_0 + \cdots + \lambda_k = 1, \lambda_0 \geq 0, \lambda_1 \geq 0, \dots, \lambda_k \geq 0\}$$

we call this the *k-dimensional simplex* containing $v_0, \dots, v_k$.



Example: Low-Dimensional Simplices

If $k = 0$ we get a point, $k = 1$ a line, $k = 2$ a triangle, $k = 3$ a tetrahedron.

Definition 23.9: Faces

If A and B are simplices and $\{\text{vertices of } B\} \subseteq \{\text{vertices of } A\}$, then B is a *face* of A.

Example: Faces of a Triangle

If A is a triangle, it has 7 faces as there are 7 nonempty subsets of $\{v_0, v_1, v_2\}$.

Definition 23.10: Simplicial Complexes

A finite collection of simplices in some $\mathbb{R}^n$ is a (finite) *simplicial complex* K if

1. When A is a simplex in the collection, so are its faces.
2. When two simplices intersect, they do so in a common face.

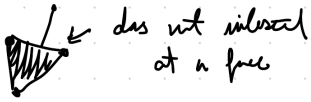
Example:



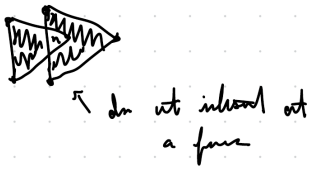
Example:



Non-Example:



Non-Example:



Note: A simplicial complex is a collection, so we write $|K|$ for the corresponding subset of $\mathbb{R}^n$.

10/21/26 Lecture 24: Triangulations, Cones, and Barycentric Subdivision

Recall: An equivalent definition for $v_0, \dots, v_n$ being in general position is that the vectors $\{v_1 - v_0, \dots, v_n - v_0\}$ are linearly independent. In this case, the vertices form an n-simplex

$$\left\{ \sum_{i=0}^{n} \lambda_i v_i \ : \ \lambda_i \geq 0, \sum_{i=0}^{n} \lambda_i = 1 \right\}$$

Now, if K is a simplicial complex, let $|K|$ be the corresponding subset of $\mathbb{R}^n$ with the subset topology.

Definition 24.1: Triangulations

A *triangulation* of a topological space X is a simplicial complex K and a homeomorphism from $|K|$ to X.

Example: Triangulating S^2

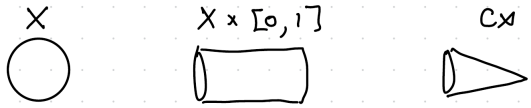
. Let $X = S^2$. Then, let $|K|$ be the boundary of the tetrahedron



This is homeomorphic to S^2.

If X is a topological space, we had a notion of the cone

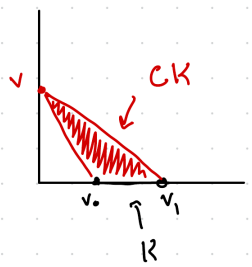
$$CX = X \times [0,1]/(x,1) \sim (x',1)$$



How do we do this simplicially? Let K be a simplicial complex in $\mathbb{R}^n$. Then, consider the embedding $K \hookrightarrow \mathbb{R}^{n+1}$. Then, let $v = (0, \dots, 0, 1) \in \mathbb{R}^{n+1}$. If A is a simplex $(v_0, \dots, v_k)$ in K, let $CA = (v_0, \dots, v_k, v)$. Then, let

$$CK = K \cup \{v\} \cup \{CA \ : \ A \in K\}$$

Example: $n = 1$



Then CK is a simplicial complex in $\mathbb{R}^{n+1}$ and $|CK|$ is homeomorphic to $C|K|$.

Definition 24.2: Isomorphisms of Simplicial Complexes

Let K, L be simplicial complexes, possibly in different Euclidean spaces. An *isomorphism* between K and L is a bijection $\phi : \{\text{vertices in } K\} \to \{\text{vertices in } L\}$, so $(v_0, \dots, v_k)$ is a simplex in K iff $(\phi v_0, \dots, \phi v_k)$ is a simplex in L.

Fact: If the above holds, $|K|$ and $|L|$ are homeomorphic, since ϕ can be defined as a linear map.

Definition 24.3: Interiors of Simplices

. If K is a simplex in $\mathbb{R}^n$ with vertices $(v_0, \dots, v_k)$, then the *interior* is

$$\text{int}(K) = \left\{ \sum_{i=0}^{k} \lambda_i v_i \ : \ \lambda_i > 0, \sum_{i=0}^{k} \lambda_i = 1 \right\}$$

Proposition 24.4

Suppose K is a finite simplicial complex in $\mathbb{R}^n$.

1. $|K|$ is compact.
2. Each point in $|K|$ lies in the interior of a unique simplex.
3. The closed subsets of $|K|$ are those that intersect each simplex in a closed subset.
4. If $|K|$ is connected, it is path connected.

Proof: **1)** Since each simplex is closed and bounded in $\mathbb{R}^n$, and as $|K|$ is a finite union of these, it is compact.

2) First, we want to show that each x in $|K|$ is in the interior of some simplex.

First, x must be in some simplex $(v_0, \dots, v_k)$ in K. Then, write

$$x = \sum_{i=0}^{k} \lambda_i v_i, \quad \lambda_i \geq 0, \ \sum \lambda_i = 1$$

Then, x must be in the interior of the simplex spanned by the vertices for which $\lambda_i > 0$ (at least one must exist).

To show this is unique, suppose $x \in \text{int}(A) \cap \text{int}(B)$, so $x \in A \cap B$, so x must lie in a common face of both A and B. However, if x is in the interior of A, all of its weights corresponding to the vertices of A must be positive, so the face must be A. Similarly, the face must be B, so $A = B$.

3) ($\Rightarrow$) If $C \subseteq |K|$ is closed, then as each simplex is closed, the intersection of C with any simplex is itself closed in $|K|$.

($\Leftarrow$) Suppose $C \subseteq |K|$ and $C \cap A$ is closed for all simplices $A \in K$. Then, $C \cap A = A \cap D$ for some closed D in $|K|$. Since A is closed, we see that $C \cap A$ is closed in $|K|$. Thus, as

$$C = \bigcup_{A \in K} C \cap A$$

is a finite union of closed sets, it is closed.

4) Given an $x \in |K|$, for some $\delta > 0$, $B(x, \delta) \cap |K|$ is path connected. Hence, $|K|$ is locally path connected, so as $|K|$ is connected, $|K|$ is path connected.

■

Definition 24.5: Barycenters

If $A \subseteq \mathbb{R}^n$ is a simplex with vertices $v_0, \dots, v_k$, then its *barycenter* is

$$\widehat{A} = \frac{1}{k+1} \sum_{i=0}^{k} v_i \in A$$

If K is a simplicial complex, then we will define its *barycentric subdivision* K' as follows:



Let's formalize this.

Notation: If A is a proper face of B, write $A < B$.

Definition 24.6: Barycentric Subdivision

If $A_0 < \cdots < A_k$ is an increasing sequence of simplices in K, then

$$(\widehat{A}_0, \widehat{A}_1, \dots, \widehat{A}_k)$$

are the vertices of a k-simplex in K'.

Example: If

$$K = \quad \begin{array}{ccc} & v_2 & \\ e_{02} & f_{012} & e_{12} \\ v_0 & e_{01} & v_1 \end{array}$$

then K' consists of

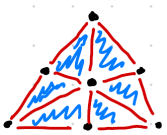
- 0-simplices: $\{\widehat{v}_0, \widehat{v}_1, \widehat{v}_2, \widehat{e}_{01}, \widehat{e}_{12}, \widehat{e}_{02}, \widehat{f}_{012}\}$
- 1-simplices:

$$\{(\widehat{v}_0, \widehat{e}_{01}), (\widehat{v}_0, \widehat{e}_{02}), (\widehat{v}_0, \widehat{f}_{012}),$$
$$(\widehat{v}_1, \widehat{e}_{01}), (\widehat{v}_1, \widehat{e}_{12}), (\widehat{v}_1, \widehat{f}_{012}),$$
$$(\widehat{v}_2, \widehat{e}_{12}), (\widehat{v}_2, \widehat{e}_{02}), (\widehat{v}_2, \widehat{f}_{012}),$$
$$(\widehat{e}_{01}, \widehat{f}_{012}), (\widehat{e}_{12}, \widehat{f}_{012}), (\widehat{e}_{02}, \widehat{f}_{012})\}$$

- 2-simplices:

$$\{(\widehat{v}_0, \widehat{e}_{01}, \widehat{f}_{012}), (\widehat{v}_0, \widehat{e}_{02}, \widehat{f}_{012}),$$
$$(\widehat{v}_1, \widehat{e}_{01}, \widehat{f}_{012}), (\widehat{v}_1, \widehat{e}_{12}, \widehat{f}_{012}),$$
$$(\widehat{v}_2, \widehat{e}_{12}, \widehat{f}_{012}), (\widehat{v}_2, \widehat{e}_{02}, \widehat{f}_{012})\}$$

so

$K' =$ 

10/26/26 Lecture 25: Simplicial Maps and Simplicial Approximation

Recall that K' is the first barycentric subdivision. We can iterate this to get more.

Definition 25.1: Iterated Barycentric Subdivision

If K is a simplicial complex, let K^m be the *mth barycentric subdivision* of K, defined by iterating barycentric subdivision.

Definition 25.2: Dimension

The *dimension* of a simplicial complex is the maximal dimension of its simplices.

Definition 25.3: Mesh

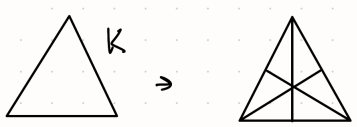
The *mesh*, $\mu(K)$, is the maximal diameter of a simplex of K. Equivalently, it is the maximal length of an edge.

Proposition 25.4

The following are true:

0. K' is a simplicial complex.
1. Each simplex of K' is contained in a simplex of K.
2. $|K'| = |K|$.
3. If $\dim(K) = n$, then $\mu(K') \leq \frac{n}{n+1}\mu(K)$.

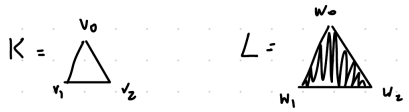
Proof: See book. For 3,



Definition 25.5: Simplicial Vertex Maps

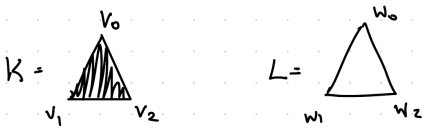
If K, L are simplicial complexes, a function $s : \{\text{vertices of } K\} \to \{\text{vertices of } L\}$ is *simplicial* if whenever $(v_0, \ldots, v_k)$ are the vertices of a simplex in K, then $\{s(v_0), \ldots, s(v_k)\}$ spans a simplex in L, possibly with redundancies.

Example:



s by $s(v_0) = w_0$, $s(v_1) = w_1$, $s(v_2) = w_2$ is simplicial.

Non-Example:



Then s as above is not simplicial in this case.

Now, given an s, we can extend it to a map $s : |K| \to |L|$ linearly on each simplex. That is,

$$s\left(\sum \lambda_i v_i\right) = \sum \lambda_i s(v_i) \in |L|$$

and we call s a *simplicial map*.

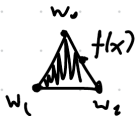
Note: If A is a simplex in K, $s(A)$ may have lower dimension. For example, if $s : K \to L$ and L is a point, s is simplicial.

Goal: Approximate any map $f : |K| \to |L|$ by a simplicial map. Thus, we can turn algebraic topology statements into combinatorial statements. To do this, we want our simplicial maps to be homotopic to the original map. Note: to do this, we will have to subdivide K.

Definition 25.6: Carriers

Say $f : |K| \to |L|$ is any map. Then, given $x \in |K|$, we know $f(x)$ is in the interior of a unique simplex in $|L|$. We call this closed simplex the *carrier* of $f(x)$.

Example:



Then the carrier of $f(x)$ is $\{w_0, w_2\}$.

Definition 25.7: Simplicial Approximations

Given a map $f : |K| \to |L|$, a simplicial map $s : |K| \to |L|$ is a *simplicial approximation* to f if, for all $x \in |K|$, $s(x)$ is in the carrier of $f(x)$.

Note: If $f(x)$ is a vertex in L, then $s(x) = f(x)$, as we need $s(x) \in \{f(x)\}$.

Lemma 25.8

If s is a simplicial approximation of f, then f and s are homotopic maps.

Proof: Consider $\mathbb{R}^n$. Let

$$F(x,t) = (1-t)s(x) + tf(x)$$

This is a homotopy in $\mathbb{R}^n$, but as $f(x)$ and $s(x)$ are in the same simplex of $|L|$, it is clear that F is also a homotopy in $|L|$! Thus, $F : |K| \times [0,1] \to |L|$ is a homotopy from s to f.

■

Definition 25.9: Stars

If v is a vertex of a simplicial complex K, let $\text{star}(v, K)$ be the union of the interiors of the simplices containing v.



Theorem 25.10: Simplicial Approximation Theorem (Alexander)

Suppose $f : |K| \to |L|$ is a map. Then, if m is large enough, there exists a simplicial approximation $s : |K^m| \to |L|$ to the map $f : |K^m| \to |L|$.

Proof Idea: Let $|L| = \bigcup_{w \text{ vertex of } L} \text{star}(w, L)$ be an open cover of $|L|$. Then, $\{f^{-1}(\text{star}(w, L))\}$ is an open cover of $|K|$. Let δ be its Lebesgue number.

Now, subdivide K so that each simplex in K^m has diameter at most $\frac{1}{2}\delta$. For any vertex v in K^m, note $\text{star}(v, K^m)$ has diameter $< \delta$, so $\text{star}(v, K^m) \subseteq f^{-1}(\text{star}(w, L))$ for some vertex w in L. Thus,

$$f(\text{star}(v, K^m)) \subseteq \text{star}(w, L)$$

for some w. Then, define $s(v) = w$, and do this for each vertex v in K^m. Then, show this is simplicial, extend linearly on simplices, and show it is a simplicial approximation to f.

■

Note: We don't subdivide L, but we may have to subdivide K many times.

Exercise: Look at Example 6.8 in the book and convince yourself that it's correct.

10/28/26 Lecture 26: Free Groups, Presentations, and Maximal Trees

Definition 26.1: Generators and Free Groups

A subset X of a group G is a *set of generators* if each $g \neq e$ can be written as

$$g = x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k} \quad \text{where } x_i \in X,\ n_i \in \mathbb{Z} \setminus \{0\},\ \text{and } x_i \neq x_{i+1}$$

We say X is a *free set of generators* if the representation is unique, and we say G is a *free group*.

Definition 26.2: Words

Given an alphabet X, a *word* is a finite product $x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ where $x_i \in X$ and $n_i \in \mathbb{Z}$. We say a word is *reduced* if $n_i \neq 0$ and $x_{i+1} \neq x_i$.

These form a group in a natural way (concatenation and reduce), called the *free group generated by* X, and we denote it $F(X)$.

Example: Free Group on One Generator

It can be easily shown $F(\{x\}) \cong \mathbb{Z}$.

Say G is a group and X is a generating set. This gives us a homomorphism $\phi : F(X) \to G$ by the natural embedding

$$\phi(x_1^{n_1} \cdots x_k^{n_k}) = x_1^{n_1} \cdots x_k^{n_k} \in G$$

Now, ϕ is surjective, so letting $N = \ker(\phi)$, we see

$$G \cong F(X)/N$$

If R is a collection of words so that N is the smallest normal subgroup containing R, we say $\langle X \mid R \rangle$ is a *presentation* of G with generators X and relations R.

Idea: We start with the free group $F(X)$ and impose that words in R become the identity.

Definition 26.3: Finitely Presented Groups

If X and R are finite, we say that the corresponding group G is *finitely presented* and write

$$G = \langle x_1, \ldots, x_n \mid r_1, \ldots, r_\ell \rangle$$

Examples:

1. $\langle x \mid \emptyset \rangle \cong \mathbb{Z}$
2. $\langle x \mid x^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$
3. $\langle x_1, x_2 \mid x_1 x_2 x_1^{-1} x_2^{-1} \rangle \cong \mathbb{Z} \times \mathbb{Z}$

Definition 26.4: Free Products

Let G, H be groups. Then, form reduced words $x_1 x_2 \cdots x_m \in G \cup H$ so that no x_i is the identity of G or H, and x_i and x_{i+1} are not in the same group. Adding the empty word as an identity element, we get a group $G * H$ called the *free product* of G and H.

Examples:

1. $\mathbb{Z} * \mathbb{Z} \cong F(\{x_1, x_2\}) =: F_2$.
2. If $G = \langle x_1, \dots, x_m \mid r_1, \dots, r_k \rangle$ and $H = \langle y_1, \dots, y_n \mid s_1, \dots, s_\ell \rangle$, then
$$G * H = \langle x_1, \dots, x_m, y_1, \dots, y_n \mid r_1, \dots, r_k, s_1, \dots, s_\ell \rangle$$
3. $\mathbb{Z}_2 * \mathbb{Z}_2 = \langle x \mid x^2 \rangle * \langle y \mid y^2 \rangle = \langle x, y \mid x^2, y^2 \rangle$. Now, consider $\phi : \mathbb{Z}_2 * \mathbb{Z}_2 \to \mathbb{Z}_2$,
$$\phi(\text{word}) = \#x\text{'s} + \#y\text{'s} \pmod 2$$
Then, you can show $\ker(\phi) = \{(xy)^n\}_{n \in \mathbb{Z}} \cong \mathbb{Z}$. We call $\mathbb{Z}_2 * \mathbb{Z}_2$ the *infinite dihedral group*.

   It turns out $\mathbb{Z}_2 * \mathbb{Z}_2$ can act on $\mathbb{R}$ by $x : u \mapsto -u$ and $y : u \mapsto 2 - u$, which are reflections around 0 and 1.

Goal: Our new goal is to compute the fundamental group of simplicial complexes.

Let K be a connected simplicial complex. Now, recall that if $f : |K| \to |L|$ is a map and s is a simplicial approximation to f, then if $x \in |K|$ and $f(x)$ is a vertex in $|L|$, $f(x) = s(x)$.

Definition 26.5: Trees

A *tree* in K is a one-dimensional simplicial subcomplex which is simply connected. A tree is *maximal* if no other tree contains it.

Lemma 26.6

A maximal tree in K contains all the vertices.

Proof: Say T is a maximal tree and suppose a vertex v is not in T. Take a vertex u in T.

Since $|K|$ is path connected as $|K|$ is connected, there exists a path $\gamma : [0, 1] \to |K|$ from u to v. Then, after barycentric subdivision, we can get a simplicial approximation $s : [0, 1] \to |K|$ such that $s(0) = u$ and $s(1) = v$. Now, as s is simplicial, s runs along edges.

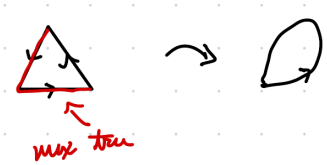
Replacing γ with s, we see s gives us a path $u, v_1, v_2, \dots, v_k, v$, so as u is in our tree, we can continue building edges along the tree until we get to v, either traveling along the tree to get to v_{i+1}, or adding that edge if it doesn't exist. Note that if we ever encounter a vertex on a path not in T, adding the edge to T gives us a bigger tree, which contradicts maximality, so we immediately find that $v \in T$.

■

Now, what we will do is find relations for $\pi_1(|K|, v)$.

Idea: Find a maximal tree, and as this is contractible, contract the tree to a point and consider $|K|/T$ (same fundamental group). Each edge not in T becomes a loop in $|K|/T$; these are the generators for $\pi_1(|K|, v)$, and relations come from 2-simplices in K.

Examples:



$$\pi_1(|K|, v) = \langle \gamma \mid \emptyset \rangle \cong \mathbb{Z}$$



$$\pi_1(|K|, v) = \langle \gamma \mid \gamma \rangle \cong \{e\}$$

10/30/26 Lecture 27: Edge Paths and the Edge Group

Definition 27.1: Edge Paths

An *edge path* is a sequence of vertices $v_0, \dots, v_k$ in K so that each $\{v_i, v_{i+1}\}$ is a simplex. We allow $v_i = v_{i+1}$, and if $v_0 = v_k = v$, we call the edge path an *edge loop*.

Definition 27.2: Equivalent Edge Paths

Two edge paths are *equivalent* if one differs from the other by a finite number of the following moves:



That is, $(v_1, v_2, v_3) \equiv (v_1, v_3)$ if $\{v_1, v_2, v_3\}$ spans a simplex, and $(v_2, v_3, v_2) \equiv (v_2)$. We write the equivalence class of $(v_0, \dots, v_k)$ as $\{v_0, \dots, v_k\}$, and the equivalence classes of edge loops at v form a group $E(K, v)$ under concatenation.

Theorem 27.3: Equivalence of $E(K, v)$ and $\pi_1(|K|, v)$

$$E(K, v) \cong \pi_1(|K|, v)$$

Proof: We will construct a homomorphism

$$\phi : E(K, v) \to \pi_1(|K|, v)$$

Given an edge loop $vv_1 \cdots v_{k-1}v$, divide $[0, 1]$ into k equal segments. Let $\alpha : [0, 1] \to |K|$ be the piecewise linear path so that $\alpha(0) = \alpha(1) = v$ and $\alpha(i/k) = v_i$ for $1 \leq i \leq k - 1$. Then, as α is a loop in $|K|$ based at v and equivalent edge loops give homotopic α's, the map

$$\phi(\{v, v_1, \dots, v_{k-1}, v\}) = \langle \alpha \rangle$$

is well defined and is a homomorphism.

We now show ϕ is surjective. Think of $[0, 1]$ as a simplicial complex L. By the simplicial approximation theorem, there is a simplicial map $S : |L^m| \to |K|$ which is homotopic to α with $S(0) = S(1) = v$.

Now, L^m has $2^m + 1$ vertices $\{i/2^m\}_{i=0}^{2^m}$, so pick $v_i = S(i/2^m)$ (S is simplicial). Then, we see

$$\phi(\{v, v_1, \dots, v_{2^m-1}, v\}) = \langle \alpha \rangle$$

We will now show ϕ is injective. Suppose $vv_1 \cdots v_{k-1}v$ is an edge loop based at v so

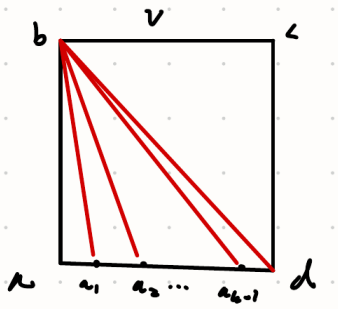
$$\phi(\{vv_1 \cdots v_{k-1}v\}) = e \in \pi_1(|K|, v)$$

so the edge loop can be continuously deformed in $|K|$ to the constant loop. We have to show that

$v, v_1, \dots, v_{k-1}, v$ is equivalent to the point edge loop v using our moves. Say α is the path corresponding to $vv_1 \cdots v_{k-1}v$. We know there exists a homotopy $F : I \times I \to |K|$ so that

$$F(s,0) = \alpha(s), \quad F(s,1) = v, \quad F(0,t) = F(1,t) = v$$

Say that L is the simplicial complex on $I \times I$ given by



where a, b, c, d are the vertices of the square and $a_i = (i/k, 0)$. Note $F(a_i) = v_i$.

Note that the edge paths $aa_1 \cdots a_{k-1}d$ and $abcd$ are equivalent in L. Subdivide L to get L^m and take the corresponding edge paths E_1, E_2, which are still equivalent in L^m.

By the simplicial approximation theorem, if m is sufficiently large, we can find a simplicial approximation $S : |L^m| \to K$ to $F : |L^m| \to |K|$. Now, as E_1 and E_2 were still equivalent, $S(E_1)$ and $S(E_2)$ are equivalent in K. Now, note S sends E_2 to the constant path v as F does.

Then, as $F([a_i, a_{i+1}]) = [v_i, v_{i+1}]$, if $x \in [a_i, a_{i+1}]$, the carrier of $F(x)$ is $[v_i, v_{i+1}]$ (or $\{v_i\}$ if they are the same). Thus, if u is a vertex in L^m between a_i and a_{i+1}, S sends u to v_i or v_{i+1}. Thus, using our equivalences,

$$S(E_1) \sim \{vv_1 \cdots v_{k-1}v\}$$

in $E(K, v)$. Thus, $\{v, v_1 \cdots v_{k-1}v\} = \{v\}$ in $E(K, v)$.

Thus, ϕ is bijective, so we have an isomorphism between $E(K, v)$ and $\pi_1(|K|, v)$.

■

Definition 27.4: The Group $G(K, L)$

Say K is a connected simplicial complex and L is a maximal tree. Let the vertices of K be $V = \{v_0, \dots, v_s\}$. Define $G(K, L)$ to be the group formed by the generators g_{ij}, one for each ordered pair (v_i, v_j) that spans a simplex, with relations

(a) $g_{ij} = 1$ if v_iv_j spans a simplex of L

(b) $g_{ij}g_{jk} = g_{ik}$ if v_i, v_j, v_k span a simplex of K

11/2/26 Lecture 28: Presentations of the Edge Group

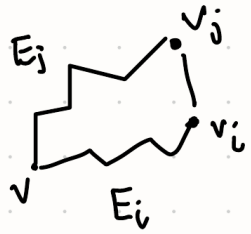
Theorem 28.1: Equivalence of $G(K,l)$ and $E(K,v)$

$$G(K,L) \cong E(K,v)$$

Proof: We will construct

$$G(K,L) \underset{\theta}{\overset{\phi}{\rightleftarrows}} E(K,v)$$

To define ϕ, join v to each vertex v_i by an edge path E_i in L



and define

$$\phi(g_{ij}) = \{E_i v_i v_j E_j^{-1}\} \in E(K,v)$$

and extend to products of generators.

We need to show ϕ respects relations. If $v_i v_j$ is a simplex in L, then $E_i v_i v_j E_j^{-1}$ is in L, but as L is simply connected, we see

$$\{E_i v_i v_j E_j^{-1}\} = \{v\} \in E(K,v)$$

Now, if $v_i v_j v_k$ span a simplex in K, then

$$\begin{aligned}
\phi(g_{ij})\phi(g_{jk}) &= \{E_i v_i v_j E_j^{-1}\} \cdot \{E_j v_j v_k E_k^{-1}\} \\
&= \{E_i v_i v_j E_j^{-1} E_j v_j v_k E_k^{-1}\} \\
&= \{E_i v_i v_j v_j v_k E_k^{-1}\} \\
&= \{E_i v_i v_k E_k^{-1}\} \\
&= \phi(g_{ik})
\end{aligned}$$

Thus, ϕ is a homomorphism.

Recall that $v_0 = v$. Then, define $\theta : E(K,v) \to G(K,L)$ by

$$\theta(\{v v_k v_\ell \cdots v_n v\}) = g_{0k} g_{k\ell} \cdots g_{n0} \in G(K,L)$$

We want to show this is well defined. This is identical to above, so we will skip this part. This is also a homomorphism.

Now, consider $\theta \circ \phi : G(K,L) \to G(K,L)$. Note

$$\begin{aligned}\theta \circ \phi(g_{ij}) = \theta(\phi(g_{ij})) &= \theta(\{E_i v_i v_j E_j^{-1}\}) \\ &= \theta(\{v_i v_j\}) \quad (E_i, E_j^{-1} \in L) \\ &= g_{ij}\end{aligned}$$

so $\theta \circ \phi$ is the identity on $G(K,L)$.

Finally, consider $\phi \circ \theta : E(K,v) \to E(K,v)$. Note that in $E(K,v)$, we always have

$$\{v v_k v_\ell \cdots v_n v\} = \{E_0 v v_k E_k^{-1}\}\{E_k v_k v_\ell E_\ell^{-1}\} \cdots \{E_n v_n v E_0^{-1}\}$$

where E_0 is the trivial edge path $v \to v$. Thus,

$$\phi \circ \theta(\{v v_k v_\ell \cdots v_n v\}) = \phi \circ \theta(\{E_0 v v_k E_k^{-1}\}) \cdots \phi \circ \theta(\{E_n v_n v E_0^{-1}\})$$

and as

$$\phi \circ \theta(\{E_i v_i v_j E_j^{-1}\}) = \phi(g_{ij}) = \{E_i v_i v_j E_j^{-1}\}$$

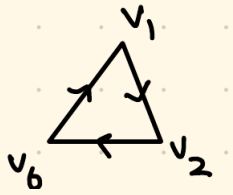
as $E_i, E_j^{-1} \in L$, $\phi \circ \theta$ is the identity on $E(K,v)$.

■

Simplified Presentation

Start by ordering the vertices $V = \{v_0, \ldots, v_s\}$ and consider the generators g_{ij} for each edge $v_i v_j$ not in L with $i < j$, with the relations $g_{ij} g_{jk} = g_{ik}$ whenever v_i, v_j, v_k are vertices of a 2-simplex (if an edge $v_a v_b$ is in L, write $g_{ab} = e$).

Example: Boundary of a Triangle



Let $L = \{v_0, v_1, v_2, \{v_0, v_1\}, \{v_0, v_2\}\}$. The only generator is g_{12}, and we have no relations. Thus,

$$\pi_1(K) \cong \langle g_{12} \mid \emptyset \rangle \cong \mathbb{Z}$$

Example: Solid Triangle



Let $L = \{v_0, v_1, v_2, \{v_0, v_1\}, \{v_0, v_2\}\}$. The only generator is g_{12} with relation $g_{01}g_{12} = g_{02} \implies eg_{12} = e \implies g_{12} = e$, so

$$\pi_1(K) = \langle g_{12} \mid g_{12} \rangle \cong \{e\}$$

Example: Boundary of a Tetrahedron



Let $L = \{v_0, v_1, v_2, v_3, \{v_0, v_1\}, \{v_0, v_2\}, \{v_0, v_3\}\}$. Our generators are g_{12}, g_{23}, g_{13}, and our relations are

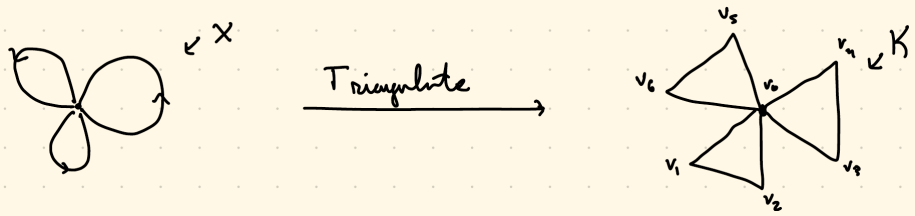
$$g_{01}g_{12} = g_{02} \implies g_{12} = e \qquad g_{02}g_{23} = g_{03} \implies g_{23} = e$$

$$g_{01}g_{13} = g_{03} \implies g_{13} = e \qquad g_{12}g_{23} = g_{13}$$

Then,

$$\pi_1(K) = \left\langle g_{12}, g_{23}, g_{13} \mid g_{12}, g_{23}, g_{13}, g_{12}g_{23}g_{13}^{-1} \right\rangle = \{e\}$$

Example: Three Circles



Let L be the maximal tree made of the edges $v_0v_1, \dots, v_0v_6$:



Then, our generators are g_{12}, g_{34}, g_{56}. There are no relations, so

$$\pi_1(X) \cong \langle g_{12}, g_{34}, g_{56} \mid \emptyset \rangle \cong F_3$$

In general, if we have n circles that intersect at a point, called a *bouquet of n circles*, then $\pi_1(X) \cong F_n$.

11/4/26 Lecture 29: Van Kampen's Theorem and Abstract Simplicial Complexes

Here are some corollaries from the results proved last time.

Corollary 29.1

If K is a (finite) connected simplicial complex, then $\pi_1(|K|, v)$ is finitely presented.

Corollary 29.2

If $K(2)$ denotes the 2-skeleton, that is, the union of the vertices, edges, and triangles of K, then

$$\pi_1(|K|, v) \cong \pi_1(|K(2)|, v)$$

Corollary 29.3

If J and K are connected simplicial complexes that intersect at a point v, then

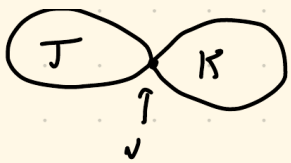
$$\pi_1(|J \cup K|, v) \cong \pi_1(|J|, v) * \pi_1(|K|, v)$$

Proof: Take maximal trees T_1 and T_2 in J and K respectively. Let $T_0 = T_1 \cup T_2$. This is a tree in $J \cup K$. Then, the generators are the edges in $J \setminus T_1$ ($\{g_{ij}\}$) and $K \setminus T_2$ ($\{h_{ij}\}$), and the relations come from 2-simplices in J ($\{r_p\}$) or K ($\{s_q\}$). Then,

$$\pi_1(|J \cup K|, v) = \langle \{g_{ij}\} \cup \{h_{ij}\} \mid \{r_p\} \cup \{s_q\} \rangle = \pi_1(|J|) * \pi_1(|K|)$$

■

Example: Two Circles



$$\pi_1(J \cup K) = \mathbb{Z} * \mathbb{Z} = \pi_1(|J|) * \pi_1(|K|)$$

Theorem 29.4: Van Kampen's Theorem

Suppose J, K are simplicial complexes, $J \cap K$ is a simplicial complex, and all of these are path connected. Choose $v \in J \cap K$. Let $j : J \cap K \to J$ and $k : J \cap K \to K$ be the inclusion maps. Then, $\pi_1(|J \cup K|, v)$ can be obtained from $\pi_1(|J|, v) * \pi_1(|K|, v)$ by adding the relations

$$j_*(z)k_*(z)^{-1}$$

where z runs over the generators of $\pi_1(|J \cap K|, v)$.

Proof: Choose a maximal tree T_0 of $J \cap K$. Extend to maximal trees T_1, T_2 in J, K respectively. Note $T_1 \cup T_2$ is a maximal tree in $J \cup K$.

Then, $\pi_1(|J \cup K|, v)$ is generated by the edges in $J \cup K$ not in $T_1 \cup T_2$, which are

$$\{\text{edges in } J \setminus T_1\} \cup \{\text{edges in } K \setminus T_2\}$$

but there are some redundancies for edges in $(J \setminus T_1) \cap (K \setminus T_2)$, so we add the relations

$$a_{ij} = b_{ij}, \quad a_{ij} \in G(J, T_1),\ b_{ij} \in G(K, T_2)$$

for edges $v_i v_j$ in $(J \cap K) \setminus T_0$.

■

Special Case: If J, K are simply connected and $J \cap K$ is nonempty and path connected, then $\pi_1(J \cup K, v)$ is a quotient of

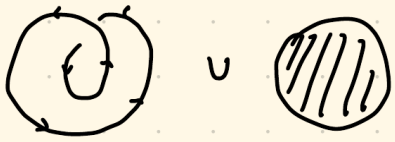
$$\pi_1(J, v) * \pi_1(K, v) = \{e\}$$

so $J \cup K$ is simply connected.

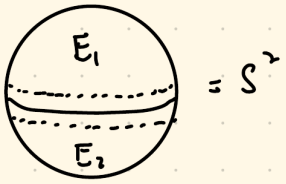
Example: The Projective Plane

Consider $P^2 = S^2/\mathbb{Z}_2$. We know $\pi_1(P^2) = \mathbb{Z}_2$ since $\pi_1(S^2)$ is trivial and $S^2 \to P^2$ is a covering map. We will show this with Van Kampen.

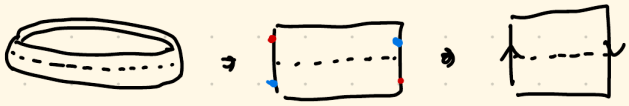
Claim: $P^2 = \{\text{Möbius strip}\} \cup_{S^1} D^2$



The reason is that if



then elements in E_1 are identified with elements of E_2, so these can be represented by $E_1 \cong D^2$. For elements of the equator, these are locally identified by the Möbius strip:



Then, let $J =$ Möbius strip, $K = D^2$, $J \cap K = S^1$, and $J \cup K = P^2$. Thus, by Van Kampen, and from the fact that the inclusion of a generator of $\pi_1(S_1)$ into the Möbious strip is a loop that runs around twice, we see

$$\begin{aligned}\pi_1(P^2) &= \frac{\pi_1(\text{Möbius strip}) * \pi_1(D^2)}{\langle j_*(z)k_*(z)^{-1}\rangle} \\ &= \frac{\langle z\rangle * \langle e\rangle}{\langle z^2\rangle} \\ &= \mathbb{Z}/2\mathbb{Z}\end{aligned}$$

Definition 29.5: Finite Abstract Simplicial Complexes

A *finite abstract simplicial complex* is a pair (V, S) where V is a finite set and S is a collection of nonempty subsets of V such that

(i) $\{v\} \in S$ for all $v \in V$

(ii) $\mathcal{P}(X) \setminus \{\emptyset\} \subseteq S$ for all $X \in S$

Theorem 29.6

Any finite abstract simplicial complex can be realized as a simplicial complex in $\mathbb{R}^n$ for some n.

Proof: Say V has k elements. Let Δ be the $(k-1)$-dimensional simplicial complex corresponding to

$$\left\{x \in \mathbb{R}^k \,:\, x_i \geq 0,\ \sum_{i=1}^{k} x_i = 1\right\}$$

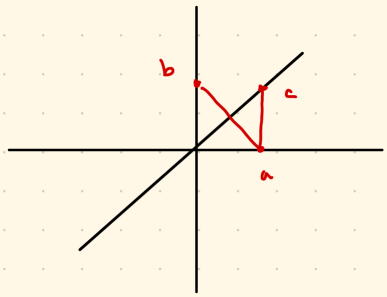
Any collection of vertices in Δ spans a simplex. Thus, if $V = \{v_1, \ldots, v_k\}$, match these to the vertices $\{e_1, \ldots, e_k\}$ of Δ, so that for each $X \in S$, we see that the vertices corresponding to the elements of X form a simplex. Do this for all $X \in S$ and we get a simplicial complex.

■

Example: Realizing an Abstract Complex

$V = \{a, b, c\}$, $S = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}\}$. Then, putting this in $\mathbb{R}^3$, we see the corresponding simplicial complex is

$$V' = \{e_1, e_2, e_3\}, \quad S' = \{\{e_1\}, \{e_2\}, \{e_3\}, \{e_1, e_2\}, \{e_1, e_3\}\}$$



11/6/26 Lecture 30: Orientations, Chains, and Boundaries

Let X be a topological space. We want invariants of X which are homotopy invariant. We have π_0 and π_1. Using π_0, we showed $\mathbb{R} \ncong \mathbb{R}^n$ for $n > 1$, and using π_1, we showed $\mathbb{R}^2 \ncong \mathbb{R}^n$ for $n > 2$.

One way to extend to $\mathbb{R}^k$ is using the *higher homotopy groups* $\pi_k(X, x)$ for $k \geq 0$, given by

$$\pi_k(X, x) = \{\text{homotopy classes of maps } S^k \to X \text{ sending the north pole to } x\}$$

Fact: This has a group structure for $k \geq 1$, and for $k > 1$, these are abelian.

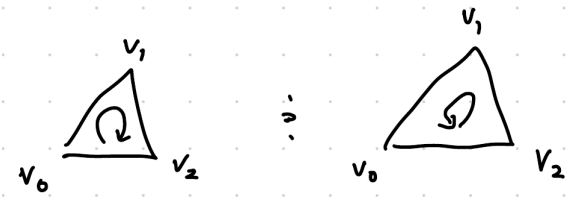
Problem: Hard to compute. We don't even know $\pi_k(S^2, x)$ for all k.

Our Approach: Simplicial Homology.

Pro: Relatively easy to compute.

Orientation of a Simplex

We want to distinguish



First case: Denote it with the added information ("orientation") (v_0, v_1, v_2) or a cyclic permutation of that.

Second case: Denote it with the added information ("orientation") (v_0, v_2, v_1) or a cyclic permutation of that.

Definition 30.1: Orientation

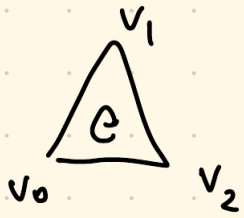
Let A be a simplex other than a point, and consider orderings of its vertices. Say two orderings are *equivalent* if they differ by an even permutation. The two equivalence classes are called *orientations* of A.

Definition 30.2: Induced Orientation

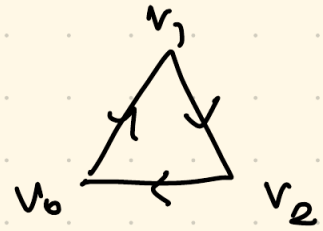
Say the vertices of A have an orientation given by $(v_0, v_1, \ldots, v_k)$. Say B is the face obtained by deleting v_i. We get an ordering $(v_0, \ldots, \widehat{v_i}, \ldots, v_k)$ of the vertices of B (the notation means remove v_i).

If i is even, say this is the *induced orientation* of B. If i is odd, it's the opposite of the induced orientation.

Example: A Triangle



The orientation of A is (v_0, v_1, v_2). Then, (v_1, v_2) is right, (v_0, v_2) is wrong so (v_2, v_0) is right, and (v_0, v_1) is right:



Say $(G, +)$ is abelian.

Definition 30.3: Generating Sets

A subset X of G is a *generating set* if every $g \neq 0$ can be written as

$$g = n_1 x_1 + \cdots + n_k x_k$$

where $n_i \in \mathbb{Z} \setminus \{0\}$, $x_i \in X$, and $x_i \neq x_j$ if $i \neq j$.

Definition 30.4: Free Abelian Groups

A generating set is *free* if such a representation as above is unique. We then call G a *free abelian group*.

Definition 30.5: Abelian Words

Given a set X, an *abelian word* is a formal finite sum $\sum n_i x_i$, $n_i \in \mathbb{Z}$, $x_i \in X$. It's *reduced* if $n_i \neq 0$ and $x_i \neq x_j$ if $i \neq j$. The reduced abelian words form a group $FA(X)$ generated by X.

Now, if G is an abelian group and X is a set of generators, we get a homomorphism

$$\phi : FA(X) \to G \quad \text{by} \quad \phi\left(\sum n_i x_i\right) = \sum n_i x_i$$

ϕ is onto, so let $N = \ker(\phi)$. Then, we see $G \cong FA(X)/N$, so we can talk about presentations and finite

presentations.

Given an abelian group G, set

$$T = \{g \in G : ng = 0 \text{ for some } n \neq 0\}$$

called the *torsion subgroup*. Note that then, G/T is torsion free (it has a trivial torsion subgroup).

If G is finitely generated, so is G/T, and it turns out we can write $G/T \cong \mathbb{Z}^n$ for some n. We call n the *rank*, $\text{rk}(G)$, of G.

Note that we can write $G \cong T \oplus G/T$, but the isomorphism is not unique.

Simplicial Complexes

Let K be a simplicial complex.

Definition 30.6: Chain Groups

Let $C_q(K)$ be the free abelian group generated by the oriented q-simplices of K, modulo the objects of the form $\sigma + \tau$ if σ and τ are the same simplex with opposite orientations. If $q = 1$, then as there is only one orientation of a vertex, $C_0(K)$ is just the free abelian group on the vertices.

Definition 30.7: Chains

An element of $C_q(K)$ is called a *q-chain*.

A q-chain is something of the form $\sum \lambda_i \sigma_i$, where $\lambda_i \in \mathbb{Z}$ and σ_i is an oriented q-simplex in K. Remember $(-\lambda)\sigma = \lambda(-\sigma)$, where $\lambda \in \mathbb{Z}$ and $-\sigma$ denotes σ with the opposite orientation.

It is clear that $C_q(K) \cong \mathbb{Z}^{n_q}$, where n_q is the number of q-simplices in K.

Definition 30.8: Boundary

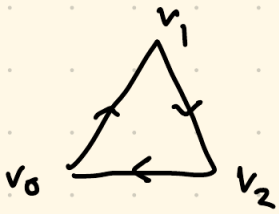
If the oriented q-simplex has vertices $(v_0, \dots, v_q)$, then the *boundary* is

$$\partial(v_0, \dots, v_q) = \sum_{i=0}^{q} (-1)^i (v_0, \dots, \widehat{v_i}, \dots, v_q) \in C_{q-1}(K)$$

Example: Boundary of a Triangle



$$\begin{aligned}\partial(v_0, v_1, v_2) &= (v_1, v_2) - (v_0, v_2) + (v_0, v_1) \\ &= (v_1, v_2) + (v_2, v_0) + (v_0, v_1) \in C_1(K)\end{aligned}$$



Now, we can extend to a map $\partial : C_q(K) \to C_{q-1}(K)$ by

$$\partial\left(\sum n_i\sigma_i\right) = \sum n_i(\partial\sigma_i)$$

If $q = 0$, we define $C_{-1}(K) = \{0\}$, so for $q < 1$, ∂ is the zero map.

Definition 30.9: Cycles and Boundaries

Let

$$Z_q(K) = \ker(\partial : C_q(K) \to C_{q-1}(K))$$

be the set of *q-cycles*, and let

$$B_q(K) = \operatorname{Im}(\partial : C_{q+1}(K) \to C_q(K))$$

be the set of *q-boundaries*.

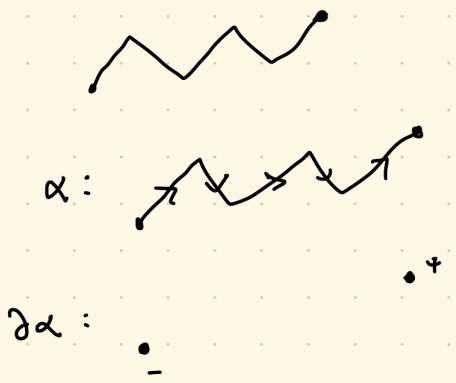
Example: Edge Paths

Say $(v_0, \dots, v_\ell)$ is an edge path (with $v_i \neq v_{i+1}$). Let

$$\alpha = (v_0, v_1) + (v_1, v_2) + \cdots + (v_{\ell-1}, v_\ell) \in C_1(K)$$

We see

$$\begin{aligned}\partial\alpha &= ((v_1) - (v_0)) + ((v_2) - (v_1)) + \cdots + ((v_\ell) - (v_{\ell-1})) \\ &= (v_\ell) - (v_0)\end{aligned}$$



If our edge path is a loop, with $v_\ell = v_0$, then $\partial\alpha = 0$, so α is a 1-cycle, so $\alpha \in Z_1(K)$.